

The background of the cover is a photograph of a beach at sunset or sunrise. The sky is filled with large, dark, textured clouds, with a bright patch of light where the sun is setting or rising. The ocean is visible in the middle ground, with white foam from breaking waves. In the foreground, two people are standing on the dark sand, looking out at the sea. The overall mood is contemplative and serene.

Looking Out to Sea

*New Zealand as a model
for ocean governance*

Raewyn Peart

Environmental Defence Society Incorporated

Looking
Out to Sea

Looking Out to Sea

*New Zealand as a model
for ocean governance*

Raewyn Peart

Environmental Defence Society

First published 2005

Published by:

Environmental Defence Society Incorporated

P O Box 95 152

Swanson

Auckland

New Zealand

Phone (09) 835 4350

www.eds.org.nz

www.rmaguide.org.nz

ISBN 0-9582641-3-9

This book is copyright. Apart from any fair dealing for the purpose of private study, research, criticism or review, as permitted under the Copyright Act, no part may be reproduced by any process without the permission of the publishers.

© Raewyn Peart, Environmental Defence Society Incorporated, 2005

Printed by ACT Printing Limited

Cover photograph by Raewyn Peart: Ocean Beach, Kawhia

*To my father,
for sharing with me
his love of the sea*

Table of Contents

Foreword	xiii
Acknowledgements	xv
Executive Summary	xix
Introduction	xix
Approaches to national oceans policy	xix
The use of New Zealand's oceans	xx
The governance of New Zealand's oceans	xx
Development of a New Zealand oceans policy	xxi
Conclusions	xxiii
Introduction	1
Part One	
Approaches to National Oceans Policy	
Introduction	3
1 What is a National Oceans Policy?	5
Definition of a national oceans policy	5
Spatial coverage of a national oceans policy	5
Elements of a national oceans policy	8
Outcomes of a national oceans policy	9
2 Why is a National Oceans Policy Needed?	11
Implementing international obligations	11
Managing environmental pressures	16
Maximising economic and social opportunities	20
3 Governance Challenges for Oceans Management	23
Large marine ecosystems	25
Fragmented jurisdictions	26
Multiple competing uses	27
Uncertain knowledge	27
Uncertain and polarised constituencies	28
Lack of established property rights regime	28
Rapid development of new uses	29
4 Governance Approaches to Oceans Management	31
Ecosystems-based management	32
Integrated oceans management	33

Adaptive management	38
Constituency building	39
5 Development of a National Oceans Policy	41
Political leadership	41
Time and resources	42
Science and policy mix	43
Legislation	43
Phasing	44
Part One Conclusions	47
Part Two	
The Use of New Zealand's Oceans	
Introduction	49
6 What Ocean Resources Does New Zealand Have?	51
Extent of the oceans	51
Living marine resources	53
Non-living marine resources	57
7 How Are New Zealand's Ocean Resources Used?	59
Commercial extractive uses	59
Transportation	66
Cultural	66
Enjoyment	68
Future uses	71
8 How Do Uses Impact on Ocean Ecosystems?	73
Marine habitat alteration and destruction	73
Harvesting of marine resources	76
Water pollution and sedimentation	81
Invasive marine species	85
Toxic blooms	86
Part Two Conclusions	87
Part Three	
The Governance of New Zealand's Oceans	
Introduction	89
9 Oceans governance framework	91
Sustainable management practices	92
Special marine management areas	93

Representative network of highly protected marine areas	93
Functioning of governance system	95
'Snapshot' of New Zealand's oceans governance system	95
10 Sustainable Management Practices	99
Fisheries management regime that addresses the impacts of fishing activity on marine ecosystems	99
Catchment management regime that effectively manages marine sedimentation and pollution from land-based sources	100
Environmental impact regime that ensures that structures and activities which are established in the marine area do not disrupt marine ecosystems	102
Marine pollution management regime that effectively manages the discharge of pollutants from ships and marine structures	103
Effective bio-invasion prevention and response capability	104
Evaluation	105
11 Special Marine Management Areas	107
Marine mammal sanctuaries	107
Areas closed to fishing activity	107
Mataitai reserves	108
Taiapure-local fisheries	108
Foreshore and seabed reserves	109
Marine parks	109
Areas protected by special legislation	109
Areas of significant conservation value	111
Areas to be avoided by shipping	111
Cable exclusion areas	111
Marine Protected Areas Policy Statement and Implementation Plan	111
Evaluation	113
12 Network of Highly Protected Marine Areas	115
Legislative provisions	115
Current marine reserves	116
Establishment of marine reserves	120
Evaluation	124
13 Functioning of Governance System	125
Connectivity	125
Adaptive management	127
Strong constituency	131
Evaluation	132

14 What Should New Zealand's Oceans Policy Contain?	133
What might a world-leading oceans governance system look like?	133
How might an oceans policy help New Zealand get there?	134
Part Three Conclusions	139
Conclusions	141
Appendices	
Comparator Country Case Studies	
A Australia's Oceans Policy	145
Context	145
Origin of national oceans policy	146
Development of policy	147
Content of policy	149
Implementation	153
Evaluation	158
B Canada's Oceans Policy	161
Context	161
Origins of national oceans policy	161
Development of policy	164
Content of policy	164
Implementation	165
Evaluation	171
C The USA's Oceans Policy	173
Context	173
Origins of national oceans policy	174
Development of policy	175
Content of policy	176
Implementation	181
Evaluation	181
D New Zealand's Oceans Policy	183
Origins of national oceans policy	183
Development of policy	184
Possible content of policy	189
Evaluation	191
References	193

List of Figures

Figure 1.1	Definition of national oceans policy	5
Figure 1.2	Different maritime jurisdictions under UNCLOS	6
Figure 1.3	Elements of national oceans policy	9
Figure 1.4	Potential outcomes of national oceans policy	9
Figure 2.1	Reasons for preparing a national oceans policy	11
Figure 2.2	Summary of key international environmental obligations	16
Figure 2.3	Coastal populations around the world are rapidly increasing – Vancouver.	16
Figure 2.4	The relationship between key ocean environmental pressures and disturbances	17
Figure 2.5	Coastal developments can result in the removal of important habitats including dunes – Omaha Beach, Tawharanui Peninsula.	18
Figure 2.6	The oceans governance challenge	21
Figure 3.1	Time line for development of national oceans policy	24
Figure 3.2	Key governance challenges	25
Figure 3.3	Definition of precautionary approach	28
Figure 4.1	Linkage between governance challenges and approaches	31
Figure 4.2	Summary of national oceans management integrative bodies	36
Figure 6.1	New Zealand's maritime jurisdictions	52
Figure 6.2	Ocean currents around New Zealand	53
Figure 6.3	Estimated number of marine species in New Zealand's EEZ	54
Figure 6.4	The endemic New Zealand sea lion is currently threatened.	56
Figure 6.5	New Zealand is the seabird capital of the world – Australasian gannet.	57
Figure 6.6	Hydrocarbon basins in New Zealand	58
Figure 7.1	Employee numbers in the seafood industry	60
Figure 7.2	Principal fisheries exports by value 2000 to 2003	61
Figure 7.3	Commercial aquaculture harvest 1983 to 2000	63
Figure 7.4	Aquaculture is a fast-growing industry – oyster farm located within the Ohiwa Harbour.	64

Figure 7.5	New Zealand receives around 3300 calls from international trading ships each year – ship departing from Port Chalmers, Dunedin.	66
Figure 7.6	Number of international visitors involved in marine activities and visiting marine attractions in New Zealand 1998 to 2004	69
Figure 7.7	Development can degrade the natural character of the coastal environment – Langs Beach, Whangarei.	70
Figure 8.1	Estimated stock levels of hoki and orange roughy as percentage of virgin biomass	77
Figure 8.2	Annual by-catch of non-target species	78
Figure 8.3	Estimated number of seabirds captured by specific fisheries during the 2000-01 fishing year	79
Figure 8.4	Number of sea lions killed in the squid fishery 1987–88 to 2002–03	80
Figure 8.5	Mangroves continue to expand in northern estuaries owing to increased sedimentation – Miranda coast, Firth of Thames.	82
Figure 8.6	Beaches are often closed in New Zealand owing to sewage contamination – Kawakawa Bay, Manukau City.	84
Figure 9.1	Elements of an oceans governance framework	91
Figure 9.2	Principles for marine reserves	94
Figure 9.3	Key elements of New Zealand's oceans environmental governance system	97
Figure 12.1	Marine reserves in New Zealand	118
Figure 12.2	Location of marine reserves and applications	119
Figure 12.3	There can be strong local opposition to marine reserve proposals – signs displayed at Nugget Point, the Catlins.	121
Figure 12.4	Time taken to process marine reserves established between 1971 and 1999	123
Figure 13.1	Processing times for regional coastal plans prepared by regional councils	131
Figure B.1	Structure of proposed ESSIM forum in Canada	169
Figure C.1	New USA ocean governance structure	181
Figure D.1	Initial process to develop New Zealand's oceans policy	184

Foreword

If one grasps a globe of the Earth and rotates it so as to maximise the amount of ocean seen on one side and the amount of land on the far side, thus dividing the world up into a water hemisphere and a complementary land hemisphere, then the island nation of Aotearoa would lie at the pole of the water hemisphere. In other words, New Zealand is surrounded by more ocean than any other nation on the planet!

So then, one might wonder why such a modest-sized country with a relatively small population should worry about protecting the vast ocean that surrounds it. Isn't the ocean virtually infinite in size, unlimited in its potential to wash away human and industrial wastes and by-products, to disperse the sediment loads derived from inland erosion and carried by swollen rivers? And aren't the living resources of the oceans so abundant that we need never worry about depleting fish stocks? And what about global warming? Is that so uncertain and the effects so minimal that we should dismiss climate change as merely another natural fluctuation in the earth's climate that has been going on for millions of years?

The answer to all these questions is a resounding no! New Zealand needs to worry because the burden caused by the cumulative effects of ocean-related activities is most acutely felt around the country's open coasts, bays, estuaries, fiords, sea straits and continental shelves. It's a bit like dealing with the dirty rim around the bathtub left concentrated after a number of grubby young siblings have had a good scrub.

The world's oceans cover 71 per cent of the planet and are divided into a number of geographic zones, where there exist distinct water masses, circulation patterns and marine ecological communities. These regions are separated by frontal zones, where water properties change very rapidly. New Zealand straddles such a major ocean boundary, the subtropical convergence, separating warm, northern subtropical water from much cooler subantarctic water to the south. Sea life is abundant in these frontal zones, which accounts in part for New Zealand's rich diversity of marine plants, fish, marine mammals and seabirds.

New Zealand lies at the boundary of the Pacific – Australian tectonic plates, a region of high seismic and volcanic activity. This gives rise to a rich and complex ocean bathymetry surrounding the country, with its fault lines, sea mounts, giant underwater landslides and sediment structures which all contribute to the steering of undersea current systems, influence habitat for fish and mammals, shape coastal features and influence offshore mineral and hydrocarbon deposits.

Raewyn Peart's treatise is a virtual encyclopaedia of New Zealand's existing laws, regulations and new initiatives pertaining to the development and implementation of a new national oceans policy. As she states, New Zealanders have always been passionate about the ocean. The sea has played a central role over periods of colonization – by Māori, Pākehā and modern-day immigrants. This love affair with the sea has helped

define the national psyche and shaped how cultural values have developed and how immigrant peoples came to terms with their great isolation from their former homelands, often tens of thousands of kilometres away.

But the country has moved well beyond the European pioneering stage, where the intent was to tame the wildness of the land, to clear vast tracts of forests to create the farms and industries that would support the growing population and communities, and to build the infrastructure as the country industrialized and prospered. Now, in the post-industrial age, there is a growing awareness and even alarm that fisheries are being seriously depleted, offshore resources of natural gas are drying up, land-use practices have given rise to serious erosion problems, and the seas and beaches around New Zealand's spectacular coastal communities are being degraded.

Planet Earth is immersed in a pivotal time of change. Global climate change cannot be dismissed but must be faced squarely. Mt Kilimanjaro in Kenya has lost its permanent ice cap for the first time in 11,000 years. Similarly, scientists report that the breaking off of the huge Larson Ice Shelf in west Antarctica is unprecedented in the Holocene period spanning the last 11 millennia. The north polar ice cap is thinning at an alarming rate. Huge deposits of frozen methane hydrate deposits line the deep continental slopes of many countries, including New Zealand. If ocean currents warm or change their paths, huge amounts of methane gas, a potent greenhouse gas, could be released into the atmosphere with dire consequences.

Oceanographers have recently analysed 10 years of ocean temperature data, measured using sensitive, modern, drifting profiling buoys. There is a very small, but measurable, increase in the world's ocean temperature which is consistent with the greenhouse effects of the burning of fossil fuels. European marine ecologists have recently learned that there are migratory shifts occurring in fish populations in the North Sea as they seek refuge by moving northward to cooler waters. New Zealand has taken a first step in signing the Kyoto Protocol to limit greenhouse gas emissions, but much more needs to be done.

Raewyn Peart's detailed analyses of the historical record, recent attempts to restart the oceans policy debate and the possible contents of such a policy are 'must' readings for elected officials, government policy planners, regional councils, NGOs, academics and their students in law, economics, marine science and ecology. *Looking Out to Sea* should be widely distributed to public, school and university libraries around the nation, its content folded into school and university curricula and debated in forums all around the country. The issues involved are so far-reaching and urgent, they need to be re-injected into the national consciousness as quickly and efficiently as possible.

Malcolm Bowman

Professor of Oceanography and Distinguished Service Professor at the State University of New York at Stony Brook and Director of the Environmental Defence Society, New Zealand

Acknowledgements

Oceans governance is a large and complex topic, and one which I would not have been able to grapple with without the help of many people involved in the field. Those I contacted for information and comment, without exception, generously gave up their time to discuss aspects of oceans management with me and/or reviewed parts of the material in this publication. I would like to warmly thank them. They include:

- Chris Anderson, Senior Policy Officer, Environment Division, Ministry of Foreign Affairs and Trade, Wellington
- Allison Arnold, Marine Conservation Officer, WWF – New Zealand, Wellington
- Sam Baird, Chief, Oceans Policy – Pacific Region, Fisheries and Oceans Canada, Vancouver
- Bill Ballantine, retired scientist, Leigh Marine Laboratory, Leigh
- Mark Bellingham, Lecturer, Massey University, Auckland
- Randall Bess, Spatial Allocations Manager, Ministry of Fisheries, Wellington
- Malcolm Bowman, Professor of Oceanography, Marine Sciences Research Center, State University of New York
- John Buckeridge, Professor; Director, Earth & Oceanic Sciences Research Centre, Auckland University of Technology, Auckland
- Jim Cannon, Conservation International, Brussels
- Maria Cassidy, Biosecurity New Zealand, Wellington
- Jackie Challis, Principal Policy Analyst, Department of Conservation, Wellington
- Ho Chang, Scientist, National Institute of Water and Atmospheric Research Limited, Wellington
- David Crawford, General Manager, Analysis and Government Relations, Maritime Safety Authority of New Zealand, Wellington
- Charles Crothers, Professor of Sociology, Auckland University of Technology, Auckland
- Danica Devery-Smith, Marine Conservation Unit, Department of Conservation, Wellington
- Chris Edkins, Department of Conservation, Wellington
- Riki Ellison, Manager, Marine and Treaty Team, Ministry for the Environment, Wellington
- Nici Gibbs, Policy Manager, The New Zealand Seafood Industry Council, Wellington
- Ben Goodsir, Senior Policy Analyst, Department of Premier and Cabinet, Hobart
- Lindsay Gow, Deputy Chief Executive, Ministry for the Environment, Wellington

- Dennis Gordon, Scientist, National Centre for Aquatic Biodiversity and Biosecurity, National Institute of Water and Atmospheric Research Limited, Wellington
- Janet Grieve, Scientist, National Institute of Water and Atmospheric Research Limited, Wellington
- Meredith Hall, Manager, South Team, National Oceans Office, Hobart
- Marcus Haward, Programme Leader – Policy, Antarctic Climate & Ecosystems Cooperative Research Centre, Hobart
- David Hill, Director, Hill Young Cooper, Auckland
- Greg Hill, Coastal Planner, Auckland
- Murray Hosking, Department of Conservation, Wellington
- Liz Jones, Policy Analyst, Biosecurity Group, Ministry of Fisheries, Wellington
- Stephen Karaitiana, Policy Advisor Resource Policy, Wellington Regional Council, Wellington
- Shane Kelly, Team Leader in Marine Ecology, Auckland Regional Council, Auckland
- Lorne Kriwoken, Senior Lecturer, Centre for Environmental Studies, University of Tasmania, Hobart
- Megan Linwood, Senior Advisor, Ministry for the Environment, Wellington
- Dominic McCarthy, Environmental Scientist, Auckland Regional Council, Auckland
- Barbara Musso, Manager, Policy Team, National Oceans Office, Hobart
- Jim Nicolson, Principal Policy Analyst, Conservation Policy Division, Department of Conservation, Wellington
- Steve O'Shea, Senior Research Fellow, Auckland University of Technology, Auckland
- Jonathan Peacey, Chief Policy Analyst, Ministry of Fisheries, Wellington
- Lisa Sheppard, Manager, New Zealand Oceans Policy Secretariat, Wellington
- Gary Taylor, Chairman, Environmental Defence Society, Auckland
- Christophe Tulou, Director, The Center for SeaChange, Washington
- Cath Wallace, Senior Lecturer in Economics and Public Policy, Victoria University Wellington
- Kathy Walls, Senior Marine Conservation Officer: Marine Protected Areas, Department of Conservation, Wellington
- Geoff Westcott, Associate Professor, School of Ecology and Environment, Deakin University, Melbourne
- Gerard Willis, Director, Enfocus, Auckland
- Rachel Wilson, Advisor, Ministry for the Environment, Wellington
- Kirsty Woods, Senior Policy Analyst, Te Ohu Kai Moana (Treaty of Waitangi Fisheries Commission), Wellington

Comments received from the above people were personal to them and do not necessarily reflect the policy of their organizations. In addition, I do not wish to imply in any way that the people who contributed agree with everything in this publication or endorse it in any way. It remains my perspective on oceans governance and any errors or omissions in it are solely my responsibility.

This publication would also not have been possible without financial support. On behalf of the Environmental Defence Society I would like to gratefully acknowledge Conservation International for financially supporting the research underpinning this publication and providing valuable peer review comments on the outcomes.

I would also like to thank Neale Wills and Margaret Peart for assisting with the publication's production.

Finally, I would like to thank my husband Charles and our daughter Tanya for their steadfast support during the many months of hard work which the project required.

Raewyn Peart
Senior Policy Analyst
Environmental Defence Society

Executive Summary

Introduction

New Zealand is blessed with a vast ocean which teems with life and is the source of valuable oil, gas and minerals. New Zealanders are passionate about their ocean. It plays an important role in the nation's social, cultural and economic well-being.

Although New Zealand has been a world leader in some aspects of oceans management, it has fallen behind in other critical areas. To address some of these weaknesses, the government commenced the Oceans Policy Initiative in 2000. This has provided an invaluable opportunity to redesign New Zealand's oceans governance system for the twenty-first century.

Much can be learnt from the experiences of other countries in developing oceans policies, as well as from the experiences of oceans managers within New Zealand. This publication reviews international thinking and experience on oceans governance. It investigates what we know about the current state of New Zealand's oceans and the effectiveness of management efforts. It concludes with recommendations on how an oceans policy could assist New Zealand to develop a world-leading oceans governance system.

Approaches to national oceans policy

A national oceans policy can be defined as 'the development of a common purpose for the management of oceans within a country's jurisdiction and the establishment of a governance system to deliver that purpose'.

The three main reasons why a country might embark on the preparation of a national oceans policy are to implement international obligations, to more effectively manage environmental threats to its ocean areas and to maximise economic and social opportunities arising from ocean use.

The effective governance of oceans within national jurisdictions has proved to be enormously difficult, as has the development of fully integrated national oceans policies. International debate on the kinds of governance approaches which are required to successfully manage ocean areas has focused on the need to develop a governance system which is ecosystems-based, integrated and adaptive, and which is supported by a strong constituency.

Practical considerations associated with the development of a national oceans policy include acknowledging the need for strong political leadership to drive the process forward, allocating sufficient time and resources to the process, ensuring an

appropriate mix between science and policy inputs, and phasing implementation while political support for more fundamental change is developed. The passing of legislation for an integrated oceans policy does not appear to determine its success, although it may help.

The use of New Zealand's oceans

New Zealand has a large and highly diverse marine area with very rich marine biodiversity. The oceans are thought to contain between one-third and three-quarters of New Zealand's indigenous species and are an international hot spot for some of these. There are significant deposits of oil, gas and minerals within New Zealand's continental shelf, only a small proportion of which is currently being exploited. These ocean resources support substantial seafood and mining industries, marine biotechnology, shipping and marine tourism. The oceans are very important to Māori culture and significantly contribute to New Zealanders' way of life and well-being.

Fishing has a major impact on marine ecosystems through trawling and dredging activities which damage habitats on the ocean floor, removing large amounts of marine life from the sea, and inadvertently catching protected marine mammals and seabirds. The discharge of land-based sediments and pollutants is also having a negative impact on marine habitats through smothering bottom-living organisms, reducing underwater light levels, and accumulating heavy metals in the food chain. The introduction of invasive exotic marine species into New Zealand is a continuing problem.

The governance of New Zealand's oceans

An optimal oceans governance framework can be thought of as consisting of three key elements:

- Sustainable management practices applied to activities impacting on the marine environment
- A network of special marine management areas incorporating sensitive and important marine environments where additional management effort is required
- A representative network of highly protected marine areas (often referred to as marine reserves or marine national parks)

Insufficient information is available to accurately assess the extent to which management practices are effective in ensuring that human activities are not seriously disrupting marine ecosystems. However, four notable gaps are evident:

- The failure of fisheries management to seriously grapple with addressing environmental impacts of fishing activity and, in particular, the impacts of trawling and dredging on benthic habitats
- The failure in many areas to reduce the levels of land-based sedimentation and pollution entering the marine area
- The lack of a consistent environmental regime to manage the environmental effects of activities outside the territorial sea
- The lack of an effective mechanism to control the arrival of invasive marine species on vessels' hulls

The system for establishing and governing special management areas is highly fragmented, with areas being established under at least nine pieces of legislation and administered by 10 different management bodies. New Zealand falls far short of adequate provision for highly protected areas, with inadequate coverage of representative areas of all habitats and important marine biodiversity hot spots.

Development of a New Zealand oceans policy

So how might an oceans policy help New Zealand develop a world-leading oceans governance system? It could do so by including the following:

Common purpose

- Provide a clear statement of the purpose of oceans management in New Zealand and establish a set of common principles to guide oceans management
- Make a commitment to progressively apply ecosystems-based approaches to oceans management planning and decision making
- Make a commitment to protect representative areas of all New Zealand's marine habitats, and all special and unique areas, in a viable network of highly protected areas

Integrative mechanisms

- Establish a permanent *Oceans Ministerial Group* to provide high-level policy direction
- Establish an *Oceans Management Group*, consisting of high-level officials of relevant central and local government departments and agencies, to coordinate activities, share information and undertake joint projects
- Establish an *Oceans Science Group* to provide independent scientific advice to the *Oceans Ministerial Group* and the public

- Establish *Oceans Stakeholder Advisory Groups*, including a *Māori Advisory Group*, to provide stakeholder input into oceans policy priorities and initiatives

Strategic actions

- Support the passage of the Marine Reserves Bill and provide substantial dedicated funding for the identification and establishment of marine reserves
- Develop a revised statutory framework for the establishment and management of special marine management areas
- Undertake a national investigation into the impacts of trawling and dredging on benthic habitats and identify effective mitigation measures to be adopted, incorporating the precautionary principle
- Prepare a *Guide to Sustainable Coastal Development* and give key provisions statutory force through incorporation into the New Zealand Coastal Policy Statement (NZCPS)
- Undertake an investigation into coastal planning under the Resource Management Act 1991 (RMA), at both regional and territorial authority levels, and identify reforms to increase both the effectiveness of planning in achieving sustainable coastal management and its ability to respond quickly to changing information and circumstances
- Develop a *National Agenda for Sustainable Estuaries Management* which scopes the problems, develops solutions, and targets resources to improve the management of estuaries and harbours of national ecological importance
- Develop an environmental regime for the management of activities outside the territorial sea
- Develop and resource the implementation of a *Marine Biosecurity Strategy*
- Establish a contestable *Oceans Constituency Fund*

Performance management systems

- Develop a performance management system for oceans management, drawing on international best practice
- Develop an *Oceans Information Strategy*
- Prepare a *State of the Oceans Report* and regularly update (at least five-yearly)
- Task the Parliamentary Commissioner for the Environment with undertaking regular (three-yearly) independent reviews of progress in implementing the oceans policy

Implementation

- Continue the role of *Oceans Minister* to provide oversight of the implementation of the initiative
- Establish an *Oceans Agency* to champion the further development and implementation of the policy
- Provide sufficient *dedicated funding* to enable successful implementation

Conclusions

New Zealand governs an expansive area of the world's oceans, which is enormously important to the country's future welfare. Current marine governance arrangements are not up to the task of managing this invaluable resource in several key areas. A national oceans policy could provide significant assistance in addressing this, through providing an improved organizational, legal and policy framework and focusing coordinated efforts and resources on key areas.

The oceans policy initiative provides New Zealand with the opportunity not only to improve outcomes for its own oceans but to become an exemplar of best practice oceans governance for the rest of the world. It is an opportunity not to be missed.

Introduction

New Zealand is a maritime nation, settled by some of the greatest seafaring peoples in the world. The country's oceans are vast. They span some 4.2 million square kilometres, 15 times the size of New Zealand's land area.

These oceans are very rich in life. Whales, dolphins, seals, sea lions, penguins, albatrosses, petrels and numerous other seabirds frequent our coasts. The country has a diverse marine environment which includes deep sea trenches, hydrothermal vents, underwater mountains, forests of kelp and extensive bryozoan coral beds, as well as a vast range of other habitats. These provide homes for the estimated 55,000 to 75,000 marine species living in New Zealand's seas. There are also significant quantities of oil, gas and valuable minerals under the seabed.

New Zealanders are passionate about their oceans. Most live close to the sea and summer holidays are traditionally spent at the beach. Many people are keen fishers and boaties. For Māori, the use and management of ocean resources is an integral part of their culture and identity.

New Zealand has been an international leader in some aspects of oceans management. It was reputedly the first country to establish a no-take marine reserve. The country's fisheries management, which implemented individual transferable quotas, is looked to from around the world as a model to learn from. Its regional planning system, whereby catchment-based agencies manage both land and marine areas on a sustainable management model, is also internationally progressive.

In many respects, however, oceans management in New Zealand has fallen behind. Only a tiny fraction of the country's oceans is protected by marine reserves. Fisheries management has yet to effectively grapple with addressing the environmental effects of fishing. Coastal management has largely failed to stem ongoing sedimentation and pollution of marine areas. There is no comprehensive environmental management regime for the bulk of New Zealand's oceans located outside territorial limits, and the country is still exposed to a considerable risk of damaging marine bio-invasions.

Problems with New Zealand's oceans governance system have not gone unnoticed. They were detailed in a report by the Parliamentary Commissioner for the Environment (PCE) released in 1999. In early 2000, the government initiated the preparation of a National Oceans Policy. Work was undertaken to scope problems and solutions but was brought to a halt in 2003 while the ownership of the foreshore and seabed was addressed. Now that this issue has been determined with the passage of the Foreshore and Seabed Act 2004, the initiative seems likely to recommence in early 2006.

The oceans policy initiative provides a valuable opportunity to redesign New Zealand's ocean governance system for the twenty-first century. It enables a critical evaluation of what a national oceans policy is, why New Zealand might need one, what

governance challenges such a policy might need to address, how it might approach addressing them and what a national oceans policy for New Zealand might contain.

The following chapters explore approaches to these issues, drawing on both international and national experience. The research undertaken for this book focused initially on reviewing international literature to develop an understanding of the role of national oceans policies in improving oceans governance. The results of this research are presented in *Part One: Approaches to National Oceans Policies* and provide the framework for the subsequent analysis of oceans governance in New Zealand.

The concepts identified in the international review were fleshed out by undertaking case studies of the development of national oceans policies in Australia, Canada and the USA, as well as documenting the process that has been undertaken in New Zealand to date. Relevant material from the case studies is incorporated into Part One and more detailed information on the individual country experiences is set out in the Appendices.

The focus of the research then shifted to the New Zealand situation. Information gathered from the literature review and interviews with key stakeholders was drawn together to develop a picture of New Zealand's oceans resources, how they are currently used, how these uses are impacting on the marine environment and how these impacts are currently being managed. Ideas were then developed as to what an oceans policy might contain in order to assist New Zealand to become an international leader in oceans governance. This material is set out in *Part Two: The Use of New Zealand's Oceans* and *Part Three: The Governance of New Zealand's Oceans*.

This material has been prepared with the aim of increasing understanding of the role that national oceans policy can play in improving oceans governance and to enrich the debate surrounding the development of New Zealand's first national oceans policy.

Part One

Approaches to National Oceans Policy

Introduction

This Part firstly develops a definition of national oceans policy, describes the key elements of such policy, and identifies some of the outcomes that could be achieved through successfully implementing an integrated national oceans policy.

Next, it goes on to explore why a country might need to develop a national oceans policy, the reasons why the effective governance of oceans has proved to be so difficult, and the range of governance challenges which an oceans policy might need to address.

Finally, approaches which are being developed to address these challenges and practical issues related to the successful development and implementation of national oceans policies are discussed.

This material provides the context and a conceptual framework for the subsequent description and analysis of New Zealand's oceans governance system contained in Parts Two and Three. More detailed case study information from Australia, Canada, the USA and New Zealand, which is summarised in this Part, is contained in the Appendices.

What is a National Oceans Policy?

Definition of a national oceans policy

Most countries, including New Zealand, have a raft of laws, public policies and management approaches which apply to the use and management of coastal and marine areas. These have often been developed in a piecemeal fashion, over a lengthy time period, by different levels of government and in response to specific sectoral or area-based issues. Collectively, these elements could be said to comprise a country's oceans policy.

The term 'national oceans policy' is used here to mean something different from this. It refers to an overarching policy that has been developed to apply to the management of a country's entire ocean area, extending from coastal land to the outward extent of the Exclusive Economic Zone (EEZ) and continental shelf. A definition is shown in Figure 1.1. Such a national oceans policy represents an attempt to bring together a disparate group of ocean regulatory and management systems within a comprehensive and integrated approach to oceans management.

A national oceans policy comprises the development of a common purpose for the management of oceans within a country's jurisdiction and the establishment of a governance system to deliver that purpose.

Figure 1.1. Definition of national oceans policy

The common purpose for oceans governance is defined through the vision, objectives and goals of the oceans policy. The governance system required to achieve the common purpose consists of the policies, laws, practices and organisations which collectively deliver oceans management.

Spatial coverage of a national oceans policy

Under the United Nations Convention on the Law of the Sea (UNCLOS), countries exercise differing jurisdictional powers over various maritime areas. These areas include the territorial sea, archipelagic waters, the contiguous zone, the EEZ and the continental

shelf, where this extends beyond the EEZ (see Figure 1.2). Each of these areas is subject to a different allocation of rights and obligations.

All countries bordering the coast are able to extend the breadth of their territorial sea up to 12 nautical miles from the 'baseline', which is normally located along the low-water mark. Waters on the landward side of the baseline are known as internal waters. In the case of a state which is constituted wholly of groups of islands, an archipelagic baseline can be used. This is constructed by joining the outermost points of the outermost islands and drying reefs. The waters enclosed by the archipelagic baselines are known as archipelagic waters. A country's territorial sea, internal waters and archipelagic waters are part of its territory and within its national sovereignty. The powers of a coastal state within its territorial sea are essentially the same as those that it can exercise over its land area.

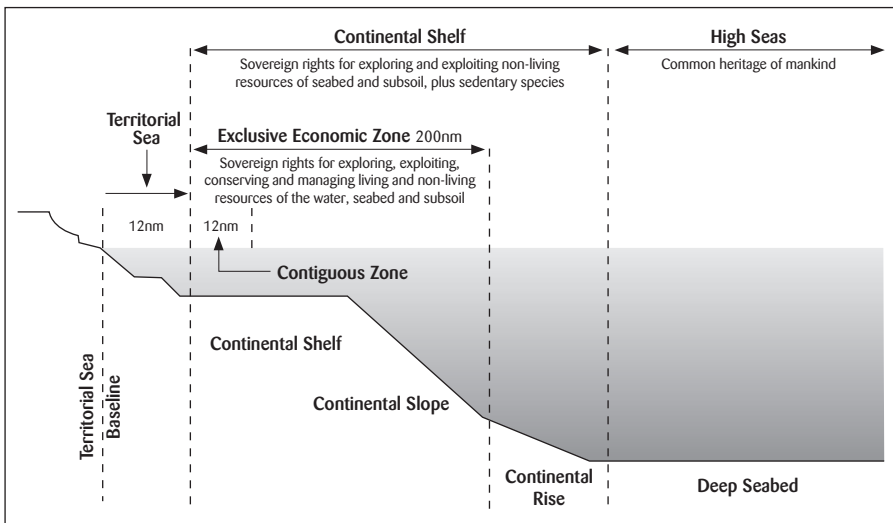


Figure 1.2. Different maritime jurisdictions under UNCLOS

Source: Adapted from Commonwealth of Australia, 1998a

A country can declare a contiguous zone over an area adjacent to the territorial sea and extending up to 24 nautical miles from the 'baseline'. This area is not part of its territory and therefore not subject to full sovereignty, but UNCLOS gives a country the right to police the area to prevent or punish infringements (of customs, fiscal, immigration or sanitary laws and regulations) that have been committed or are threatening to be committed in the territorial sea.

The Exclusive Economic Zone is usually the largest of a country's maritime areas, extending from the outer edge of the territorial sea up to 200 nautical miles seawards,

thereby overlapping the contiguous zone. In New Zealand and some other countries, this area extends over the edge of the continental shelf into deep ocean areas. Although the EEZ is not part of a state's territory and thereby not within its sovereignty, countries have sovereign rights within their EEZs for the purpose of exploring, exploiting, conserving and managing all of its living and non-living natural resources, including marine life, oil, gas and minerals, and producing energy from the water, currents and winds. Other countries retain the freedom of navigation and overflight, and are able to lay submarine cables and pipelines within all EEZs.

UNCLOS imposes on countries an obligation to ensure that the living resources in their EEZs are not endangered by over-exploitation and to maintain or restore populations of harvested species to levels that can produce the 'maximum sustainable yield'. This requires a country, among other things, to determine the allowable catch of living resources within their EEZ. Moreover, as part of the obligation to promote the objective of sustainable utilisation, where a country does not have the capacity to harvest the entire allowable catch, other countries must be given access to the surplus.

Where the outer edge of a country's continental shelf (including the continental slope and rise) extends beyond the EEZ, it is entitled to sovereign rights for the purpose of exploring and exploiting the natural resources of the continental shelf. These natural resources consist of mineral and other non-living resources within the seabed and subsoil as well as living organisms belonging to sedentary species. Payments to the International Seabed Authority established under UNCLOS are required in respect of mining resources from this part of the continental shelf.

Most countries, including Canada, the USA, Australia and New Zealand have declared a territorial sea, contiguous zone and EEZ to the maximum limits allowable under UNCLOS. National oceans policy initiatives usually cover all these areas. In some places, the boundaries of these areas overlap between countries and are disputed.

New Zealand has recently signed a treaty with Australia to settle the maritime boundaries between the two countries. This involves the Tasman Sea and adjacent areas of the south-western Pacific Ocean where the EEZs or continental shelves which could be claimed by each respective country overlap. New Zealand has still to conclude maritime treaties with Fiji and Tonga and to delimit the maritime spaces offshore from its Antarctic territory (Land Information New Zealand, 2005).

Although the maritime boundaries within which an oceans policy applies can be relatively easy to define, determining the appropriate landward extent of the policy can be a trickier issue. Activities on land areas well inland can affect oceans through their impact on rivers, which ultimately flow into estuaries and the sea, but it can be difficult in practice to integrate their management with marine areas. Environmental issues in these water catchments are often addressed through catchment management planning. Activities on land situated adjacent to the coast can have a more direct impact on marine areas through destroying coastal habitats and polluting the sea. Coastal

management strategies typically focus on the land–water interface and can be more readily integrated into ocean policy initiatives. In practice, hard inland boundaries are rarely drawn for ocean policies and the spatial area of interest differs, depending on the management issue being addressed.

In both Canada and the USA, where Oceans Acts have been passed, the legislative focus is on the marine environment, and coastal land is not explicitly included. In Australia and New Zealand, no legislation has yet been passed. In practice, all these initiatives have considered land-based impacts on the marine area.

Elements of a national oceans policy

From the analysis of the national oceans policies in the four case-study countries, five key elements of such policies can be identified (see Figure 1.3):

- *A common purpose* for the management of oceans, which is often articulated in the form of a vision for the oceans, and high-level management objectives and principles which are to apply to oceans management
- *Integrative mechanisms* to jointly harness the actions of different oceans managers to achieve the common purpose. This may be through joint committees, working bodies, advisory bodies and/or specialist oceans agencies with coordinating roles. Integrative mechanisms may also consist of processes developed to bring decision makers together on joint issues and to build consistency in approaches and the application of tools and principles
- *Strategic actions* which are required to implement the policy. These often focus on addressing gaps and conflicts in the current oceans management system
- *Area-based planning* which applies the principles of the policy to a specific marine ecosystem. This planning can bring together a range of stakeholders to jointly focus on ocean management issues within a specific ocean region. It may identify a range of management areas including those to be fully protected (such as marine reserves) and those to be managed for a range of outcomes and uses
- *Performance management systems* which include the ongoing monitoring of the health of the oceans and regular review of the effectiveness of management actions. Performance management is important when working in a context of uncertainty, as it enables progressive ‘learning by doing’ and the adaptation of management approaches in light of new information or changes in technologies, markets and community values

Some or all of these elements may be encapsulated in an Oceans Act and/or in policy documents approved by government.

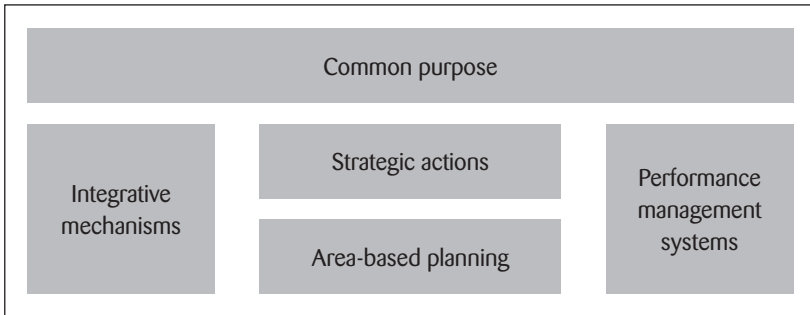


Figure 1.3. Elements of national oceans policy

Outcomes of a national oceans policy

The benefits of successfully implementing an integrated national oceans policy can include a number of mutually reinforcing outcomes (see Figure 1.4):

- *Improved decision making* on ocean management issues – through considering the impacts of the proposed course of action in an integrated manner and more carefully focusing scientific research on joint management needs
- *Reduced conflict* over oceans issues – through clarifying jurisdictional boundaries and responsibilities, engaging stakeholders in joint planning processes to address oceans issues, and providing mechanisms to proactively address potential conflicts before they arise
- *Improved environmental outcomes* – through higher-quality decision making and the management of marine ecosystems as entire systems
- *Improved economic and social outcomes* – through the maintenance of the natural resource base, greater clarity over rights to the access and use of marine resources, and the reduction of lengthy conflict resolution processes

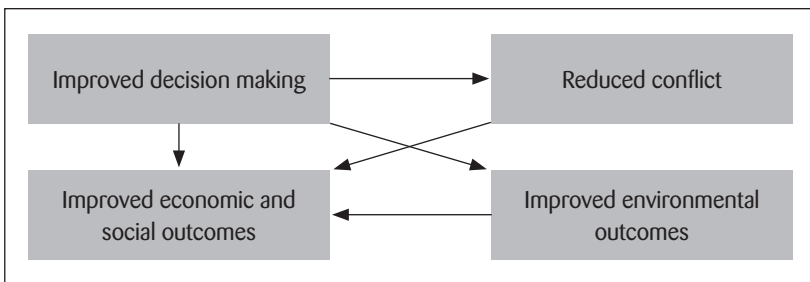


Figure 1.4. Potential outcomes of national oceans policy

2

Why is a National Oceans Policy Needed?

There are three main reasons why a country might embark on the preparation of a national oceans policy (see Figure 2.1):

- To implement international obligations
- To more effectively manage environmental threats to its ocean areas
- To maximise economic and social opportunities arising from ocean use

Other reasons for preparing a policy, which are not considered further here, include a desire to exert sovereignty over a spatial area and a desire to increase defence capability and security over that area.

The reasons for oceans policy preparation are highly interrelated. International obligations commonly arise as a result of perceived environmental threats as well as economic opportunities. Economic uses of the oceans can cause environmental threats, which, in turn, can threaten economic opportunities.

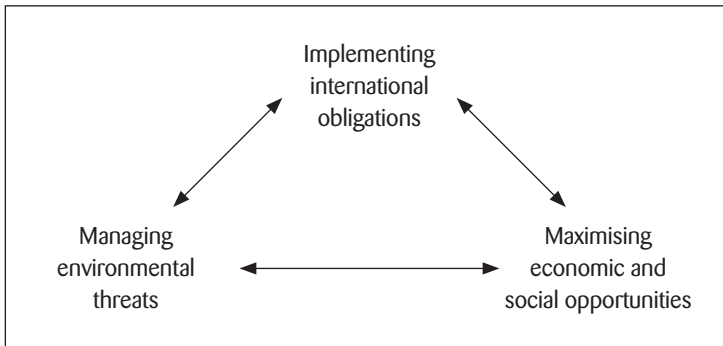


Figure 2.1. Reasons for preparing a national oceans policy

Implementing international obligations

There are numerous international conventions and non-legally binding documents that apply to the management of ocean areas within national jurisdictions. Of these,

the following are particularly significant for the development of national oceans policies:

- UNCLOS and related agreements
- The Convention on Biological Diversity (CBD)
- Agenda 21
- The Global Programme of Action for the Protection of the Marine Environment from Land-based Activities
- The Food and Agriculture Organisation of the United Nations (FAO) Code of Conduct for Responsible Fisheries
- The Plan of Implementation of the World Summit on Sustainable Development

United Nations Convention on the Law of the Sea

All countries which have ratified UNCLOS are subject to the general legal obligation to protect and preserve the marine environment. They are also required to take all measures necessary to prevent, reduce and control pollution of this environment. These steps include adopting laws and regulations to address pollution from land-based sources, seabed activities, dumping and the atmosphere. Countries must endeavour, as far as practicable, to observe, measure, evaluate and analyse the risks or effects of pollution on the marine environment and to publish reports on the results obtained. There is also an obligation to carry out an assessment of the potential effects of planned activities which may cause substantial pollution of, or significant and harmful changes to, the marine environment. One hundred and forty-five states, including New Zealand, Canada and Australia but not the USA, have now ratified UNCLOS. The Bush Administration has recently indicated an intention to seek USA ratification of UNCLOS.

Fish Stocks Agreement

In 1995 an agreement was adopted for the implementation of certain provisions of UNCLOS in relation to the conservation and management of straddling and highly migratory fish stocks (Fish Stocks Agreement). The Agreement builds on the provisions of UNCLOS by making them broader, stronger and more detailed. Among other things, the Agreement establishes a set of general principles which are to be applied to the management and conservation of straddling and highly migratory fish stocks. These include:

- Applying a precautionary approach

- Protecting the biodiversity of the marine environment
- Minimising pollution, waste, discards, catch by lost or abandoned gear, catch of non-target species and impacts on associated or dependent species
- Taking measures to prevent or eliminate overfishing and, where necessary, adopting conservation and management measures for species belonging to the same ecosystem

The Agreement strengthens obligations on coastal and high-seas fishing states to cooperate in respect of straddling and highly migratory fish stocks. Such cooperation is preferably to take place within regional fisheries management organisations. The Agreement came into force in 2001 and New Zealand, Australia, Canada and the USA are all parties to it.

Convention on Biological Diversity

The Convention on Biological Diversity (CBD) is aimed at the conservation of biological diversity, the sustainable use of its components, and the fair and equitable sharing of the benefits arising out of the utilisation of genetic resources. The Convention also applies to the marine environment, even though its obligations on components of biodiversity are not applicable beyond the limits of national jurisdiction (EEZ and outer continental shelf). Obligations on the parties include:

- Establishing a system of protected areas
- Promoting the protection of ecosystems and natural habitats and the maintenance of viable populations of species in natural surroundings
- Preventing the introduction of, and controlling or eradicating, those alien species which threaten ecosystems, habitats or species
- Integrating the consideration and sustainable use of biological resources into national decision making
- Introducing appropriate arrangements to ensure that the environmental consequences of programmes and policies that are likely to have significant adverse effects are taken into account

The convention was opened for signature at the 1992 United Nations Conference on Environment and Development (UNCED) and came into force in 1993. It currently has 188 parties, including New Zealand, Australia and Canada but not the USA.

Agenda 21

Agenda 21 is a non-legally binding document endorsed by governments, including New Zealand, at UNCED in 1992. It sets out a vision and action plan for the sustainable development of natural resources. Chapter 17 of Agenda 21 deals with the protection of oceans and coastal areas and the protection, use and development of their living resources.

Countries are, among other things, urged to commit themselves to the integrated management and sustainable development of coastal areas and the marine environment under their jurisdiction. This includes:

- Providing for an integrated policy and decision-making process
- Applying preventive and precautionary approaches to project planning and implementation
- Providing opportunities for consultation and participation in planning and decision making at appropriate levels

This may be achieved through the implementation at appropriate levels of integrated coastal and marine management plans and programmes and the integration of sectoral programmes.

Effective management is to be underpinned by good information, and countries are encouraged to conduct regular assessments of the state of the environment of their coastal and marine areas and to maintain databases and profiles.

With respect to the management of fisheries, many of the provisions of Agenda 21 reflect the requirements under UNCLOS. As an action plan, however, Agenda 21 goes further. Among other things, it promotes the establishment of marine protected areas by requiring nations to identify marine ecosystems exhibiting high levels of biodiversity and other critical habitat areas and providing necessary limitations on use in these areas. Priority is to be afforded to coral reef ecosystems, estuaries and temperate and tropical wetlands (including mangroves, seagrass beds and other spawning and nursery areas).

Global Programme of Action for the Protection of the Marine Environment from Land-Based Activities

This programme of action, formally adopted in 1995, urges countries to develop comprehensive, continuing and adaptive programmes of action within the framework of integrated coastal area management. These are to include providing for the identification and assessment of problems, establishing priorities, setting management

objectives for priority problems, and identifying, evaluating and selecting strategies and measures and evaluating their effectiveness.

Food and Agriculture Organisation of the United Nations Code of Conduct for Responsible Fisheries

This voluntary code, also adopted in 1995, covers a wide range of matters related to fisheries, including fisheries management and operations, aquaculture, the integration of fisheries with coastal management, post-harvest practices and trade, and fisheries research. The principles set out in the code acknowledge that 'the right to fish carries with it the obligation to do so in a responsible manner so as to ensure effective conservation and management of the living aquatic resources' (paragraph 6.1). Amongst many other things, the code promotes the integration of fisheries into coastal area management planning and development, and the establishment of procedures and mechanisms to resolve conflicts within the fisheries sector and between that sector and other users of the coastal area.

Plan of Implementation of the World Summit on Sustainable Development

This plan, adopted at the 2002 World Summit on Sustainable Development in which New Zealand participated, is designed to speed up progress in reaching the goals set out in Agenda 21. The many provisions of the plan include the establishment of a series of target dates for oceans management. These include:

- Substantial progress on protecting the marine environment from land-based activities by 2006
- Encouraging the application of an ecosystem approach by 2010
- Establishment of representative networks of marine protected areas by 2012
- Action to maintain or restore fish stocks to their maximum sustainable yield by 2015

A summary of the key international environmental obligations placed on countries by the above agreements is shown in Figure 2.2.

- Protect the biodiversity of the marine environment
- Ensure marine species are not over-exploited
- Control pollution of the marine environment
- Protect ocean ecosystems
- Establish a system of protected areas
- Prevent the introduction of alien species
- Carry out environmental impact assessments of activities which may have significant negative impacts on oceans
- Provide for an integrated system of policy making, planning and decision making for oceans
- Regularly monitor the state of the marine environment

Figure 2.2. Summary of key international environmental obligations

Managing environmental pressures

Pressures on the oceans are rapidly increasing as more people choose to live on the coast and technological advancements enable greater exploitation of ocean resources. Almost one-third of the world's population lives within 100 kilometres of the sea and two-thirds of all cities with over 2.5 million inhabitants are located on the coast (The Joint Group of Experts on the Scientific Aspects of Marine Environmental Protection (GESAMP), 2001: 19). Coastal populations both around the world and in New Zealand are rapidly increasing. At the same time, technological advancements have enabled deep-sea fishing and the exploitation of oil, gas and minerals from the continental shelf and deep ocean floor.



Figure 2.3. Coastal populations around the world are rapidly increasing – Vancouver

The environmental health of the world's oceans is deteriorating. In its review of the state of the oceans, GESAMP found that:

Fisheries are a shambles, grossly mismanaged and exploited everywhere. Coasts are overdeveloped. Habitats are being increasingly destroyed. And the introduction of species, either intentionally or accidentally, to habitats far from their own is now taking place on a large scale, often disrupting both ecosystems and economies (2001: 2).

When literature on oceans management is reviewed, four key environmental disturbances stand out as being largely responsible for the environmental degradation of the oceans (see Figure 2.4)¹:

- Destruction or alteration of marine habitats
- Over-harvesting of marine resources
- Water pollution and eutrophication
- Invasive species

The most significant pressures causing these disturbances appear to be:

- Poorly planned coastal development
- Poorly managed agricultural and fishing activities
- The transport of marine organisms by inter-regional shipping
- The impacts of climate change

Key environmental pressures	Key environmental disturbances			
	Destruction or alteration of marine habitats	Over-harvesting of marine resources	Water pollution and eutrophication	Invasive species
Coastal development				
Agriculture				
Fishing				
Shipping				
Climate change				

Note: The shaded areas indicate where a particular environmental pressure can result in a specific environmental disturbance.

Figure 2.4. The relationship between key ocean environmental pressures and disturbances

1 A much more detailed analysis of the relationship between specific activities and marine disturbances can be found at National Oceans Office (2002).

Similar environmental disturbances are being experienced, to varying degrees of severity, by coastal states around the world, including New Zealand.

Many marine habitats are being substantially altered through a variety of activities. Coastal developments can result in the removal of important habitats through the drainage of wetlands, destruction of mangrove forests, and removal of dunes. The building of dams and the extraction of water reduce the flow of rivers and the amount of sedimentation flowing down them to the sea. This can cause major changes to coastlines and has led to serious coastal erosion in many parts of the world. On the other hand, vegetation clearance and earthworks can increase the amount of sediment flowing into the ocean, smothering coral reefs and communities on the sea floor (GESAMP, 2001).



Figure 2.5. Coastal developments can result in the removal of important habitats including dunes – Omaha Beach, Tawharanui Peninsula.

Bottom trawling by fishing vessels can result in the destruction of seabed marine communities and these can take many years to recover, particularly in the deep ocean. Rising sea levels, the increasing temperature of sea water and the resultant changes in the flow of major currents caused by global warming are likely to profoundly change key natural habitats such as wetlands, estuaries, mangroves and coral reefs. Already, over half the world's wetlands and mangrove forests have been lost and 70 per cent of coral reefs worldwide are threatened (GESAMP, 2001).

The over-harvesting of marine resources through fishing activities, and unintended impacts resulting from the by-catch of non-target fish species, seabirds, turtles and marine mammals, is having a significant negative effect on marine ecosystems as well as coastal economies. Other likely negative ecological effects, arising from the removal of large amounts of sea life from the middle ranks of the food chain and the discard of by-catch, are still poorly understood.

Substantial reductions in the size of economically valuable fish stocks have occurred in New Zealand as well as in countries such as Canada and the USA. In New Zealand, the economically important western hoki fishery quota was recently cut from 110,000 to 40,000 tons, with the stock being estimated at only 13 to 22 per cent of its original size (Morgan, 2004:C1). Canada's Atlantic fisheries landings declined by nearly 50 per cent between 1990 and 1994, putting 25,000 fishers and plant workers out of work (Mitchell, 1998:403). In New England, traditional cod, haddock and flounder fisheries collapsed, resulting in the loss of an estimated 20,000 jobs (US Senate, 2000:3).

Water pollution and the resulting eutrophication are increasingly evident in coastal ecosystems. Eutrophication, which is a consequence of large increases in the amount of phytoplankton or algae in sea water, can deplete oxygen over large areas and dramatically alter ecosystems. This process also makes the water less transparent, reducing the penetration of sunlight. Causes of eutrophication include sewage and agricultural chemicals and wastes washing into sea water, and nitrogen oxides produced by the burning of fossil fuels falling out onto the water. Algal blooms are becoming more common in urbanised estuaries around the world, causing significant economic losses to fisheries and aquaculture operations (GESAMP, 2001:8).

Alien species are being introduced to ocean ecosystems at an increasing rate. It is estimated that, every day, 3000 species of animals and plants are being transported around the world in the ballast water of ships or on the hulls of ships and recreational vessels. Over 140 exotic marine species are known to have been accidentally introduced into New Zealand (Oceans Policy Secretariat, 2003a:2). Such species can out-compete other marine organisms for food and explode in numbers. For example, a jellyfish invasion of the Black Sea caused the collapse of fish stocks and the loss of at least US\$300 million from falling fishery revenues (GESAMP, 2001:13).

These environmental disturbances can interact with each other to magnify the negative effects. Degradation of marine ecosystems can make them more susceptible to invasive species. Land-based pollution and changing weather patterns resulting from climate change can put additional pressure on ecosystems already under stress from overfishing. It is the synergistic nature of these disturbances and their effects that has prompted a move towards integrated ecosystem approaches to ocean management and multiple-use risk assessment.

Despite the pressures on ocean ecosystems and the resultant environmental degradation, promising improvements have been made in some areas. For example, the rights-based system of fisheries management in New Zealand appears to be producing better outcomes for some commercially important species than traditional fisheries management approaches, marine pollution from land-based point sources has reduced in many countries, and integrated oceans management is showing promise in such diverse areas as the Great Barrier Reef in Australia and Chesapeake Bay in the USA.

Maximising economic and social opportunities

Many economic and recreational activities are linked with the oceans and the extent of these is likely to increase with growing populations and economies and the development of new technologies. Economic opportunities associated with marine areas include fishing, aquaculture, biotechnology, tourism, shipping, extraction of oil, gas and minerals, energy generation through waves and currents, and the laying of submarine cables and pipes.

Oceans are also the location of many recreational activities such as fishing, boating, swimming, diving and the enjoyment of the beach and seaside. Oceans are the focus of strong cultural associations, particularly for indigenous peoples. Conflicts frequently arise between these different activities. As so little of the ocean is yet explored, the extent of future economic and social opportunities is largely unknown, making planning for them difficult.

Environmental disturbances, as previously discussed and summarised in Figure 2.3, can negatively impact on a range of such activities. Governance failures can also have a negative impact including difficulties in:

- Resolving multi-use conflicts
- Clearly defining rights of access to and use of marine resources
- Generating and applying high-quality knowledge to management issues
- Adapting to changes in knowledge, technology, markets or community values

Different activities are affected to varying extents by environmental disturbances and governance failures. Those which are particularly susceptible to environmental disturbances include fishing, aquaculture, biotechnology, tourism, recreational activities and cultural practices. Stakeholders involved in such activities have a vested interest in ensuring ongoing ecosystem health.

Stakeholders involved in other activities, such as shipping, extraction of non-living resources, energy generation and communication, are less reliant on the living component of the oceans and therefore have less interest in ecosystem management approaches. However, such activities are more susceptible to poor governance, which can include unresolved conflicts, poorly defined rights and poor knowledge management. Such stakeholders are therefore more likely to be interested in an integrated approach to oceans management if it provides greater certainty by clearly defining rights.

The considerable challenge for a national oceans policy is to establish a governance system which can undertake the very complex task of, simultaneously, implementing international obligations, managing environmental threats and maximising economic and social opportunities, over a vast spatial area about which very little is known.

Figure 2.6. The oceans governance challenge

Governance Challenges for Oceans Management

Although interest in developing comprehensive and integrated oceans policies began in the 1960s, surprisingly little progress has been made (see Figure 3.1). In the USA, interest in applying a systems approach to the management of the country's oceans was first articulated in 1966 (US Senate, 2000:2). Despite considerable efforts to achieve that goal, it could be argued that the USA still lacks a fully integrated national oceans policy, although it passed an Oceans Act in 2000 and released an Ocean Action Plan at the end of 2004. This slow progress may be owing to strong opposition to adopting an integrated approach from some sectoral interests.

Canada first articulated the need for a national oceans policy in 1973, released an oceans policy document in 1987 which failed to gain traction, and released a second oceans policy 'discussion document' in 2002 (Herriman et al., 1997; Hildebrand and Chircop, 2000; Government of Canada, 2002a). Although Canada adopted an Oceans Act in 1996, over eight years later it still lacks a comprehensive national oceans policy formally endorsed by government. The failure of the Canadian federal government to provide additional resources to prepare and implement the policy has contributed to this slow progress.

Australia and New Zealand were relatively late starters in the game, with Australia beginning oceans policy development in 1996 and New Zealand in 2000. Australia has made the most rapid progress, with the Australian Commonwealth (federal) government formally endorsing the development of a national oceans policy in 1996. The final policy was released in 1998 and the first regional marine plan developed under the policy was released in 2004. However, this relatively rapid progress has been at the cost of failing to obtain endorsement of the policy by State governments which control coastal land and waters extending out to three nautical miles (Westcott, 2000; National Oceans Office, 2004).

The New Zealand government's oceans policy initiative commenced in 2000, but it has since been put on hold because of controversy over the ownership of the seabed and foreshore, which was contested by iwi. This issue has now been partially resolved through the passage of new legislation. Ocean policy proposals have yet to be endorsed by government and are likely to be released for public consultation during early 2006 when the initiative is expected to recommence.

Country	1960	1970	1980	1990	2000
USA	Marine Resources & Engineering Act (1966) 'Stratton Report' (1969)	National Oceanic and Atmospheric Administration created (1970) Coastal Management Act (1972) Fishery Conservation and Management Act (1976)			Oceans Act (2000) US Commission on Ocean Policy established (2001) Commission's Final Report (2004) U.S. Ocean Action Plan (2004)
Canada		Minister announces Oceans Policy (1973) Department of Fisheries and Oceans established (1979)	Oceans Policy released (1987)	Oceans Act (1996)	Discussion Paper on Ocean Strategy released (2002)
Australia				Development of Oceans Policy commenced (1996) Oceans Policy released (1998) National Oceans Office established (1999)	South-East Regional Marine Plan completed (2004)
New Zealand					Development of Oceans Policy commenced (2000)

Figure 3.1. Time line for development of national oceans policy

There are seven key aspects of oceans governance (discussed in more detail below) which make it a particularly difficult enterprise (see Figure 3.2):

- The need to manage large, interrelated and dynamic ecosystems
- The fragmented jurisdiction that different government bodies and agencies exercise over the ocean
- The need to manage multiple competing uses of the ocean and its resources
- The lack of solid scientific understanding of the dynamics of ocean ecosystems on which to base management decisions
- The lack of a strong constituency to support an integrated approach to oceans management
- The lack of an established property rights regime
- The rapid development of new uses

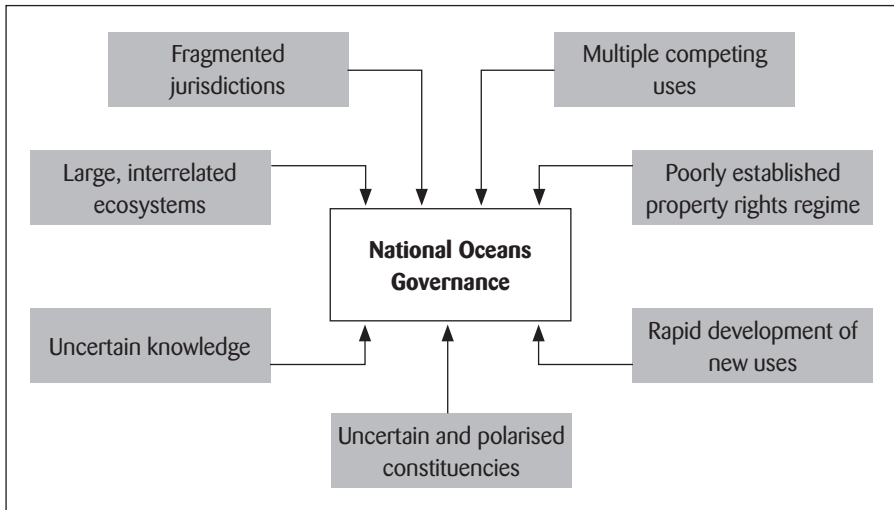


Figure 3.2. Key governance challenges

Large marine ecosystems

The oceans can be conceptualised as comprising large, interrelated ecosystems which are dynamic over space and time as a result of the movement of major currents and species. These large marine ecosystems (LMEs) can typically encompass 200,000 square kilometres or more and include coastal areas from river basins and estuaries out to the seaward boundary of continental shelves and coastal current systems (Juda, 1999:1). They

are areas which each have unique hydrographic attributes and submarine topography and a characteristic biological community structure.

Forty-eight large marine areas, extending over continental shelves, have been identified throughout the world's oceans by the American Association for the Advancement of Science, including one surrounding New Zealand called the New Zealand Shelf (Vallega, 2001b:195–196).

LMEs typically cut across many government departmental and jurisdictional boundaries and can also extend beyond the jurisdiction of the coastal nation state. Within these large areas can be nested a range of smaller more local marine ecosystems.

Fragmented jurisdictions

Typically, many levels of government, government departments and agencies are involved in activities which impact on the oceans. Fragmentation occurs horizontally, whereby different government departments deal with different sectoral issues, and vertically whereby different levels of government have jurisdiction over different spatial portions of the ocean and coastal land, or over different activities impacting on them.

For example, in the USA more than 140 federal laws are relevant to the oceans and the coast. They involve at least six departments of federal government and dozens of federal agencies in day-to-day management (Pew Oceans Commission, 2003:26). In addition, state governments have jurisdiction over coastal land and the area of ocean extending seawards for three nautical miles. In Canada, more than 23 federal departments have mandates that impact on oceans policies and outcomes, and state governments have jurisdiction over coastal lands and seabed (Government of Canada, 2002b). In New Zealand, there are at least 18 statutes relevant to the management of the oceans administered by 14 government agencies (Hodgson, 2003:3).

These fragmented jurisdictions create a particularly difficult governance environment. A high degree of interagency coordination is required to manage large marine ecosystems effectively. Government departments and agencies can, however, be loath to surrender jurisdictional prerogatives and 'turf wars' can break out between competing departments (Juda, 2003b). In addition, sectoral managers can have narrowly defined jobs where, for example, fishing management efforts are focused on calculating sustainable yields for specific stocks rather than addressing broader ecosystem management issues (Young, 2003).

Multiple competing uses

The oceans are used for many activities and these often compete for resources and space. The large and interrelated nature of marine ecosystems means that the effects of one activity can have significant negative impacts on other activities. For example:

- Aquaculture can occupy water space, which is also in demand for recreational boaties and fishermen
- Commercial fishers can reduce stocks also targeted by recreational fishers and vice versa
- Deep-sea mining can disturb ecosystems important to maintaining fish stocks
- Coastal development can lead to the destruction of marine habitats, sedimentation and water pollution, which impact on fishing and aquaculture
- Shipping can introduce invasive species, which impact on fishing and aquaculture

Conflicts between competing uses need to be mediated and this process inevitably results in the redistribution of the costs and benefits of oceans use. The perceived 'losers' of this process can provide strong opposition to change (Juda, 2003b).

Uncertain knowledge

Comparatively little is known about marine ecosystems in terms of their components, functioning, interactions and spatial location. Only a very small percentage of the oceans has been explored. Problems keep arising that scientists know little about, such as an unidentified pathogen which killed tens of thousands of lobsters in Long Island Sound (US Senate, 2000:4). It is now thought that the deep seas, once considered to have almost no life, may have more species than on all of earth's landmass (GESAMP, 2001:12).

A 1999 review of the adequacy of environmental information on the ocean in New Zealand found significant gaps in oceans information. These included gaps in knowledge of what species live in the oceans, the functioning of the biophysical processes of marine ecosystems, the linkages between species and populations, and the impacts of human activities on marine environments (PCE, 1999:76).

Although more scientific information would be helpful, oceans management will always be carried out in the context of scientific uncertainty. Oceans are such large and complex systems that they are unlikely to ever be entirely understood. Management techniques, such as the application of the precautionary principle and adaptive management, have been developed to assist decision making where information is incomplete.

The precautionary approach seeks to limit damage if the wrong decision is made because of lack of information. It includes avoiding decisions which have irreversible consequences, which could seriously threaten the resource base over the long term and which could foreclose options for future generations.

Figure 3.3. Definition of precautionary approach

Source: Cicin-Sain and Knecht, 1998:175

Uncertain and polarised constituencies

Much of the ocean is away from the public eye and this can result in weak public support for a holistic and integrated management approach. Small but highly motivated and well-organised interest groups, such as those involved in the harvesting and processing of marine living resources, are well practised at protecting their 'patch'. Close relationships can arise between industry representatives, officials located in relevant management agencies and legislators who make decisions about activities occurring in the ocean. The alliance between these groups can protect the immediate interests of a relatively small group in society and override broader but more diffuse concerns about the adequacy of current management approaches (Young, 2003).

Diverse and entrenched political positions can make reaching a consensus on oceans policy amongst all participants extremely difficult. Stakeholders have different interests and often bring very different viewpoints to the table. For example, in the USA, representatives of marine-related industries have opposed a fully integrated national oceans policy. This is in part because they fear it may circumscribe their operations, but also because they fear that it will result in more regulation and new government bodies which will increase their costs of compliance. Environmental activists, on the other hand, are often in favour of centralised government and tougher regulation, on the basis that industry operations must be circumscribed. In contrast, local government politicians and officials generally oppose handing power to federal authorities, and argue that solutions need to be tailor-made to local circumstances and are thus best designed, in a participatory and democratic way, at the local level (Cannon, 2004).

Lack of established property rights regime

Most countries have a well-established property rights regime for land-based resources, which has developed over many years and in some cases centuries. Rights to marine resources are often much less clear, particularly within EEZs over which

countries have only relatively recently acquired jurisdiction. Although some countries have implemented rights-based regimes for marine resources, such as in the case of commercial fisheries in New Zealand, others rely on regulatory regimes which manage use rather than define rights. Lack of clarity over rights in the marine area can magnify multi-use conflicts and act as a disincentive to investment.

Rapid development of new uses

Advances in technology and changes in market demand are enabling the rapid establishment of new uses of the oceans, sometimes in advance of effective governance systems. For example, advances in fishing technology have enabled the exploitation of deep-sea species, such as orange roughy, well ahead of the availability of scientific knowledge to ensure the sustainable management of these species.

4

Governance Approaches to Oceans Management

International debate on the kinds of governance approaches which are required to successfully manage oceans within national jurisdictions has been focusing on the following areas:

- The application of ecosystems-based management to large marine areas
- Mechanisms to integrate the management effort to achieve a common purpose
- Adaptive management mechanisms
- The development of a strong constituency to support the oceans governance efforts

The relationship between these approaches and the governance challenges identified in the previous section is shown in Figure 4.1 and is described in more detail in the sections below.

Governance challenges	Governance approaches			
	Ecosystems-based management	Integrated management	Adaptive management	Constituency building
Large marine ecosystems				
Fragmented jurisdictions				
Multiple competing uses				
Uncertain knowledge				
Uncertain constituency				
Rapid development of new uses				

Note: The shaded areas indicate where a particular governance approach has been developed to address a particular governance challenge.

Figure 4.1. Linkage between governance challenges and approaches

Ecosystems-based management

The ocean is increasingly being perceived as a complex system that requires consideration and management as a whole, through an ecosystems-based approach (Vallega, 2001a:400). Such a management approach begins with developing an understanding of how the natural systems function, and then identifying how human activities are impacting on those systems. The protection of ecosystems is to be given first priority, and changes to human behaviour are sought to accommodate, rather than undermine, those systems (Juda, 2003b). Ecosystems-based management aims to ensure the maintenance of (Commonwealth of Australia, 1998a:10):

- Ecological processes, including water and nutrient flows, community structures, food webs and ecosystem links
- Marine biological diversity, including the capacity for evolutionary change
- Viable populations of all native marine species in functioning biological communities

Tools which can be used to achieve these aims include planning for the management of large marine ecosystems as a whole and the establishment of a system of marine protected areas, which includes representation of marine ecosystem types across their natural range of variation.

Both Australia and Canada have explicitly sought to adopt an ecosystems-based approach to managing their oceans. This has been primarily through the application of regional marine planning.

Australia's first regional marine plan was prepared for the south-east region of the country's oceans, which covers an area of 2 million square kilometres at the junction of three major oceanic systems and bordering the coastline of four states. The scale of the planning exercise was unprecedented anywhere in the world and the process was far more difficult and complicated than anticipated.

In Canada, one of the first pilot regional planning projects initiated by the federal government under the Oceans Act was the Eastern Scotian Shelf Integrated Management Project (ESSIM), which covers an area of approximately 325,000 square kilometres. Progress since the initiative was announced in 1998 has been slow.

The US Commission on Ocean Policy identified ecosystems-based management as one of the guiding principles for a USA oceans policy. It also recommended the establishment of regional oceans councils for large marine areas, but did not go so far as to support regional marine planning. The Bush Administration's response in the U.S. Ocean Action Plan indicated an intention to work towards an ecosystems-based approach in making decisions but did not adopt the recommendation regarding regional ocean councils.

The publicly available material outlining the New Zealand approach does not explicitly adopt an ecosystems-based management approach as a key principle underlying oceans management. Details of the policy development work programme do, however, identify 'holistic management' as a key issue to be addressed and this is described as including the development of a comprehensive ecosystems-based approach to managing the marine environment (Risk, 2002c:6).

The New Zealand initiative also does not appear to have embraced regional marine planning as a tool to bring together different agencies and stakeholders involved in oceans management to jointly plan for large marine ecosystems. Regional councils currently undertake a limited form of marine planning with the development of coastal plans, which cover only marine areas, and in some cases regional coastal environment plans, which also include policies relating to land. These plans only extend 12 nautical miles seawards, are not able to address important marine sectors such as fisheries, and are poorly linked with marine protected areas. Whether the scope of regional planning will be expanded to encompass large marine ecosystems, associated water catchments and all relevant marine sectors is not yet clear but, based on information currently available, seems unlikely.

Integrated oceans management

Integrated oceans management aims to limit the conflicts which occur when each use is governed by a separate authority and decisions are made without sufficient reference to other uses managed by other agencies (Juda, 1999:100). There are five aspects of integration which can be considered when addressing oceans management (Cicin-Sain and Knetch, 1998:45):

- Intersectoral integration whereby different sectors (such as fisheries, mineral exploration, coastal development, agriculture and shipping) are integrated
- Intergovernmental integration whereby activities amongst different levels of government, including national, provincial, regional and local, are integrated
- Spatial integration whereby management of coastal land, coastal water and the EEZ are integrated
- Science-management integration whereby science provides information which informs management, and management issues help drive the research agenda
- International integration whereby ocean management efforts by neighbouring countries are coordinated over joint issues such as transboundary pollution and straddling fish stocks

Five main mechanisms have been adopted to increase integration within the oceans management system: establishing a common purpose, undertaking regional marine

planning, establishing integrative bodies and processes and providing incentives to encourage more integrated actions.

Establishing a common purpose

A common purpose for all oceans managers may be expressed as a vision and a set of policy goals and/or objectives. These help establish a joint inspirational purpose. They are usually expressed at a high level in very general terms and in a manner which represents 'all things to all people' so that, in themselves, they are unlikely to impact significantly on management decisions. For example, the Australian Oceans Policy states the following vision:

Healthy oceans: cared for, understood and used wisely for the benefit of all, now and in the future.

For the preparation of its oceans policy, the New Zealand government seems likely to adopt a similar vision:

Healthy oceans: wisely managed for the greatest benefit to New Zealand now and in the future.

Regional marine planning

Regional marine planning seeks to bring together all the relevant stakeholders from different levels of government and government departments to work together to plan for a particular region. It seeks to plan across different jurisdictional boundaries and to include waters both in the coastal areas and in the EEZ where these form part of one marine system. Regional marine planning has also been used as a mechanism to identify management needs for scientific information and to develop management priorities for research.

Integrative bodies

Integrating and coordinating bodies can include intra-governmental and inter-governmental committees, coordinating agencies and stakeholder advisory groups (see Figure 4.2).

In Australia, the Oceans Policy (1998) provided for the following integrative bodies:

- National Oceans Ministerial Board of Commonwealth Ministers
- National Oceans Advisory Group

- National Oceans Office
- Regional Marine Plan Steering Committees

In practice, these bodies have not performed as effectively as was hoped. In the early stages of implementation, the Ministerial Board became bogged down with operational issues rather than focusing on determining strategy. As a result, an Oceans Board of Management comprising heads of relevant Commonwealth government departments was subsequently established to deal with operational issues. This Board was supported with scientific advice through the establishment of an Oceans Policy Science Advisory Group.

The National Oceans Advisory Group and the South-East Regional Marine Plan Steering Committee consisted of a mix of experts and stakeholders and were torn between the distinct roles of providing advice and serving as a sounding board for stakeholder reactions to government proposals.

The National Oceans Office (NOO), initially established as a small non-statutory executive agency in a location remote from the capital and political power, struggled to effectively engage with other Commonwealth departments. However, it developed a strong core of skilled and committed people with expertise in integrated oceans management approaches and has worked at the cutting edge of planning for integrated oceans management. It has more recently been incorporated into the Department of Environment and Heritage.

State governments were not formally represented in any of these bodies, reflecting the failure of the federal government to obtain buy-in at this level. Notwithstanding some of these teething problems, these initiatives have resulted in significant progress being made in encouraging cross-sectoral thinking about oceans issues.

In Canada, an existing department, the Department of Fisheries and Oceans (DFO) was given the role of developing and implementing an oceans policy. It did not receive any additional funding for this task, perhaps reflecting a relatively low level of government commitment and, as a result, progress has been somewhat slower than in Australia. The Canadian Department has struggled to integrate the oceans functions of other federal departments, with the Senate review in 2000 finding that oceans management had become even more fragmented since the passing of the Oceans Act in 1996.

Type	Australia	Canada	USA	New Zealand (initial indications) ²
Intra- and inter-governmental coordinating bodies	National Oceans Ministerial Board of Commonwealth Ministers Oceans Board of Management	Federal-provincial-territorial Oceans Task Group	Committee on Ocean Policy (cabinet level) Interagency Committee on Ocean Science and Resource Management Integration (with subcommittees) National Security Council Policy Coordinating Committee	Ad Hoc Ministerial Group Oceans Policy Steering Group
Coordinating agencies	National Oceans Office	Department of Fisheries and Oceans		Oceans Agency
Advisory groups	National Oceans Advisory Group Oceans Policy Science Advisory Group	Ministerial Advisory Council on Oceans	Ocean Research Advisory Panel	Māori advisory groups Other stakeholder groupings
Multi-stakeholder regional management bodies	Regional Marine Plan Steering Committees (as plans developed)	Regional plan multi-stakeholder forums (pilots)		Regional Councils

Figure 4.2. Summary of national oceans management integrative bodies

Less emphasis has been placed on establishing coordinating bodies in Canada. In 2001, a federal-provincial-territorial Oceans Task Group was established to provide an inter-governmental forum for discussing oceans issues of national significance. A year earlier, a Ministerial Advisory Council on Oceans was established to provide independent

2 The fully developed proposals for New Zealand's oceans policy have yet to be made public and may differ from the integrative bodies shown here.

advice on oceans policy matters. Work is currently being undertaken to establish a multi-stakeholder forum for the ESSIM initiative.

In the USA, a cabinet-level Committee on Ocean Policy has been established to advise the president and agency heads on the establishment and implementation of ocean policies. It will be supported by an Interagency Committee on Ocean Science and Resource Management Integration consisting of high-level officials. This committee in turn will be assisted by two subcommittees, one focusing on the management of ocean resources and the other focusing on ocean science and technology. The Committee will also be informed on international issues by the National Security Council Policy Coordinating Committee. An Ocean Research Advisory Panel consisting of a range of stakeholders will advise the Committee.

Proposals being canvassed in New Zealand include the establishment of an Oceans Agency, which could prepare and administer an Oceans Act and Plan. It is not yet clear whether the two bodies established to oversee the policy development process, the Ad Hoc Ministerial Group and the Oceans Policy Steering Group comprised of government officials, will continue to oversee the implementation of the policy. It is also not yet clear if stakeholder groupings other than Māori will have a formal role in the implementation of the policy. Regional councils currently perform a somewhat circumscribed role in regional coastal management and this looks likely to continue.

As demonstrated by Figure 4.2, the types of integrative bodies adopted or proposed by the case-study countries show much similarity. There are also some notable differences. Canada has failed to provide a dedicated coordinating agency, rather allocating the function to an existing department, and this may have contributed to the slow pace of implementation. The USA has chosen not to identify a coordinating agency or to establish regional management bodies. New Zealand is not currently proposing a formalised multi-stakeholder advisory body, but this may reflect the smaller size of the country and the greater informal interaction between government and stakeholders.

Integrative processes

Where difficult oceans issues are encountered, formalised integrative processes can assist with their resolution. In Australia, proposals for an Integrated Oceans Process have been developed. The process is intended to apply in situations where (National Oceans Office, 2003:6):

- There is a lack of clarity on who is responsible for an issue
- There is a new ocean use for which no management arrangements are in place
- New policies are being developed that could impact on Commonwealth marine waters and/or the high seas

- Policies or large-scale projects have considerable cross-jurisdictional and/or cross-sectoral implications
- There are issues with potential interactions/conflicts with no established process to bring together all the relevant interests
- There is a potential for high-risk impacts

Once an issue is identified as being appropriate for the integrated process, a lead agency is appointed to coordinate scoping, risk analysis and strategy development, with the involvement of all relevant agencies and stakeholders.

Integrative incentives

‘Business as usual’ is well entrenched and there may need to be strong incentives in place to encourage people to change their behaviour in favour of an integrative management approach. Incentives can be provided in various forms. Legislation can compel behaviour change, but needs to be carefully designed and enforceable and may not be politically feasible. None of the countries studied have yet passed legislation specifically for the implementation of a national oceans policy, as opposed to its development.

Budgetary allocations dedicated to integrative oceans processes can provide strong financial incentives for change and this approach was effectively adopted by the Australian government, which allocated significant dedicated funding to the initial implementation of its oceans policy.

Adaptive management

Adaptive management seeks to ensure that the management system is responsive to unintended outcomes and new information. It involves establishing a monitoring system, undergoing frequent reviews to determine whether or not the desired outcomes are being achieved, and the timely adaptation of the management approach in light of new information.

In Australia, the government commissioned an independent and comprehensive review of the oceans policy process in 2002 (Norton, 2002). This provided much valuable information, which guided subsequent changes in the initiative.

The National Oceans Office has also designed an adaptive management system, which will be applied to the implementation and review of its first Marine Plan. This system will continually improve management by establishing a comprehensive performance-monitoring system, improving methods for evaluating different management actions, and monitoring and reporting on the health of the ocean and economic and social outcomes.

In Canada, the Standing Committee on Fisheries and Oceans reviewed the Oceans Act in 2001. This appears to be a much less rigorous review than the one undertaken in Australia, being based largely on submissions from stakeholders. It found lack of tangible evidence of implementation of the Act, a finding which was subsequently challenged by a rather defensive Department of Fisheries and Oceans (Government of Canada, 2002c).

Constituency building

The successful implementation of an integrated oceans policy may ultimately depend on the development of a group of stakeholders which support adopting such an approach as opposed to 'business as usual'. This is important, as well-entrenched sectoral interests do not necessarily see an integrated and ecosystems-based approach to oceans management as being to their benefit. Developing a strong constituency keeps the issue alive and provides politicians with a greater measure of comfort about adopting new approaches.

The endorsement by key stakeholders of the policy process and outcomes can be critical to the successful implementation of oceans policy, but can prove hard to achieve. Key stakeholders include politicians and officials from all levels of government and leaders of key sectoral interest groups. Stakeholders are unlikely to be supportive unless they can see a clear benefit to themselves and/or their sector from participating in the initiative. This makes it important to clearly identify and articulate the benefits of an integrated approach to oceans management. To retain credibility and stakeholder support it can also be important to have clear processes and outcomes, including clarity around how the aspirations expressed in the policy will be implemented in practice.

Mechanisms to develop such stakeholder support can include establishing and/or strengthening stakeholder and researcher networks, disseminating high-quality and accessible information and convening multi-stakeholder forums at which people from a range of viewpoints can discuss common issues.

In Australia, a strong constituency was built around the ocean policy process through the establishment and utilisation of the Marine and Coastal Community Network. The Network was established in 1993 with Commonwealth government funding to employ regional coordinators. It currently has around 7000 members and makes a significant contribution to increasing interest in oceans issues. In Canada, an Oceans Management Research Network was launched in 2001 with over 200 academic researchers.

The wide circulation of information, and the convening of multi-stakeholder meetings at which issues of mutual concern are discussed, can help develop a shared understanding of issues. In Australia, a significant amount of information was collected, analysed and circulated during the preparation of the South-East Regional Marine Plan. As a result, both National Oceans Office staff and stakeholders derived a vastly

improved knowledge of the ecosystems and their uses within the region, as well as broader ocean management issues.

In New Zealand, the information-gathering process prior to commencing the development of the policy was primarily qualitative and based on the input of key stakeholders. These key stakeholder groupings were involved in a series of briefing and feedback sessions. However, very little quantitative information was produced to inform the discussion and this may have limited the development among stakeholders of a well-informed and shared understanding of the issues.

Development of a National Oceans Policy

Practical considerations associated with the development of a national oceans policy include the need for strong political leadership to drive the process forward, the allocation of sufficient time and resources to the process, developing the appropriate mix between science and policy, the consideration of whether to legislate, and phasing of the initiative while political support for its fuller implementation is developed.

Political leadership

The chances of successfully developing a national oceans policy are likely to be considerably enhanced if the process is driven by strong political leadership. The oceans policy initiatives in New Zealand, Australia and Canada were all initiated by strong government ministers who seized upon the opportunity to make a difference in a relatively new policy field. Subsequent changes in ministerial portfolios can, however, lead to a loss of impetus.

In New Zealand, the then Minister of Fisheries (who was also the Minister of Energy and Science and Technology), the Honourable Pete Hodgson, drove the process from its inception in 2000. When the initiative lost impetus during 2002, the minister established a dedicated, cross-departmental Secretariat and instructed it to deliver a policy within nine months. However, in June 2003, the initiative was put on hold owing to possible political fallout from dispute over Māori ownership of the seabed and foreshore. Subsequently the fisheries portfolio was transferred to the Honourable David Benson-Pope who is also the Associate Minister for the Environment, but with Minister Pete Hodgson retaining overall responsibility for the oceans policy. It is not yet clear if this change will have any significant impact on the impetus behind the policy initiative when it is recommenced.

In Australia, the oceans policy initiative was driven by the Honourable Robert Hill, the first Minister for the Environment in the Liberal government elected in March 1996. He retained control of the Environment portfolio until November 2001 and brought to the job a legal background and an interest in international law (Westcott, 2004).

In Canada, an Oceans Act was passed in record time under the guidance of the Minister of Fisheries and Oceans, the Honourable Brian Tobin. Work to draft the bill commenced in November 1994, cabinet papers were prepared in February and May

1995, and the bill was tabled in Parliament in June 1995, only seven months from its inception (Baird, 2004). The Act was subsequently passed by Parliament in December 1996 and came into force in January 1997. Brian Tobin resigned from government in January 1996, however, so was not able to oversee the subsequent implementation of the Act, which was hindered by a lack of funding.

Time and resources

Developing and implementing a national oceans policy is a difficult enterprise and its success requires a significant allocation of time and resources. It is likely to take several years to collect relevant information, consult with stakeholders and develop a broad consensus on the way forward.

In Australia, the federal government committed A\$50 million over a three-year period to implement initiatives identified in the policy and established a specialised agency with around 40 staff to focus exclusively on its implementation. The preparation of the South-East Regional Marine Plan has cost over A\$9 million. The availability of these significant resources and focused attention appear to have played a large part in the positive progress made to date. Even with these dedicated resources, the policy initiative has taken many years. A final policy was released in 1998, just under three years after the initiative commenced in early 1996. Preparation of a regional marine plan commenced in April 2000 and was completed just over four years later in May 2004.

In Canada, no additional funds were made available to develop and implement the oceans policy. The job was given to an existing federal department, the Department of Fisheries and Oceans, which had traditionally focused on fisheries management. The Department was forced to reallocate resources from existing activities and, unsurprisingly, less progress has been made than by the Australian initiative. A final policy has yet to be released seven years after the initiative was launched and no regional marine plan has been prepared.

In New Zealand, about two years into the process – after a significant proportion of the budget had been spent on a widespread public consultation process – a dedicated Secretariat of approximately 10 staff was allocated approximately NZ\$1 million and given nine months to produce a policy. The modest resources and the short time frame provided precluded the collection of data on the state of New Zealand's marine environment and the activities and uses occurring within it. As a result, the input to identifying key issues to be addressed by the policy was primarily obtained from the contributions of different stakeholders. Over four years after the initiative commenced, a draft policy has yet to be released for public comment.

In the USA, the Oceans Act came into force in January 2001 and three-and-a-half years later, in July 2004, the US Commission on Ocean Policy released its final report.

This was somewhat later than the 18 months proposed in the Senate Report on the Act for the production of the final report (US Senate, 2000:6) and the two-and-a-half years provided for in the Oceans Act itself (section 3(f)(1)). In December 2004 the Bush Administration released its response to the Commission's recommendations in the form of the U.S. Oceans Action Plan. The Commission had estimated that the federal government will need to make US\$1.5 billion available in the first year of implementation of the policy, rising to US\$3.9 billion in subsequent years. In contrast, however, the Bush Administration announced new funding of only US\$2.7 million to implement the Action Plan and then subsequently, in the 2006 budget, proposed cuts of US\$391 million to the National Oceanic and Atmospheric Administration's (NOAA) ocean and coastal programmes.

Science and policy mix

The successful development of an oceans policy requires the careful assembly of natural and social sciences information and its integration with policy analysis, so that the resulting policy is technically robust, politically feasible and implementable and will achieve the outcomes sought.

Getting the right mix is not necessarily easy. In Australia, staff of the National Oceans Office had predominantly natural-science backgrounds and were relatively inexperienced in policy analysis. As a result, the process of developing the South-East Regional Marine Plan focused heavily on information gathering, analysis and mapping, with insufficient policy thinking up front to develop a clear policy framework. In New Zealand, staff of the Ocean Secretariat consisted almost entirely of policy analysts rather than scientists, resulting in a high-level policy exercise with little scientific input to inform the process.

A stand-alone interdisciplinary scientific advisory body, such as that established by the US Commission on Ocean Policy, can assist with assembling the best available scientific information to support the policy development process. Multidisciplinary policy development teams can then utilise this information in the policy analysis process.

Legislation

Legislation can provide a national oceans policy with greater political certainty and teeth. It can, however, be difficult to garner sufficient political support to pass oceans legislation. In the four case-study countries analysed, legislation does not appear to have played a key role in the success or otherwise of national oceans policy initiatives. Oceans Acts have been passed in the USA and Canada, but not in Australia or New

Zealand. Australia has made the most progress to date on the implementation of a national oceans policy.

In the USA, the Oceans Act passed in 2000 provided for the establishment of a commission to make recommendations for a national ocean policy. The Act set out a number of objectives for the policy (section 2) but did not prescribe the approach to be taken in the policy or its content. It therefore represented a very preliminary stage in the policy development process.

In Canada, the Oceans Act passed in 1996 also did not contain the elements of a policy but provided for its preparation by the Minister of Fisheries and Oceans. It did state a set of principles on which the policy was to be based and also provided for its implementation through the preparation of integrated management plans and the establishment of a national system of marine protected areas and marine environmental quality guidelines. The Act also strengthened the Department of Fisheries and Oceans through a merger with the Canadian Coast Guard, clarified the responsibilities of the Minister for Oceans and established Canada's maritime zones. Despite the Act, the development and implementation of the national oceans policy has been a very slow process.

In Australia, no legislation has been passed to support the national oceans policy. This appears to be largely owing to a lack of agreement between the federal and state governments over their respective roles in developing and implementing the policy. Despite this lack of a statutory basis for action, considerable progress has been made.

Similarly, in New Zealand no legislation has been passed to support the development of an oceans policy, although it seems likely that an Act will be developed to implement it. Because of New Zealand's unicameral political system, passing legislation is less problematic than in other more complex jurisdictions, particularly if there is a strong minister to promote it.

Phasing

Where sufficient political support is lacking to fully adopt a new approach to oceans management, a gradual and phased approach to policy implementation can get small things done while building the support and momentum required for more significant changes. The downside of this approach is that hard issues can be side-stepped and the process can be 'captured' by mobilised stakeholders who have more time to influence the policy direction.

In Australia, it was originally envisaged that regional marine planning would be the mechanism through which user conflicts could be mediated, areas for marine protected areas would be identified and long-term security for marine-based industries would be provided. Preparation of the plan proved much harder than anticipated and the final plan failed to meet these expectations. There have, however, been many positive

process outcomes from the planning exercise. The engagement of stakeholders, and the generation of much-improved information on which to base management, is helping generate a stronger constituency for ongoing planning initiatives. The 'learning by doing' approach has enabled significant progress to be made in the face of considerable political and scientific uncertainty.

In the USA, the government has adopted a phased approach to implementation of the US Commission on Ocean Policy recommendations, with the U.S. Ocean Action Plan listing a range of immediate actions and the Committee on Ocean Policy tasked with developing an 18-month work plan to address further recommendations of the Commission. It is not yet clear whether this approach will enable momentum to be built to effectively tackle 'hard' issues.

Part One Conclusions

A national oceans policy can be a useful tool to assist a country in meeting its international obligations, managing environmental threats to its oceans and maximising the economic and social opportunities which the oceans provide. Oceans governance is a complex task and new approaches to it are continuing to evolve based on the concepts of ecosystems-based management, integrated management, adaptive management and constituency building. As well as applying these concepts, there are a number of practical issues which may also need to be considered in order to successfully develop and implement a national oceans policy. A national oceans policy is more likely to succeed if adequate resources are made available for implementation. Legislation is desirable but resourcing appears more determinative of success.

Part Two

The Use of New Zealand's Oceans

Introduction

Part Two focuses on New Zealand's ocean resources. It describes the numerous and diverse living and non-living resources in New Zealand's oceans. It details how these resources are used. The Part finally describes the impacts which these uses are having on the marine environment.

This information sets the scene for Part Three, which describes and evaluates New Zealand's ocean governance system in the context of the environmental impacts of human activity.

6

What Ocean Resources Does New Zealand Have?

Extent of the oceans

New Zealand consists of an archipelago of over 330 islands³ surrounded by the world's largest ocean, the Pacific (Taylor, 2000:16). New Zealand's coastline is over 19,000 kilometres long and the country has jurisdiction over some 4.2 million square kilometres of ocean area (see Figure 6.1).

New Zealand's EEZ is large, varied and relatively deep with only a quarter being shallower than 1 kilometre. The deepest point, about 10 kilometres under water, is located within the Kermadec Trench. Because New Zealand sits astride two colliding tectonic plates, it has spectacular underwater features including chains of underwater volcanoes and deep ocean trenches and ridges (Batson, 2003:20, 24 & 29).

The extent of New Zealand's continental shelf is still being determined, but it extends far beyond the EEZ. This is because New Zealand consists of the higher ground of a drowned continent, a fragment of ancient Gondwana which broke off from Australia and Antarctica around 55 million years ago (see Figure 6.1; Batson, 2003:20; Wilson, 2004:4).

At least five major water masses, separate bodies of water with their own temperature and salinity, are found within New Zealand's EEZ. These include two surface waters: the Subtropical Surface Water, which originates from the central Pacific Ocean via the eastern coast of Australia, and the Subantarctic Surface Water, which originates from the Southern Ocean (Batson, 2003:56).

These two waters, moving in an easterly direction, meet along a Subtropical Front, which is located along the same latitude as Fiordland to the west of New Zealand and along the Chatham Rise to the east (Batson, 2003:57; see Figure 6.2). The meeting area along the Subtropical Front is a highly productive area, particularly over the Chatham Rise (Arnold, 2005:16).

Below these surface waters lie three vertically stacked water masses, the Antarctic Intermediate Water, which derives from the Southern Ocean, the Pacific Deep Water, which circulates around the world between the North Atlantic and the Pacific Ocean, and Bottom Water (Batson, 2003:56–57). Global warming is likely to result in changes in these oceanic and coastal currents as well as a rise in sea levels (Bell et al., 2001:20).

3 Larger than five hectares

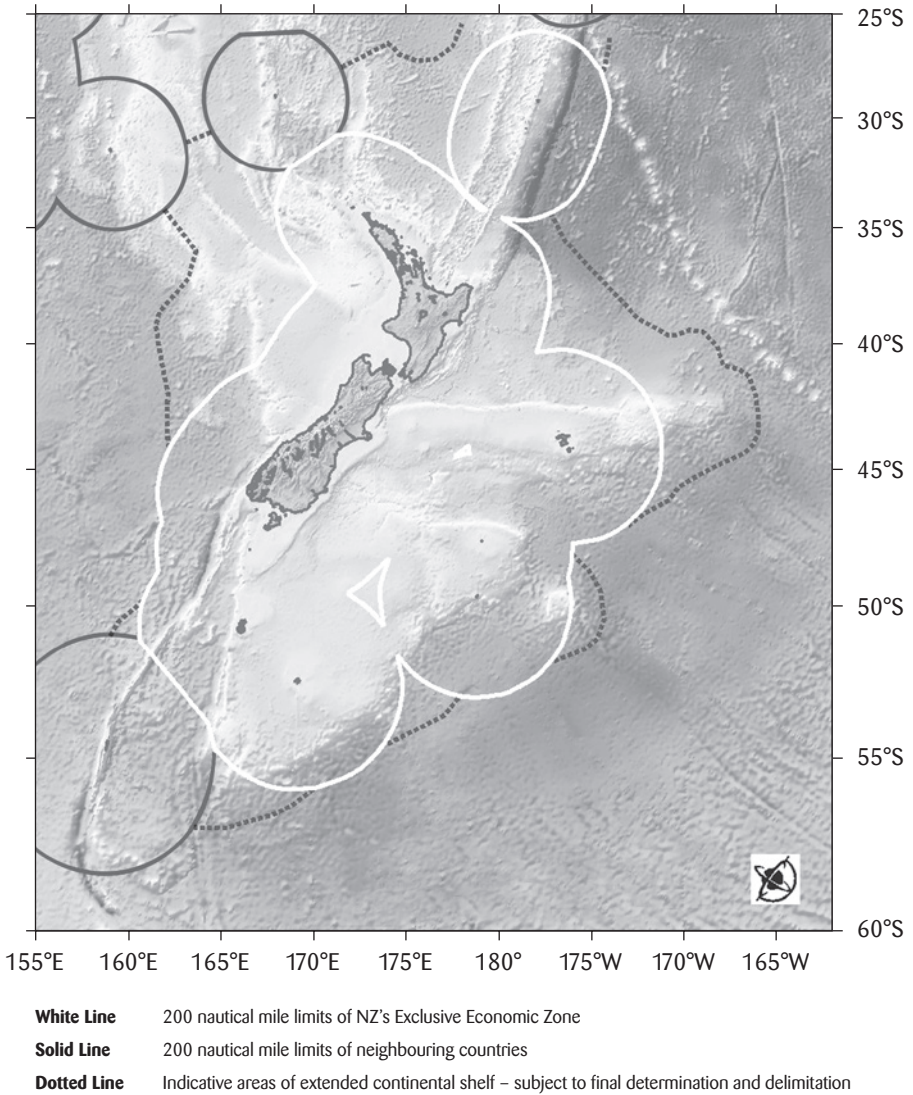


Figure 6.1. New Zealand's maritime jurisdictions

Provided courtesy of the Institute of Geological and Nuclear Sciences

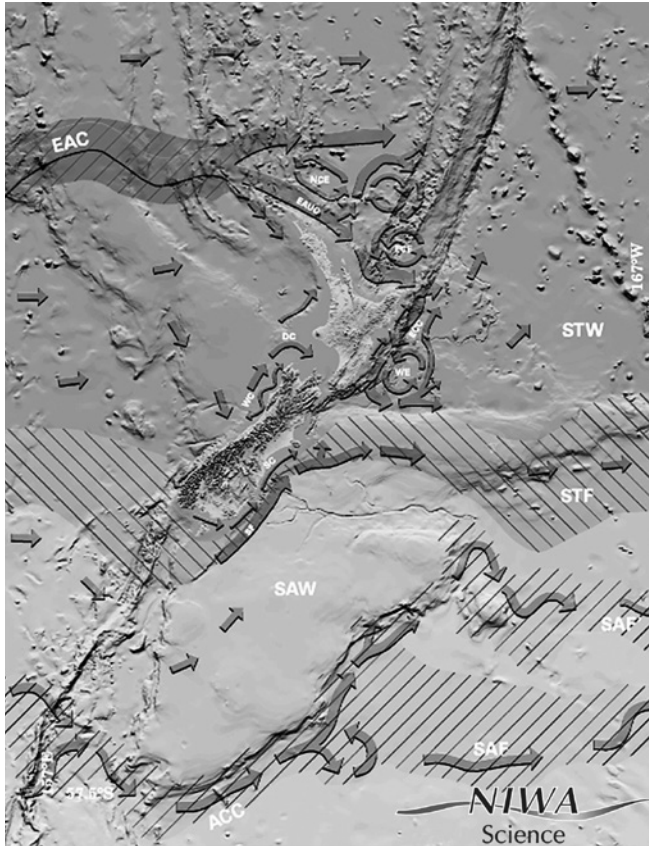


Figure 6.2. Ocean currents around New Zealand

Provided courtesy of the National Institute of Water and Atmospheric Research Limited

Living marine resources

New Zealand's marine ecosystems and species are highly diverse owing to the country's geological history, its isolation for millions of years, the range and complexity of habitats and the influence of major ocean currents (Department of Conservation (DoC), 2000a:55). The greatest fish and invertebrate biodiversity in the country is located to the north-east of the North Island owing to the presence of many warm-water species (Andrew and Francis, 2003:34).

Only about 1.5 square kilometres of the seabed have been sampled using standard oceanographic gear, representing less than one two-millionth of the EEZ (Gordon, 2000:42). About another 9,800 square kilometres have been sampled by fisheries research trawls. The government recently launched Ocean Survey 20/20, a project aimed at completing a survey of New Zealand's oceans by 2020, and providing high-quality information to support oceans decision making.

The oceans are thought to contain between one-third and three-quarters of all New Zealand's indigenous species. It is estimated that there are between 54,600 and 75,700 marine species within New Zealand's EEZ, of which only 11,202 or 15 per cent have been described (see Figure 6.3). At the present rate of new species description, of around 100 a year, it will take between 430 and 645 years to complete the task of inventorying all New Zealand's marine biota (Gordon, 2004). This is likely to be far behind the pace of technological developments facilitating its exploitation.

Group	Described species	Known unrecorded species	Estimated unknown species	Totals rounded
Protists, seaweeds, fungi, plants	2,700	175	35,000–55,000	37,900–57,900
Invertebrates	7,453	3,502	4320–5440	15,300–16,400
Vertebrates	1,050	200	160	1,410
Totals	11,202	3,877	39,500–60,600	54,600–75,700

Figure 6.3. Estimated number of marine species in New Zealand's EEZ

Source: Gordon, 2004

A large proportion of New Zealand's bottom-dwelling (benthic) marine species are endemic to New Zealand (DoC, 2000a:57). These are also the species about which the least is known and of which the smallest percentage has been discovered or described. The most biodiverse habitats within New Zealand's oceans are calcareous gravels and seamounts (Gordon, 2004).

Calcareous gravels include the shelly remains of molluscs and bryozoans (lace corals) and are the location of thriving bryozoan colonies or thickets. Bryozoan thickets are a rare habitat type by global standards and are relatively uncommon in New Zealand waters. They are thought to play an important role in the life history of several commercially exploited species including juvenile snapper and tarakihi. New Zealand has one of the largest bryozoan faunas in the world. Nearly 1000 species are

found within the EEZ, most of which are endemic (Batson and Probert, 2000:4 & 11; Batson, 2003:32–33).

The most biodiverse calcareous gravels are found on the Three Kings Platform area, extending from Spirits Bay on the northern tip of the North Island to the north-west, and on the Snares Platform, extending from the southern coast of the South Island. Other areas with prominent bryozoan communities include the coast off Otago, Foveaux Strait and Separation Point in Tasman Bay. In 1999, a scientific assessment found more than 300 species of bryozoans at Spirits Bay, representing nearly one-third of the entire New Zealand bryozoan fauna (Parliamentary Commissioner for the Environment, 1999:34; Andrew and Francis, 2003:35).

Seamounts are submerged inactive volcanoes which rise steeply from the deep-sea ocean floor and which often occur in chains or clusters. They can support very diverse and distinctive faunas with high levels of endemism, including deep-water corals, which grow up to 15 metres tall. Around 800 seamounts⁴ have been identified so far within New Zealand's EEZ and they cover about 3 per cent of its surface area. The exploitation of such features for commercial fisheries has proceeded well ahead of scientific research and understanding (Clark and O'Driscoll, 2003:442), with the result that species yet unknown to science may have already been lost.

Many of New Zealand's fish (vertebrate) species are also found in other countries, with only 11 per cent (110) being endemic to New Zealand waters. Less than half of these live in the open sea with most being confined to coastal rock pools. Eleven rock-pool species are listed as threatened, all of them endemic to New Zealand. Only two of New Zealand's sea-dwelling fish species are known to be threatened: the great white shark and the basking shark. Neither is endemic to New Zealand (Taylor and Smith, 1997:9.95).

Rocky reefs are widespread in nearshore waters around New Zealand. They are primarily located on exposed coasts where rocks are resistant enough to withstand wave attack and off embayed coasts where the softer rocks have been eroded to leave the harder rock as headlands and reefs. Rocky reefs are often covered with luxuriant forests of large brown algae and are home to some 250 species of reef fishes, of which about 28 per cent are endemic (Andrew and Francis, 2003:20 & 34).

There are about 300 estuarine systems located around New Zealand's coastline, covering at least 100,000 hectares. They are recent coastal features in geological terms, primarily caused by the flooding of valleys and bays about 6500 years ago when the sea level rose some 150 metres. Many are filling with sediment (Hume and Swale, 2003:24; Taylor and Smith, 1997:7.27; Hume, 2003:191).

New Zealand has two species of endemic marine mammals – the New Zealand sea lion, and the Hector's and Maui's dolphins, which are subspecies of *Cephalorhynchus hectori* – New Zealand's only endemic cetacean. They are all threatened.

4 Defined as a discrete hill-like feature with a vertical elevation of 100 metres or more above the seafloor. This definition differs from the commonly used international criterion of 1000 metres vertical elevation and, if this measure were used, New Zealand would have far fewer seamounts.

Only 11,000 to 15,000 New Zealand sea lions survive. They are at risk because of their limited number of breeding sites. There are currently around 7400 Hector's dolphins. Maui's dolphins, which are the smallest dolphins in the world, are only found along the west coast of the North Island. There are as few as 100 of this critically endangered subspecies left (Suisted and Neale, 2004:20 & 27).

The New Zealand fur seal is shared with southern Australia and is currently expanding in population and recolonising areas. Several species of dolphins and whales are common in New Zealand waters but are not endemic to the area (Suisted and Neale, 2004:36; Donoghue, 1998:182 & 184; Taylor and Smith, 1997:9:125–133).



Figure 6.4. The endemic New Zealand sea lion is currently threatened

New Zealand is a haven for seabirds, and has been termed the 'seabird capital of the world'. Eighty-four species of seabirds breed in New Zealand, including 35 endemic species. This is the greatest diversity of seabirds anywhere in the world. Nearly a quarter of the world's seabird species breed in New Zealand and almost 10 per cent breed only in the New Zealand region. Common species include penguins, albatrosses, petrels, shearwaters and shags. Four species, the Chatham taiko, Chatham petrel, Chatham albatross and New Zealand fairy tern, are critically endangered. A further eight species are classified as endangered, including the erect-crested penguin and the white-flipped penguin (Taylor, 2000).



Figure 6.5. New Zealand is the seabird capital of the world – Australasian gannet

Non-living marine resources

There are significant deposits of non-living marine resources within New Zealand's continental shelf, only a small proportion of which are currently being exploited. These resources include (Centre for Advanced Engineering (CAE), 2003):

- *Oil and gas*, which have been discovered in several parts of New Zealand's EEZ and which are potentially present in most of New Zealand's offshore basins (see Figure 6.6). Extraction is only currently occurring in the Taranaki Basin
- *Gas hydrates*, which are located in at least two large areas within the EEZ: the offshore margin of the east coast of the North Island and offshore Fiordland
- *Marine sands and aggregates*, which are widely distributed throughout New Zealand. Marine sands are currently being extracted from the Kaipara Harbour on the west coast of the North Island and the east coast of Northland
- *Titanomagnetite ironsand deposits*, which occur along 480 kilometres of coastline extending southwards from the Kaipara Harbour to Wanganui. Ironsands are currently being mined south of Auckland
- *Gold*, which is concentrated with other heavy minerals into lenticular (biconvex) beach placers on the west and south coasts of the South Island and also offshore around the Coromandel Peninsula and Hokitika

- *Phosphate nodules*, which have been identified over extensive areas on the Chatham Rise to the north-east of the South Island
- *Polymetallic nodules*, which contain several minerals including copper, nickel and cobalt. These occur at various deep-sea localities through New Zealand's EEZ, the largest concentration being along the south of the Campbell Plateau, 100 kilometres south of New Zealand.
- *Volcanic massive sulphides*, which are deposited by active submarine volcanoes including those along the 1200-kilometre long Kermadec arc. These include some deposits with high concentrations of base metals and gold.

New Zealand's EEZ, therefore, provides considerable potential for the extraction of oil, gas and valuable minerals. In addition to the above resources, energy in the marine area can be harnessed through a range of methods including offshore wind farms, underwater 'windmills' to capture marine currents, barriers to capture tidal currents and devices to capture wave and swell energy. None of these methods is currently being utilised in New Zealand because of the lack of technical and/or financial feasibility (CAE, 2003:7-9).

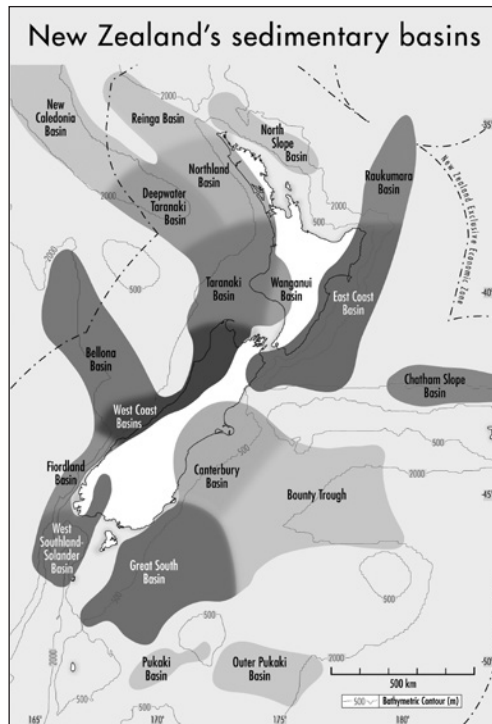


Figure 6.6. Hydrocarbon basins in New Zealand

Provided courtesy of the Ministry of Economic Development

How Are New Zealand's Ocean Resources Used?

New Zealand's ocean resources are used in many ways: for the commercial extraction of seafood, oil, gas, minerals and genetic resources; for the transportation of people and goods; for cultural purposes; and for enjoyment through lifestyle, recreation and tourism activities. Each of these activities potentially impacts on the oceans and, in turn, may be sensitive to the impacts of other ocean uses as well as climate change.

Commercial extractive uses

Seafood industry

New Zealand has a significant seafood industry, with around 600,000 tonnes of wild seafood caught commercially each year, and over 80,000 tonnes of seafood produced through aquaculture. Over 90 per cent of seafood products are exported, contributing significantly to New Zealand's foreign-exchange earnings. In 2004, the seafood industry exported NZ\$1.13 billion of seafood, comprising 3.8 per cent of New Zealand's total exports. Seafood was the country's sixth largest exported product, being outperformed only by dairy, meat, wood, fruit and machinery⁵ (Statistics New Zealand, 2005:21). In international terms, though, New Zealand is a small player, accounting for less than 2 per cent of the world seafood trade (CAE, 2003:9).

The seafood industry directly employs around 8,000 people in fishing, aquaculture and seafood processing (see Figure 7.1). An economic analysis, completed in 2000, estimated that 26,620 full-time equivalent jobs were directly and indirectly created by the seafood industry (McDermott Fairgray Group Limited, 2000:2).

Māori are dominant players in the fisheries industry, owning around 40 per cent of the quota and having an additional involvement in another 20 per cent (MFish, 2004:66). Māori own half of Sealord Products Limited, New Zealand's largest fishing company, which holds just over 20 per cent of New Zealand's fishing quota, including around 30 per cent of the quota for the valuable hoki and orange roughy species⁶. As a result of a 1992 Deed of Settlement with the Crown, Māori are to receive 20 per cent of all new species placed under the quota management system. They are also to receive 20 per cent of allocated marine farming space.

⁵ Based on provisional 2004 figures

⁶ See www.sealord.co.nz

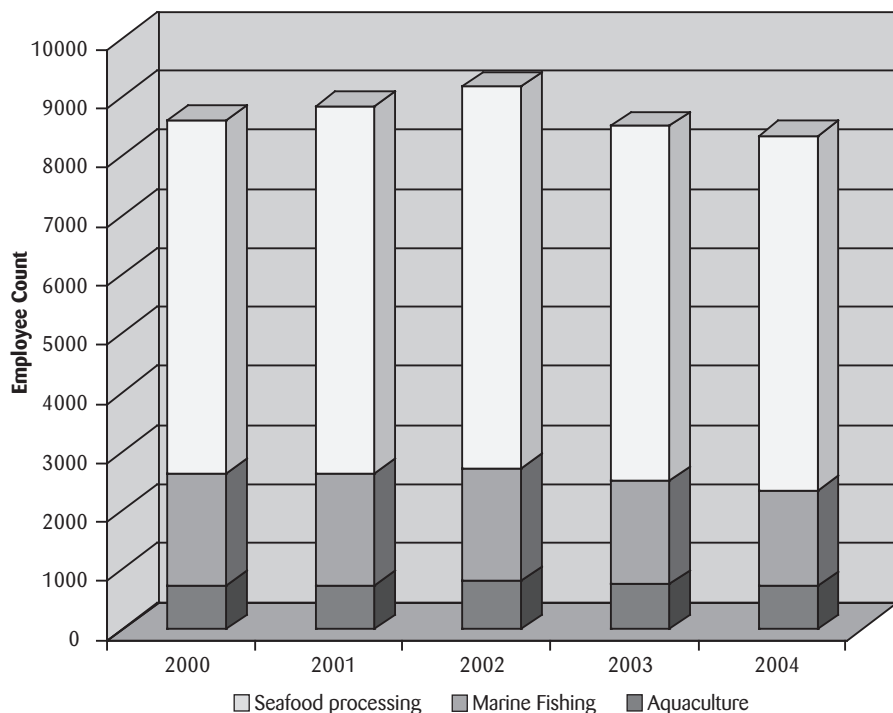


Figure 7.1. Employee numbers in the seafood industry⁷

Source: Statistics New Zealand

The most commercially valuable wild seafood species are hoki, rock lobster, orange roughy, arrow squid, paua and ling (see Figure 7.2). Although they rose during the early 2000s, the export returns for these top six export species dropped significantly in 2003 owing to reductions in the volume of hoki, orange roughy and arrow squid exported, as well as a high New Zealand dollar.

Seafood is extracted from inshore, mid-water and deep-water areas of New Zealand's oceans through a variety of modes, including those for⁸:

- *Hoki*, which is mainly harvested off the west coast of the South Island, in Cook Strait and on the Chatham Rise by trawling at depths of between 300 and 600 metres
- *Rock lobster*, which is harvested around the coast of New Zealand and the Chatham Islands through the use of baited pots in shallow inshore areas at between 5 and 275 metres

⁷ These figures are sourced from Statistics New Zealand's Business Demographic surveys, which are carried out in mid-February of each year and cover all economically significant enterprises.

⁸ See www.sealord.co.nz

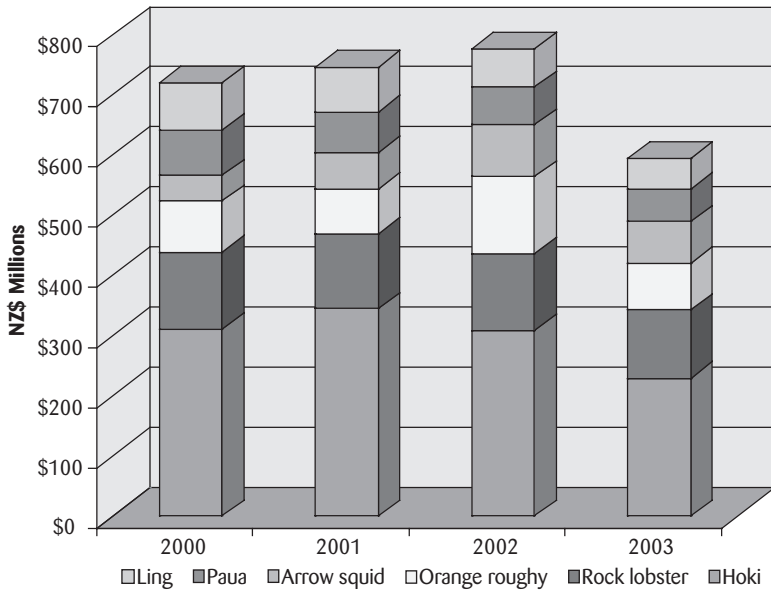


Figure 7.2. Principal fisheries exports by value 2000 to 2003

Source: New Zealand Seafood Industry Council

- *Orange roughy*, which is mainly harvested by bottom trawling on the Chatham Rise, around the southern part of the west coast of the South Island and on some parts of the Challenger Plateau, in depths of between 750 and 1200 metres
- *Arrow squid*, which is mainly harvested around the coast of the South Island and off the Campbell and Auckland Islands by mid-water trawling and sometimes by jigging down to depths of 500 metres
- *Paua*, which is harvested from the South Island, Chatham Island, Stewart Island and the southern coast of the North Island, by free diving using a hand knife in shallow coastal waters of between 1 and 15 metres
- *Ling*, which is mainly harvested around the bottom of the South Island and over the Campbell Rise through bottom longlining and bottom trawling in depths of between 300 and 700 metres

As existing fish stocks become fully exploited, significant growth in the wild seafood industry is likely to require considerable investment in adding value to the current catch rather than increasing the tonnage caught. The recent launch of *Seafood Innovations*, a joint-venture company owned by the Seafood Industry Council and Crop and Food

Research, is responding to this need. The new company will fund research into ways of adding value to the fish catch (Hembry, 2005:C4).

Commercial fishing activity can have significant impacts on the ocean environment. For example:

- Bottom trawling can substantially alter benthic marine habitats
- Over-harvesting can result in drastic reductions in the populations of target fish
- Other non-target species, including fish, seabirds and marine mammals, can be killed as a result of fishing activity
- The removal of large numbers of fish, and the discard of large quantities of fish offal into the sea, can significantly alter food chains

The fishing industry is also sensitive to changes in the ocean environment. In particular, the industry can be negatively affected by:

- Land-based sedimentation and pollution, which can degrade nursery grounds of fish and shellfish beds and contribute to algal blooms
- The introduction of invasive species, which can out-compete and/or damage commercially significant species
- Changes to weather patterns and ocean currents caused by climate change, which can affect nutrient levels and resultant fish recruitment as well as the location of fish
- Recreational fishing targeting commercially significant species
- Other activities occupying the marine environment, such as aquaculture and mining, which reduce the spatial area that can be fished.

Aquaculture has been a very fast-growing industry (see Figure 7.3). Exports peaked in 2002 at NZ\$241 million and then declined to NZ\$186 million in 2003, partly because of algal blooms prevented the movement of spat, as well as the high New Zealand dollar. The main species farmed are mussel, salmon, oyster and paua. The industry is currently concentrated in the Marlborough Sounds, the Firth of Thames, the Coromandel Peninsula, Banks Peninsula, Stewart Island and Northland. In 2003, aquaculture occupied around 47.25 square kilometres of seabed and associated water space (CAE, 2003:11).

Growth of the industry has been hindered more recently by a moratorium on new farms. The moratorium was legislated in 2002, and was subsequently lifted at the end of 2004, after a new regime for allocating ocean space to marine farms and managing their effects was put in place. There is considerable potential for future growth (CAE, 2003:11). Such growth will be affected by the ability of the sector to obtain rights to

exclusively occupy suitable areas of marine space and to maintain high water quality in their areas of operation.

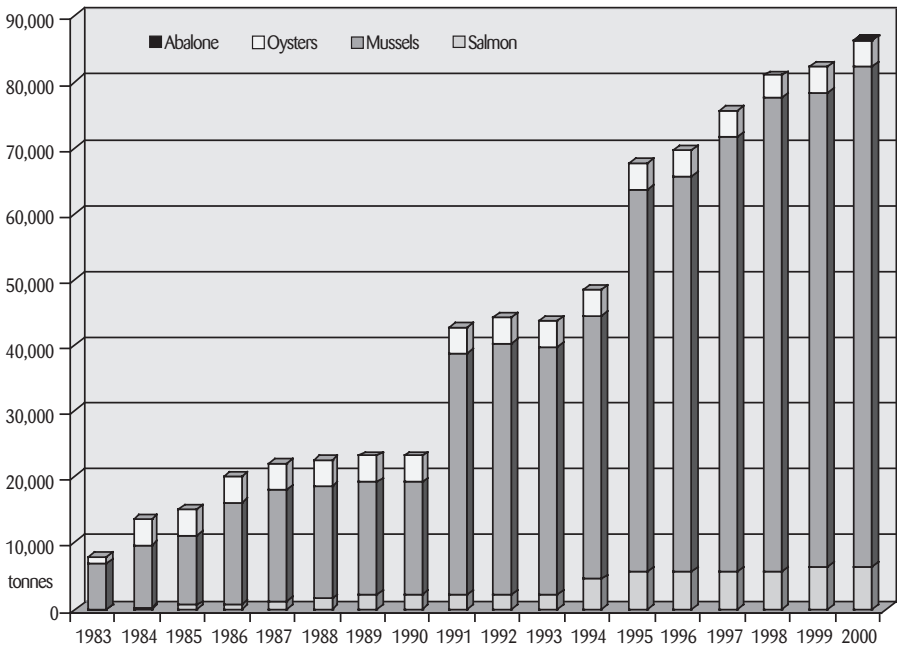


Figure 7.3. Commercial aquaculture harvest 1983 to 2000

Source: Statistics New Zealand, 2000

Mussels are by far the major component of the industry. Mussel farms consist of a series of buoys held together by long lines attached to each side and anchored to the sea floor at each end. A series of weighted ropes hang down from these lines and these are where the mussels grow. Juvenile mussel spat is wrapped around the ropes and left to grow. When the mussels have reached the desired size the ropes are lifted and the shellfish stripped from them⁹.

The impacts of aquaculture on the ocean can include (adapted from McCarthy, 2002:55; Hauraki Gulf Forum, 2005:113):

- Excluding other ocean uses, particularly fishing, shipping and recreational boating, from an area of the ocean
- Reducing the available marine area for other marine species such as dolphins (see Markowitz et al., 2004)

9 See www.seafood.co.nz/business/fishaqua/species/mussels.asp

- Reducing the natural character of the coast through introducing structures into largely unspoiled aquatic environments
- Depleting plankton levels in the water, which can alter local food webs and affect populations of macro invertebrates, fish and birds
- Smothering benthic organisms and changing the seabed habitat beneath the farms through the discharge of pseudofaeces, shell and organic material
- Changing the chemical composition of the sea water through the release of nutrients from shellfish excretion

On the positive side, marine farms can provide human-made reefs and food for fish such as snapper.

Marine aquaculture is dependent on the existence of unpolluted coastal waters, being particularly sensitive to:

- Land-based pollution, particularly contamination of sea water by sewage, which can make the seafood unsafe to eat
- Toxic algal blooms, which can also make seafood unsafe to eat
- The introduction of invasive species which can damage farmed species

Because of this sensitivity, the presence of marine farms in an area can drive initiatives to clean up land-based pollution.



Figure 7.4. Aquaculture is a fast-growing industry – oyster farm located within the Ohiwa Harbour.

Oil and gas exploitation

Oil and gas are currently exploited from 22 wells, drilled from two offshore platforms, located within the Maui gas field 35 to 50 kilometres off the Taranaki coast. There are 33 kilometres of pipelines extending from the Maui field to onshore Taranaki (CAE, 2003:3-4; DoC, 2000b:15). The production value in 2002 was about NZ\$370 million and the oil and gas extraction industry employs around 600 people¹⁰. The Taranaki Basin continues to be the focus of exploration and production. As the Maui field runs down, new offshore fields in the Taranaki Basin are being appraised (including Tui) and developed (Pohokura and Maari). Neptune Minerals plans to mine sulphide deposits around New Zealand and has licences covering large areas of the Kermadec Ridge (New Zealand Press Association, 2005:C3).

The industry has generally suffered from a lack of overseas investment, with the high costs of exploration and development, the real risk of not finding commercially exploitable resources, and the country's relatively small market making New Zealand a relatively less attractive proposition than other locations around the world. The government has recently announced a package of incentives in order to attract more investment into the industry (Ministry of Economic Development, 2004).

In 2000, petroleum exploration permits extended over 121,684 square kilometres of seabed with applications covering a further 11,680 square kilometres. In addition, mineral prospecting permit applications lodged extended over around 90,000 square kilometres of the Kermadec Ridge (DoC, 2000:15).

Oil and gas exploitation can impact on the ocean environment through the disturbance and smothering of benthic habitats in the area where drilling is taking place, pollution of sea water through discharges of oil and other toxic substances, and the reduction in the natural character of the marine area through introducing large structures into an unspoilt environment.

Marine biotechnology

Marine biodiversity is increasingly being identified as a rich source of new and potentially improved products in many areas. Marine biotechnology is currently a small sector within New Zealand, with only six organisations involved in 2001, but with significant growth potential based on New Zealand's unique marine biodiversity (Industry New Zealand, 2001:22).

This industry will benefit in the future from the maintenance of New Zealand's rich marine biodiversity and the effective management of activities which could potentially result in biodiversity loss.

10 Sourced from Statistics New Zealand's Business Demographic survey 2004

Transportation

Almost 99 per cent of New Zealand's imports and exports by volume are transported by sea. The country receives around 3300 calls from international trading ships each year (Maritime Safety Authority, 2003:8). There are currently 13 major commercial ports in the country and around 5190 full-time equivalent jobs in sea transport-related industries.



Figure 7.5. New Zealand receives around 3300 calls from international trading ships each year – ship departing from Port Chalmers, Dunedin.

Ships can transport marine organisms around the world in ballast water and in the fouling on hulls and intake chambers in the hulls of vessels. In 2002, New Zealand ports received 4.4 million metric tonnes of ballast water. Over the past four years merchant vessels visiting New Zealand have increased by 10 per cent a year and changing patterns of trade are exposing the country to additional species not previously capable of reaching New Zealand. (Hewitt et al., 2004:431–432). Ships can also strike whales, and this is recognised as a serious problem in the Hauraki Gulf (Hauraki Gulf Forum, 2005:116).

Increased shipping and larger ships drive demand for the expansion of port facilities and the dredging of harbours, which can also have significant impacts on the marine environment through occupying greater areas of marine space, destroying and smothering benthic organisms, and disrupting natural coastal processes.

Cultural

For Māori, relationships to natural resources, including the oceans, are principally exercised through kaitiakitanga (exercise of guardianship). Māori endeavour to manage

resources so that some are used, some are developed and some are rested to ensure that the mauri (life force) of all resources is kept in balance. Permanent preservation without utilisation is not a feature of Māori resource management. Māori use the resources of the sea for subsistence, ceremonial and commercial purposes, and their ability to access these resources is an important part of their culture and identity. Where direct and active involvement in the use of resources is discontinued, the knowledge Māori associate with those resources, and therefore part of their culture, is lost (Te Ohu Kai Moana, 2001:8–9).

Kaitiakitanga

The concept of kaitiakitanga contains many elements including:

- Mahi tapu – god-given and handed down through tūpuna (ancestors)
- Founded in whakapapa (genealogy) – the relationship between everything and everybody in the natural world
- Exercised on behalf of, and for the benefit of, all who are related through whakapapa
- A set of inalienable responsibilities, duties and obligations that are not able to be delegated or abrogated
- A web of obligations with kaitiaki (those exercising guardianship) to provide for everyone and ensure everyone benefits
- Independent of 'ownership' in a European sense
- Seamless and all encompassing – making no sharp distinction between moana (the sea) and whenua (the land)
- Given effect at whānau (extended family) and hapū (subtribe) levels
- Expressed in ways that are appropriate to the place and to the circumstances, according to tikanga (custom)
- Given practical effect by exercising control over access to resources and sharing the benefits of the use of those resources
- Enabled through rangatiratanga (chiefly authority), which includes the authority that is needed to control access to and use of resources and to determine how the benefits will be shared

(Adapted from Aquaculture Steering Group, 2003, 8–9)

Enjoyment

Tourism

The tourism industry is now New Zealand's top foreign-exchange earner, contributing 18.5 per cent of New Zealand's total exports in 2004. In 2004, tourism generated an all-time high of NZ\$7.4 billion in foreign exchange from around 2 million international visitors (Statistics New Zealand, 2005a; Tourism Research Council New Zealand, 2005:1). It is not clear what proportion of this expenditure is related to marine tourism, but a significant number of international tourists are involved in a range of marine-related activities (see Figure 7.5).

The Tourism Research Council's annual international visitor survey for 2004 indicates that the most popular marine activities for tourists are visiting beaches (undertaken by 650,000 visitors) and taking scenic cruises (560,000 visitors). The number of international visitors partaking in these activities has increased significantly since 2002. There has also been a significant increase over the past seven years in those visiting penguin or seal colonies, watching whales or viewing and interacting with dolphins. Over 100,000 international visitors viewed penguin colonies, seal colonies and/or dolphins during 2004 (Tourism Research Council New Zealand, 2005).

Tourism can impact on the marine environment in a wide variety of ways including (see Parliamentary Commissioner for the Environment, 1997 and 2003):

- Driving increased demand for infrastructure and buildings on the coast, including accommodation, visitor centres, toilets, car parks, jetties, shops and improved roads
- Increasing visitor numbers in relatively unspoilt areas, which can result in the loss of the wilderness experience through overcrowding and noise from vehicles and aircraft overflights, and in environmental impacts through track erosion and vegetation trampling
- Increasing boating activity, including the frequency of cruise ships, which increases the risk of marine pollution, biosecurity breaches and collision with marine mammals, and heightens water turbidity
- Driving behavioural changes in wildlife, such as seabirds and marine mammals through viewing and interacting with them

Tourism is likely to be sensitive to the quality of the marine and coastal environment, and the growing interest of international visitors in marine nature tourism may encourage members of the tourism industry to seek improved environmental management of these areas.

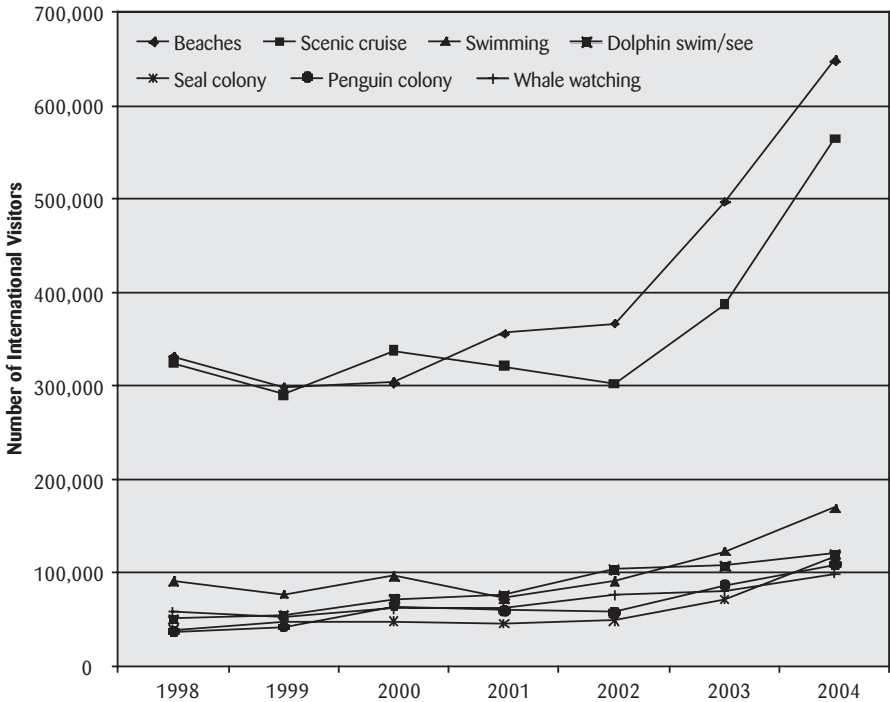


Figure 7.6. Number of international visitors involved in marine activities and visiting marine attractions in New Zealand 1998 to 2004

Source: Tourism Research Council New Zealand 2005

Lifestyle

All New Zealand residents live within 130 kilometres of the sea and more than 80 per cent live on or close to the coast. 'The ocean is central to the identity and well-being of New Zealanders. Most feel some personal connection to the sea ...' (Ministerial Advisory Committee on Oceans Policy (MAGOP), 2001:5).

'For many New Zealanders, free and open access to the sea and coast is regarded as a fundamental democratic right ...'

(MAGOP, 2001:26)

New Zealanders use the marine environment for a variety of non-economic reasons including fishing, diving, swimming, surfing, boating, playing on the beach, walking along the coastline and viewing.

The desire to live and pursue recreational activities on the coast has driven strong demand for coastal development, both for houses and holiday homes, and for commercial activities such as shops and cafes which service the inhabitants. This in turn drives the need for improved infrastructure, including roads and sewage-treatment works. If poorly managed, such development can have a negative impact on the coastal and marine environments through increasing land-based sedimentation and pollution of marine areas, reducing the natural character of the coast, obstructing natural coastal processes and destroying historic and cultural heritage located on the coast.



Figure 7.7. Development can degrade the natural character of the coastal environment – Langs Beach, Whangarei.

On the positive side, more people settling on degraded coastal farmland can result in the rehabilitation of natural systems through indigenous revegetation, weed and pest control and the upgrading of sewage treatment and disposal systems thus reducing marine pollution.

There are more than 270,000 recreational vessels within New Zealand including ocean-going yachts, high-speed-powered craft and canoes (Maritime Safety Authority, 2003:9). Recreational fishing is a major activity, with an estimated one in four adult New Zealanders (674,300) going fishing each year. Sailing is also popular with an estimation that just under 150,000 adults go sailing each year¹¹.

11 Sourced from *Sport and Recreation New Zealand Facts Series* 1997 to 2001, which contains the combined results of surveys carried out during 1997, 1998 and 2000.

Recreational fishers spend, collectively, nearly a billion dollars annually on their sport. Probably only a small proportion of these fishers, about 15 per cent, is responsible for almost two-thirds of the recreational catch. Recreational fishing extracts a significant amount of fish and shellfish from the ocean. For example, it is estimated that during the 1999–2000 year over 7.8 million snappers, 2.5 million rock lobsters, 8 million cockles and 22 million pipi were harvested by recreational fishers. In some areas the recreational catch is significantly larger than the commercial catch for species such as snapper and blue cod (Ministry of Fisheries (MFish), 2003:66; Ecologic Foundation, 2001:6; Boyd and Reilly, 2002).

Future uses

Technology is rapidly developing and future uses of marine areas may include the exploitation of gas hydrates under the sea floor, the development of offshore wind and marine farms and the deployment of structures designed to harness marine currents, tides and waves (CAE, 2003). The impacts of these may be significant but are not yet fully understood.

8

How Do Uses Impact on Ocean Ecosystems?

As described in Chapter 2, activities can disturb the marine environment through destroying or altering marine habitats, over-harvesting marine resources, causing water pollution and eutrophication and introducing invasive species. The nature of these disturbances within New Zealand's oceans is explored in the following sections.

Marine habitat alteration and destruction

The dragging of fishing gear such as trawls and dredges across the sea floor can destroy benthic habitats by removing topographic features and living organisms. Such activity also has impacts through suspending sediment in the water and changing water chemistry.

'There is now good evidence that commercial fishing has a profound effect on marine ecosystems.'
(Thrush and Dayton, 2002:450)

In the areas involved, these effects can in turn lead to reduced species diversity, the loss of important nursery areas for fish and, potentially, a reduction in the productivity of marine areas owing to the disruption of nutrient recycling by benthic organisms. The effects of such activity have been compared to the impacts of the clear-cutting of indigenous forests, because of the significant changes in habitat and resultant reduction in biological diversity that occur (Watling and Norse, 1998).

The long-term impact of trawling and dredging is largely dependent on the extent, frequency and weight of trawls and dredges and whether the biota has the opportunity to re-establish itself between disturbances. The effects of trawling are thought to be greater on habitats which suffer less natural disturbance because the organisms living in these habitats are less adapted to change (Kaiser et al., 2002:118).

Extent and nature of trawling activity

A significant amount of trawling activity takes place in New Zealand waters with around two-thirds of annual fishing revenue being derived from trawling operations (Bathgate,

2004:C2). Although the full extent and frequency of trawling activities within New Zealand's oceans has yet to be publicly documented, the level of disturbance of the sea floor appears significant in some areas. For example:

- In Foveaux Strait oyster dredging disturbs very localised areas of oyster beds at least 12 to 15 times a year (Cranfield et al., 1999:480).
- In the inner Hauraki Gulf, research into the extent of bottom trawling, Danish seining and dredging undertaken during 1993 indicated that dredging disturbed some parts of the area more than five times a year (Thrush et al., 1998:869). Commercial trawling is now banned in the Hauraki Gulf (Froude and Smith, 2004:64).
- In Quota Management Area 1 (QMA 1) off the Bay of Plenty, trawlers fishing for scampi, gemfish, hoki or tarakihi in water between 200 and 600 metres deep made an annual average of approximately 2000 tows and swept over about 2100 square kilometres of continental slope. Although this comprised only 20 per cent of the upper continental shelf in QMA 1, fishing tended to concentrate in certain areas with the effect that the main fishing depths for scampi, gemfish and hoki (300 to 500 metres) were swept, on average, once to several times each year (Cryer et al., 2002:1829).
- The number of seamounts being exploited by deepwater fisheries has increased significantly since the late 1970s and, by the 1999–2000 year, 248 seamounts had each been fished with more than 10 tows. The most rapid exploitation occurred during the early 1990s when more than 20 new seamounts were fished each year. This rate levelled off in the later 1990s when fewer new seamounts were being discovered. Almost 80 per cent of seamounts with a depth at the peak of between 500 and 1000 metres are now fished (Clark and O'Driscoll, 2003:446–7).

Impact of trawling activity

Little research has been undertaken on the impact of trawling and dredging in New Zealand and there are no long-term monitoring studies of the effects of fishing activities on benthic communities in the EEZ (Froude, 1998:12). There are, however, documented historical examples of trawling and dredging activity fundamentally and possibly irreversibly modifying benthic habitats, such as with the destruction of mussel beds in the Firth of Thames and the extensive destruction of bryozoan growths in Tasman Bay and Foveaux Strait (Turner et al., 1999:404–405, Cranfield et al., 1999).

More recent research has confirmed that trawling has significant immediate effects on benthic habitats and species. For example:

- An analysis of the impact of a scallop dredge in sand at Mercury Bay found that, after five parallel tows, 50 per cent of the macrofauna showed significant reductions in abundance, the community composition differed and the effects continued for at least three months after the dredging took place (Thrush et al., 1995)
- Analysis of seabed samples, collected from 18 sites within the Hauraki Gulf in depths ranging from 17 to 35 metres, indicated broad-scale changes in benthic communities directly related to fishing. With decreasing fishing pressure, increases in the density of echinoderms, long-lived surface dwellers, the total number of species and individuals and diversity were observed (Thrush et al., 1998)
- A study of the impacts of bottom trawling on eight seamounts found clear evidence of substantial impact on the benthic fauna. Those which had been heavily fished, having been the recipient of 600 or more tows, were characterised by large areas of barren sea floor with exposed basement rock, coral fragments or soft sediment and few live invertebrates. Coral was found on only 2 to 3 per cent of the area photographed on these seamounts, compared with 100 per cent cover often recorded on the lightly fished seamounts (10 or fewer tows) (Clark and O'Driscoll, 2003:452). Slow growth rates of deep-water corals and other species indicate that recovery times for seamount communities could be 200 to 400 years (Probert, 1999:601–602; Turner et al., 1999:407)

As well as identifying the direct effects of fishing activity on benthic habitats, research has focused on the flow-on effects within marine ecosystems from disturbance of the sea bottom. One study has linked the presence of habitat structures on the soft sediments of the sea floor in Kawau Bay to an abundance of juvenile snapper, indicating that a reduction in such structures through trawling and dredging activity may ultimately impact on fish stocks (Thrush et al., 2002:279).

A second study has linked the disturbance and simplification of seabed habitat in Fouveau Strait, as a result of oyster dredging, to slower growth in juvenile blue cod. This was thought to be owing to the greater diversity of food and shelter from predators provided by more complex habitats (Carbines et al., 2004:500).

The implications of this work for the fishing industry are that it may be important to identify which habitats are significant for juvenile species of commercial species and to provide these with some protection.

What is not yet fully understood is the significance of these documented localised effects within the broader picture of New Zealand's oceans and the extent to which affected species recover over time. Some questions we need to urgently answer are:

- How much of each major marine habitat type is being impacted by fishing activity?
- Where and to what extent is fishing activity causing significant and/or permanent change to benthic habitats?
- Are there any species and/or habitats that we are losing or may lose as a result of fishing activity?
- If we are not sure, are we sufficiently applying a precautionary approach to managing the effects of fishing?

Harvesting of marine resources

Harvesting of marine resources includes the catching of species targeted by the fisher, and the incidental or by-catch of species which are caught during fishing.

Targeted species

Around 100 marine species are harvested commercially. Fishing activity can reduce the size of fish stocks significantly, as shown in the summary of estimates of the current status of stocks for the two most commercially important fish species, hoki and orange roughy, presented in Figure 8.1.

Recreational fishing also has significant impacts on fish stocks, targeting about 40 marine species and harvesting an estimated 10 to 20 per cent of the total catch of most inshore species. For some species, the proportion of catch harvested by recreational fishers is much higher, reaching 80 to 90 per cent for blue cod in the Marlborough Sounds and 40 per cent for snapper in the Hauraki Gulf (DoC, 2000b:14).

There is no clear benchmark to establish when a stock has been reduced to a level that threatens its sustainability. However, a rule of thumb adopted by stock assessment scientists is that, once a stock is reduced to less than 20 per cent of its virgin biomass, there is an increased risk that recruitment to the fish stock will not suffice to balance removals from the fishery (Controller and Auditor-General, 1999:71).

There are ecological flow-on effects from significantly reducing fish populations. Fishing preferentially removes larger and older fish, changing the size and age structure of exploited populations and reducing their genetic diversity. A significant loss of genetic diversity was observed in a population of orange roughy within New Zealand's EEZ when a 70 per cent reduction of the population occurred after six years of intensive fishing (Goni, 1998:39–44). A similar loss of genetic diversity was found amongst the heavily fished Tasman Bay snapper where the total biomass declined from an estimated 23,000 tonnes to less than 1600 tonnes (Smith et al., 2003:7).

Species	Fish stocks	% of virgin biomass
Hoki	Western stock	13–22
	Eastern stock	= or > 40
Orange Roughy	Northern North Island	15
	East Coast	24–25
	Mid-East Coast	17–32
	North-West Chatham Rise (ORH 3B)	17–20
	North-East Chatham Rise	34–54
	South Chatham Rise	29–41
	Puysegur	7–25
	Challenger Plateau	3
	West Coast South Island	17–45

Figure 8.1. Estimated stock levels of hoki and orange roughy as percentage of virgin biomass

Source: Annala et al., 2004

The impacts of the selective removal of a target species can also cascade through the entire marine ecosystem, as another species replaces the niche now left largely unoccupied. For example, because top predators such as fish and crayfish have been extensively fished in coastal areas around New Zealand, their major prey, kina (sea-egg), has increased dramatically. This has in turn resulted in a general decrease in the abundance of kelp on which kina feed (Andrew and Francis, 2003:45). Fishing can also reduce food sources on which other species are dependent. The full extent and significance of these broader ecosystem impacts of fishing activity are largely unknown.

Impacts on non-target species

A large number of non-target species, including fish, invertebrates, birds and mammals, can become caught in fishing gear.

Trawl fisheries, in particular, can catch a significant amount of non-target species, which, in some cases, such as in the scampi fishery, can significantly exceed the target

species catch (see Figure 8.2). Species caught unintentionally may be vulnerable to overfishing, especially when populations are small (Livingston et al., 2003:4).

Much of the non-target catch does, however, consist of other commercially exploited species and this unintentional harvesting is taken into account in establishing the total allowable catch for those species. Both target and non-target marketable species are generally retained for sale whereas species for which there is no market are discarded.

Target species	Target species caught annually (tonnes)	Non-target species caught annually (tonnes)	Discards annually (tonnes)
Hoki	245,000	15,000–40,000	6,000–12,000
Orange Roughy	19,000	7,000–15,000	900–2,000
Arrow squid	17,000–31,000	10,900–18,800	2,200–4,300
Scampi	8,00–1,000	3,200–6,800	1,400–5,300
Jack Mackerel	13,000–22,000	11,600–15,500	1,550–2,600
Oreo	9,500–17,700	470–2,900	230–1,900
Southern blue whiting	17,500–76,000	300–2,500	140–1,200

Figure 8.2. Annual by-catch of non-target species

Source: Anderson et al., 2001; Anderson, 2004a; Anderson, 2004b

There is, currently, insufficient information to identify the overall impact of fishing activity on seabirds. These birds, particularly albatrosses and petrels, are vulnerable to longlining because they take baited hooks and can get hooked or tangled in the line and drown. Squid, scampi and hoki trawl fisheries can also impact on seabirds, which collide with or become entangled in fishing equipment. Thirty species of seabirds, of which 16 are endemic to New Zealand, have been reported caught in longline and trawl fisheries. Eleven of the species reported being caught have a restricted range, six are in decline and two have been classified as nationally vulnerable by DoC (MFish and DoC, 2004a:55–56).

A recently released research report has attempted to quantify the number of seabirds captured by different New Zealand fisheries (Baird, 2004). The estimates provided in the report are based on the findings of observers on fishing vessels, but as only some boats carry observers some of the time, it was not possible to provide robust estimates for all fisheries. Some of the estimates produced are shown in Figure 8.3 and indicate that at least several thousand seabirds are captured each year.

Fishery	Estimated number of birds captured (2000–01 fishing year)
Autoline bluenose longline	7
Chartered tuna longline	16
Squid trawl	433
Hoki trawl	1065
Autoline ling longline	2367

Figure 8.3. Estimated number of seabirds captured by specific fisheries during the 2000–01 fishing year

Source: Baird, 2004

Seabird mortality can be reduced significantly through adopting modified fishing procedures. For example, in the joint-venture tuna fishery the incidental by-catch of seabirds peaked in 1992–93, when an estimated 891 birds were caught, but this was subsequently reduced to 40 birds in the 1999 to 2000 year with 14 of those being released alive (MFish and DoC, 2004:8–9). However, not all fishers are willing to adopt such measures voluntarily. It was recently reported that two-thirds of the squid fishing fleet of 35 vessels were failing to adopt practices agreed between MFish and the Squid Fishery Management Company designed to reduce the number of seabirds killed in the fishery (Beston, 2005:1).

Ninety-five per cent of sea lion breeding occurs on two sites in the Auckland Islands. Rather unfortunately, breeding and pup-rearing occur at the same time as the squid trawl fishery operates on the Auckland Shelf. In attempts to eat squid, sea lions get caught in the trawl nets and drown. In the 16 years between 1987/88 and 2002/03, it is estimated that 1153 sea lions have been killed in the squid fishery (see Figure 8.4). The New Zealand sea lion is still recovering from historic harvest by early Polynesian settlers and by the sealing industry, and is classified as range-restricted and threatened owing to its limited number of breeding sites and foraging ranges (Suisted and Neale, 2004:27).

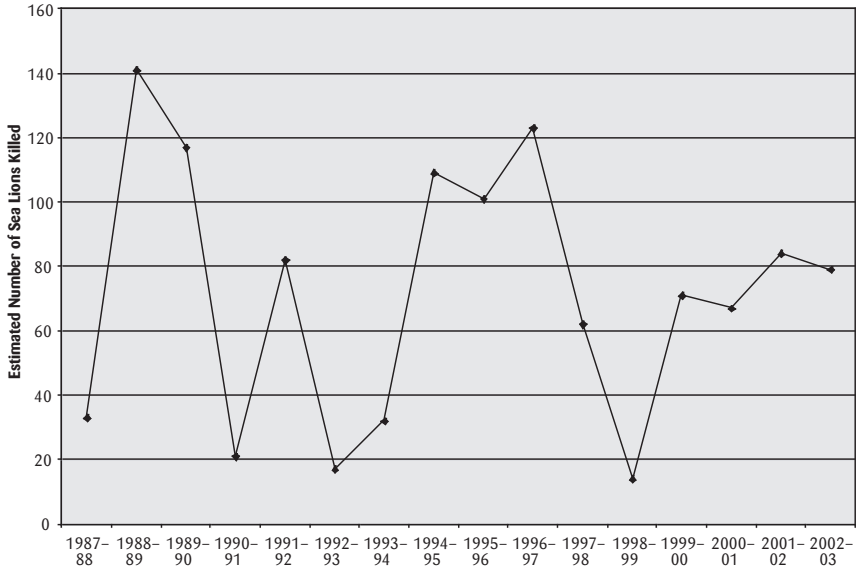


Figure 8.4. Number of sea lions killed in the squid fishery 1987-88 to 2002-03.

*Source: Squid Fishery Management Company Limited v Minister of Fisheries and
Chief Executive of Ministry of Fisheries CA39/04*

The major deep-water trawl fisheries for hoki, hake, southern blue whiting and squid capture and drown a large number of fur seals, with possibly between several hundred and 1000 being drowned annually (Donoghue, 1998:182). Fur seals are not regarded as a threatened species and are expanding their range. However, the population is still recovering from the severe impacts of harvest by the early sealing industry and DoC is endeavouring to reduce the inadvertent killing of these marine mammals (Suisted and Neale, 2004:36).

The nationally vulnerable Hector's dolphins and nationally critical subspecies the Maui's dolphins are susceptible to entanglement in set nets used by recreational and commercial fishers. Hector's dolphins live in shallow murky waters around the South Island and are caught in the nets of these fishers. Maui's dolphins are found only on the west coast of the North Island. In the 20 months up to January 2003, six dead Maui's dolphins were recovered, the death of at least four caused by nets (DoC and MFish, 2003a:2).

Water pollution and sedimentation

Land-based activities can have a significant impact on the marine area through their impact on run-off into the sea. Vegetation clearance and earthworks increase the amount of sediment flowing into the ocean. Sewage, polluted stormwater and agricultural run-off increase the nutrients and pollutants in sea water and marine sediments. Many of these impacts are heightened in sheltered estuaries and harbours, areas which provide important marine habitats, and which are the location of much human settlement in New Zealand.

Sedimentation

The discharge of sediment into the oceans is thought to be the most damaging land-based impact in New Zealand and the main cause of reduced water clarity in coastal areas. Around 209 million tonnes of sediment washes from the mainland into the sea each year. The areas of highest yield are East Cape and south Westland (Hicks et al., 2003:26–27). Deforestation and the urbanisation of catchments have increased the volume of sediment washed into estuaries significantly, typically up to 10 times greater than pre-human times.

One study at the urbanised Pauatahanui Inlet found that almost a tonne of sediment per hectare per month ran into the estuary, 15 times more than in a neighbouring non-urbanised catchment. Sediment loss rose to 70 times greater when a flood struck (Williamson et al., 2003:238). In the Mahurangi Harbour, which is surrounded by a largely rural catchment with areas of exotic forestry, sedimentation rates on intertidal flats currently average about 7 millimetres a year compared to pre-1840 rates of 0.3 to 0.8 millimetres a year (Hauraki Gulf Forum, 2005:111).

Sediment can have profound and irreversible effects on marine ecosystems, through changing benthic habitats, smothering bottom-living organisms and reducing the clarity of the water. Sedimentation has been found to result in a decrease in the biodiversity of bottom-living organisms and can result in the loss of habitats, such as seagrass, which are important nursery areas for some species of fish. Sediments can also carry toxins and nutrients which have additional negative effects. Impacts are more significant in low-energy, enclosed areas where sediment cannot escape to the wider marine environment such as estuaries and harbours (Williamson et al., 2003:239; Honeywill, 2002:39).



Figure 8.5. Mangroves continue to expand in northern estuaries owing to increased sedimentation – Miranda coast, Firth of Thames.

Examples of specific concerns about the impacts of sedimentation include:

- A decline in marine species (such as seagrass and algae) which require light for photosynthesis, due to reduced water clarity from sedimentation. New Zealand has only small areas of seagrass beds and they provide an important habitat for some species, including wading birds. Seagrass has been declining around New Zealand's coasts over past decades. For example, in Whangarei Harbour seagrass once covered around 1400 hectares but is now reduced to small pockets. In Whangamata Harbour, seagrass currently occupies only 60 hectares or 60 per cent of the area it covered 40 years ago (DoC, 2000a:15; Church, 1998:36; Hauraki Gulf Forum, 2005:110; Thompson, 2005:A13)
- An increase in mangroves in estuaries in the northern half of the North Island, which are replacing other intertidal habitats and reducing the amenity and accessibility of marine areas for human activities. For example, on the Coromandel Peninsula the area covered by mangroves in the Manaia Estuary increased by 195 per cent between 1971 and 1995 and that in the Tairua Estuary by 215 per cent between 1983 and 1996 (Hauraki Gulf Forum, 2005:110). Mangroves in the Firth of Thames now cover over 10 square kilometres, whereas 60 years ago there were only small stands at the mouths of the Waihou and Piako Rivers (Brownell, 2004:33–34). Mangroves do play important ecological roles, being a key source of organic material and nutrients for the marine area, providing a habitat for fish

species and wading birds, improving water quality and reducing coastal erosion and storm damage (Nicholls and Ellis, 2002:24).

- The negative impacts of suspended sediment on shellfish such as cockles, pipi and horse mussels. These organisms obtain their food by filtering water through their gills and at the same time also take in non-food particles, such as sediment, suspended in the water (Hancock and Hewitt, 2004:16).

So what is the significance of this progressive degradation of estuaries? Estuaries play several important roles in marine ecosystems (Arnold 2005: 38):

- They are a critical habitat for fish species which migrate between fresh and salt waters
- They are the sole habitat for a range of small species, including estuarine triplefin and gobies, some of which are endemic
- They are an essential nursery habitat for many commercially important fish species such as snapper, rig, red gurnard, flounder and mullet

In terms of ecosystem services, estuaries are thought to provide the greatest value per hectare of all biomes (Costanza et al., 1997:256).

Research being undertaken by the National Institute of Water and Atmospheric Research (NIWA) is currently endeavouring to establish which estuaries, and which habitats within them, are important for adult fish production. Initial results confirm that coastal fish populations are supported by juvenile nurseries within estuaries and shallow coastal embayments. These sheltered areas assume more central importance on exposed coasts such as the high-energy west coast where there is no shelter from island chains (Morrison et al., 2004).

Nutrient enrichment

Sewage is a significant source of nutrient inputs into many harbours and coastal areas adjacent to settlements, through inadequate sewage treatment, leakages and overflows, and the failure of septic tanks (Taylor and Smith, 1997:7.44). Nutrient enrichment can also be caused through farming run-off. It was recently estimated that more than half of Northland's septic tanks did not work properly and almost all the waterways in lowland areas of the North Island have levels of faecal bacteria that exceed Ministry of Health guidelines for swimming (Collins, 2004).

It is not uncommon for swimming to be restricted at beaches due to sewage contamination, especially after heavy rainfall. Most local authorities in the Hauraki Gulf advise against swimming for 24 to 48 hours after heavy rain (Hauraki Gulf Forum, 2005:48).

The impacts of nutrient enrichment on marine ecology are largely unknown, although they may be significant. Observed impacts have included blooms of sea lettuce in Tauranga Harbour and recurrent toxic algal blooms, which may be partially land-based nutrient related (Church, 1998:35).



Figure 8.6. Beaches are often closed in New Zealand owing to sewage contamination – Kawakawa Bay, Manukau City.

Heavy metal accumulation

The accumulation of heavy metals in marine sediments near large urban areas is a significant problem. About half of the Auckland region's 3500 hectares of coastal sediment has excessive concentrations of lead, zinc and copper. Concentrations are rapidly increasing and, if accumulation continues at current rates, it is estimated that zinc concentrations will double in the Whau Estuary within 13 years and in some locations in the Tamaki Estuary within 17 years. This contamination can negatively affect the growth and reproduction of marine organisms and result in increased death rates (Hauraki Gulf Forum, 2005:58 & 60).

The main source of these contaminants is motor vehicles, with lead from leaded petrol (which has now been phased out), zinc from tyres, copper from vehicle wiring and polycyclic aromatic hydrocarbons from vehicle exhausts (Taylor and Smith, 1997:7.74). There are currently over 600,000 cars in the Auckland region and, based on current trends, the use of cars could double over the next 20 years (Auckland Regional Council, 2003:28) potentially resulting in significantly increased pollution. Run-off from iron roofs has also been identified as a major (and possibly the major) source of zinc in the Auckland region coastal marine area (Auckland Regional Council, 2004).

The Auckland Regional Council has indicated that heavy metals will almost certainly continue to accumulate in the harbour sediments and that their management efforts are only slowing the rate of degradation (Willis, 2003:23).

Invasive marine species

The introduction of exotic marine species is a continuing problem, with 148 known to have been accidentally introduced into New Zealand, some of which have caused serious problems. About 70 per cent probably arrived as a result of fouling on the hulls of vessels (Oceans Policy Secretariat, 2003a:2). Large ports are at the greatest risk owing to the large number of visiting overseas vessels. For example, a 1997 study of the Waitemata Harbour found more than 60 new species of marine organisms, which appeared to have arrived since a previous study during the 1950s (Hayward, 1997).

Introduced species which are causing problems include (DoC, 2000a:15):

- The Pacific oyster (*Crassostrea gigas*), which forms very dense beds that exclude other native shellfish including pipis, cockles and black mussels
- The Asian date mussel (*Musculista senhousia*), which has displaced cockle beds in the Waitemata Harbour and other harbours north of Auckland, negatively affecting traditional Māori food-gathering areas
- The highly invasive Japanese kelp (*Undaria*), which arrived in the mid-1980s and which has displaced native seaweed communities. It may also decrease the recruitment of paua through displacing native coralline algae and reducing the diversity of sessile organisms
- A mudworm (*Polydora haswelli*), which bores holes into green-lipped mussels deforming their shells (Read and Handley, 2004:30)
- Some species causing toxic algal blooms, which may have been introduced through ballast water

Toxic blooms

Toxic blooms have been increasingly documented along the New Zealand coast since the early 1990s. The first outbreak of recorded shellfish poisoning from toxic algal bloom occurred in the summer of 1992 and, by the end of May 1993, 187 cases of poisoning had been diagnosed. The country's entire coastline was closed to shellfish-gathering and 1000 workers in the shellfish industry were laid off for some months. Thousands of dead shellfish were washed up along the coast. Deaths of little blue penguins, gulls and shags were also recorded (Taylor and Smith, 1997:7.73).

Since that outbreak, frequent algal-bloom occurrences have been documented, including one in the late autumn of 2002 when a bloom consisting of several species killed thousands of fish north of Auckland, with at least 10,000 being washed up on Orewa, Hatfields and Red beaches over four days. About 8500 paua at a marine farm at Kennedy Bay on the eastern Coromandel Peninsula were reported killed, even though sea water used in the land-based aquaculture facility was pre-filtered to remove algal cells. The bloom destroyed the gills of the fish, causing them to suffocate. One of the algae responsible, *karenia concordia*, was found to be new to science and capable of killing cells in marine organisms.

Around 70 per cent of algal blooms occur within the Hauraki Gulf or along the north-east coast of the North Island. Research carried out by NIWA has indicated that the main episodes of blooms are associated with the El Niño weather pattern. During an El Niño episode, north-westerly winds are more frequent and these induce an upwelling of nutrients to the sea-water surface. At the same time, algal species which cause the blooms can be carried towards the New Zealand coast along with the warm water contained in the subtropical East Auckland current. This current of warm water has been observed intruding into the Hauraki Gulf when blooms occur. Nutrient run-off from land is also likely to have a causative impact (Chang, 2004).

Part Two Conclusion

New Zealand has jurisdiction over an extensive marine area, which is rich in marine life and non-living resources. These resources are used in many ways including for commercial extraction, for the transportation of people and goods, for cultural purposes and for enjoyment. Information about the impacts of these activities on the marine environment is patchy, but available evidence indicates that the effects of fishing and land-based activities on marine habitats and species are likely to be significant in some areas. The accidental introduction of exotic invasive marine species, primarily by ocean-going vessels, is also an ongoing problem.

Part Three

The Governance of New Zealand's Oceans

Introduction

This part begins by describing what an ideal oceans governance framework might look like including its key elements: sustainable management practices across the marine environment, a network of special marine management areas and a network of highly protected marine areas.

Secondly, it describes and evaluates the New Zealand oceans governance system within the three key elements of the framework outlined, while also assessing how well these elements function together. The purpose of the evaluation is to help identify areas of weakness which a national oceans policy could help address.

Finally, this section addresses the issue of what a New Zealand oceans policy should contain, drawing on the information and analysis already presented.

Oceans Governance Framework

So what might an oceans governance framework, which incorporates ecosystems-based and integrated approaches, look like? It is likely to consist of three key elements, which have been adapted from the recent work carried out by an international technical expert group, operating under the auspices of the Convention on Biological Diversity (Secretariat of the Convention on Biological Diversity 2004: 15), and which are illustrated in Figure 9.1:

- *Sustainable management practices* applied to all human activities which impact on the marine environment to ensure that such activities do not undermine marine ecosystems
- A network of *special marine management areas* incorporating sensitive and important marine environments where additional management effort is required
- A representative network of *highly protected marine areas* where natural processes within the marine environment are able to operate largely undisturbed

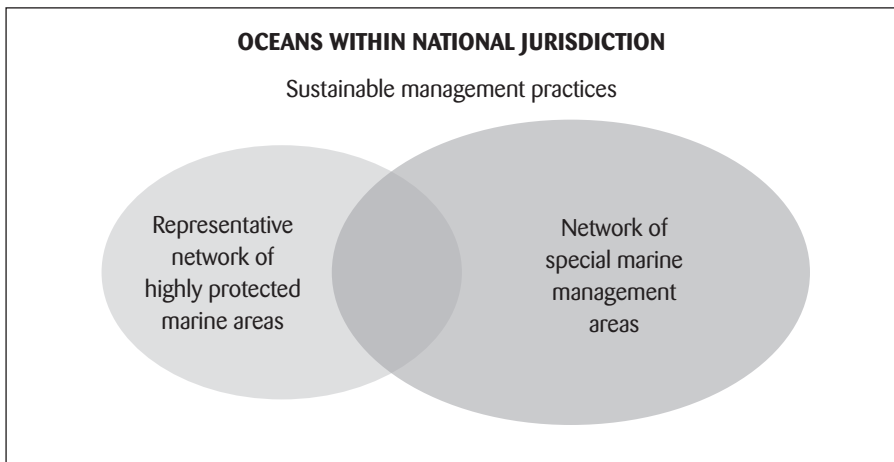


Figure 9.1. Elements of an oceans governance framework

Sustainable management practices

The first element of the oceans governance framework consists of sustainable management practices which extend over the wider marine environment. These typically include spatial planning and environmental constraints on activities such as mining, shipping, fishing and coastal development. Such management practices could be expected to incorporate an ecosystems-based management approach, whereby human activities are modified to ensure that they do not undermine marine ecosystems.

Sustainable management practices help maintain healthy marine ecosystem functioning throughout the marine area. They serve to maintain the connective processes operating between special management areas and highly protected areas and therefore link together the 'nodes' of the 'network'. They also help address human-induced impacts which are difficult to manage through spatially-specific measures (adapted from Secretariat of the Convention on Biological Diversity, 2004:20).

A governance framework promoting sustainable management practices might have several key components including:

- A *fisheries management regime* that identifies and effectively manages the impacts of fishing activity on marine ecosystems
- A *catchment management regime* that effectively manages marine sedimentation and pollution from land-based sources
- An *environmental impact assessment regime* that ensures that structures and activities which are established in the marine area – such as reclamations, wharves, minerals exploration, aquaculture and power generation facilities – do not disrupt marine ecosystems
- A *marine pollution management regime* that effectively manages the discharge of pollutants into the marine area from ships and marine structures
- An effective *bio-invasion prevention and response capability* to ensure that ecologically and/or economically damaging species are not introduced and established in New Zealand

Ensuring that sustainable management practices are adopted by activities impacting on the marine environment is important, because such practices will influence the environmental health of the vast bulk of the oceans. Only a small percentage of the marine area is ever likely to be managed within special management or highly protected areas, and spatially-based management of marine areas is not an effective tool to manage a range of impacts including bio-invasion, land-based sedimentation and pollution, all of which themselves can impact on specially managed or protected areas.

Special marine management areas

The second component of an oceans governance framework is a network of special management areas incorporating sensitive and important marine environments where additional management effort is required. This may include placing additional controls on activities to reduce their environmental impacts to an acceptable level. It may involve applying more focused management attention to a specific area including the development of specific policies, strategies and plans as well as monitoring change over time. Where there are multiple threats to, and/or conflicting uses of, a marine area it may involve developing specific mechanisms to help integrate the activities of different management agencies. International examples of special management areas include the Great Barrier Reef Marine Park in Australia and National Marine Sanctuaries in the USA.

These special management areas are often large, may contain highly protected areas, and can help buffer such highly protected areas from the impacts of intensive human activities (Secretariat of the Convention on Biological Diversity, 2004:18).

Representative network of highly protected marine areas

A key element of an oceans governance framework is a representative network of highly protected areas, where extractive uses are prevented and other significant human pressures removed or at least minimised. These areas are often referred to as marine reserves or marine national parks. The aim of the highly protected network is to enable the integrity, structure, functioning and exchange processes of and between ecosystems to be maintained or recovered. To achieve this aim, the network should where possible encompass a full range of marine ecosystems, including both representative areas, and those that are unique or special.

There is no simple 'rule of thumb' to determine when enough has been highly protected, as this depends on local conditions such as the variability in habitats, but a few small highly protected areas are unlikely to be sufficient. Recommendations on the minimum area needed to achieve adequate coverage have ranged from 10 to 75 per cent of the total marine area. At least five governing entities or initiatives overseas have adopted targets ranging from 20 to 30 per cent (Secretariat of the Convention on Biological Diversity, 2004:16).

The network of highly protected areas has the two key roles of (adapted from Secretariat of the Convention on Biological Diversity, 2004:15):

- *Providing areas in which natural processes are able to operate undisturbed.* This establishes baselines against which impacts of human activities in other areas can be assessed. It provides areas where marine research can be undertaken to increase our understanding of the marine environment. Such undisturbed areas

also provide opportunities for members of the public to learn about and enjoy the marine environment as well as providing recognition of the intrinsic value of marine ecosystems

- *Providing protection for representative examples of all biodiversity.* This helps ensure that management failures in other marine areas do not result in irreversible biodiversity loss

Highly protected areas can also produce several spin-off economic benefits including increasing fishery yields and tourist returns.

Dr Bill Ballantine, a New Zealand marine biologist and world-renowned proponent of marine reserves, has developed eight key principles for the establishment of highly protected areas (see Figure 9.2).

- The reserves are fully protected
- The reserves are permanent
- Each reserve aims for the ability to maintain itself
- Examples of all major habitats are included in reserves
- There are several spatially-separate examples of each habitat
- The reserves are spread throughout the region (a network)
- Public interest is actively encouraged
- More information about the marine life is actively sought and made available

Figure 9.2. Principles for marine reserves

Source: Ballantine, 2004

The term 'marine protected areas' is often used to refer both to highly protected areas, such as New Zealand's marine reserves, as well as to areas where some activities are permitted but where there are additional controls, such as where there are restrictions on trawling or commercial fishing.

In this publication, two separate categories have been used (highly protected areas and special management areas), on the basis that they perform different functions in the overall oceans governance system and are therefore not interchangeable. As already indicated, highly protected areas are kept undisturbed (to the extent possible) by human activities, and as well as providing important areas for marine research and public education, they act as a biodiversity insurance policy against management failures elsewhere. Special management areas (less highly protected marine areas), on the other hand, typically allow a range of human activities but in the context of more dedicated management and stricter controls. They are essentially a mechanism through which sustainable management practices, rather than total protection, are applied to an identified spatial area which has special characteristics.

Functioning of governance system

In order for an oceans governance system to function effectively, there needs to be connectivity between the individual components so that they create a coherent whole, not simply a collection of individual marine protected areas and regulatory controls (Secretariat of the Convention on Biological Diversity, 2004: 20). As discussed in Chapter 4, it is also important that the system is adaptive, so that it can respond in a timely manner to new information and changing situations, and that it is supported by a strong constituency which endorses the work of oceans policy makers and managers.

‘Snapshot’ of New Zealand’s oceans governance system

The key elements of New Zealand’s oceans governance structure are shown in Figure 9.3. This ‘snapshot’ identifies a myriad of mechanisms, implemented by various agencies over different spatial areas, which contribute to oceans governance. The extent to which they jointly comprise a cohesive and effective oceans governance system is explored in the chapters following.

Elements of oceans governance framework	Mechanism	Empowering legislation	Area	Managing body
<i>Sustainable management practices</i>	Sustainability mechanisms	Fisheries Act 1996	Territorial sea and EEZ	MFish
	Regional plans and resource consenting	Resource Management Act 1991	Territorial sea and land	Regional councils
	District plans and resource consenting	Resource Management Act 1991	Land	Territorial authorities
	Marine protection rules	Maritime Transport Act 1994	Territorial sea and EEZ	Maritime Safety Authority
	Licensing	Continental Shelf Act 1964	Continental shelf	Ministry of Energy
	Protected species	Wildlife Act 1952 Marine Mammals Protection Act 1978	Territorial sea and EEZ	DoC
	Import Health Standard	Biosecurity Act 1993	Territorial sea and EEZ	Biosecurity NZ
<i>Special marine management areas</i>	Marine mammals sanctuaries	Marine Mammals Protection Act 1978	Territorial sea and EEZ	DoC
	Areas closed to fisheries	Fisheries Act 1996	Territorial sea and EEZ	MFish
	Mataitai reserves	Fisheries Act 1996	Traditional fishing grounds	Tangata tiaka (Māori Committee)
	Taiapure-local fisheries	Fisheries Act 1996	Estuarine or littoral coastal waters	Management Committee (nominated by local Māori community)

Elements of oceans governance framework	Mechanism	Empowering legislation	Area	Managing body
<i>Special marine management areas (continued)</i>	Foreshore and seabed reserves	Foreshore and Seabed Act 2004	Foreshore and territorial sea	Foreshore and seabed reserve board
	Marine parks	Harbours Act and fisheries regulations	Areas prescribed by regulation	Regional councils and MFish
	Special legislation	Sugar Loaf Islands Marine Protected Area Act 1991	Area prescribed by legislation	DoC
		Hauraki Gulf Marine Park Act 2000	Area prescribed by legislation	Hauraki Gulf Forum (integrative body only)
		Fiordland (Te Moana o Atawhenua) Marine Management Act 2005	Area prescribed by legislation	Fiordland Marine Guardians (integrative and advisory body only)
	Areas of conservation value	Resource Management Act 1991	Territorial sea	Regional councils
	Areas to be avoided by shipping	Maritime Transport Act 1994	Territorial sea and EEZ	Maritime Safety Authority
	Protected areas	Submarine Cables and Pipelines Protection Act 1996	Territorial sea and EEZ	Ministry of Transport
<i>Network of highly protected areas</i>	Marine reserves	Marine Reserves Act 1971	Territorial Sea	DoC

Figure 9.3. Key elements of New Zealand's oceans environmental governance system

Sustainable Management Practices

There is a wide range of mechanisms which provide for broader sustainability within New Zealand's marine area. These include sustainability measures applied to fishing activities, sustainable management mechanisms under the Resource Management Act 1991, the establishment of marine protection rules to address marine-based pollution, the licensing of petrochemical and minerals exploitation on the continental shelf, protecting individual marine species, and managing biosecurity under the Biosecurity Act 1993. These mechanisms are discussed below within the governance framework promoting sustainable management practices set out in Chapter 9.

Fisheries management regime that addresses the impacts of fishing activity on marine ecosystems

The sustainability measures which are applied to the management of fisheries under the Fisheries Act 1996 include setting a total allowable catch for each commercially exploited stock, restricting the by-catch of certain protected species and restricting fishing methods and fishing seasons.

Catches by recreational fishers are controlled by MFish through regulations on the catch number, the size of fish which can be caught, and the method, time and place of fishing. Controls on aggregate catches by recreational fishers are only indirect and estimates of the total recreational catch typically have high levels of uncertainty. There has been an ongoing initiative to better define and manage recreational fishing activity, because of impacts on stock depletion, as well as ongoing conflicts between recreational and commercial fishers targeting the same stocks.

Catches by customary fishers are largely controlled by the Kaimoana Customary Fishing Regulations 1998 and the Fisheries (South Island Customary Fishing) Regulations 1998. These regulations establish a system where iwi, hapū or whānau manage customary fishing in traditional tribal fisheries. Authorised kaitiaki (caretakers) permit customary non-commercial fishing within their areas.

Measures implemented under the Fisheries Act 1996, to reduce the accidental capture of protected marine wildlife during fishing activities, include setting annual catch limits for protected species such as sea lions and dolphins.

The Ministry of Fisheries and DoC have recently prepared a National Plan of Action to Reduce the Incidental Catch of Seabirds in New Zealand. For fisheries with known interactions with seabirds, it is proposed to implement voluntary codes of practice as

well as mandatory input controls and by-catch limits for individual seabird species (MFish and DoC 2004a). As a result of substantial non-compliance with voluntary codes of practice in some sectors of the fishing industry, the Ministry of Fisheries has indicated an intention to introduce regulations to require the use of equipment to scare seabirds away from vessels and to control the discharge of offal (Benson-Pope 2005).

Voluntary measures adopted by the fishing industry have included (Seafood Industry Council, undated):

- The development of a code of practice by the hoki fishery to reduce levels of seal by-catch
- The development of sea lion excluder devices to reduce sea lion deaths in the sub-Antarctic squid fishery
- The development of a code of practice, and trialling of a device to deter Hector's dolphins from nets, by inshore gill net fishers in the South Island
- The development of a code of practice by trawl fishers to reduce the by-catch of other dolphins
- The adoption of codes of practice and mitigation devices in the tuna and ling fisheries to reduce seabird by-catch

Another significant voluntary initiative is the establishment of the Southern Seabird Solutions in 2003. This is an alliance between government, fishers and environmental groups to help develop and promote the adoption of seabird-safe fishing practices (Southern Seabird Solutions, undated).

For some years, MFish has been developing a Strategy for Managing the Environmental Effects of Fishing, which has been recently released (MFish, 2005). The core of the Strategy is the establishment of environmental standards which will specify the limits of acceptable effects of fishing on the marine environment. The development and implementation of fisheries plans, which are to be prepared by the Ministry, will be the main mechanism through which fishing activity is managed to ensure that the standards are met. The development of such plans will be informed by an environmental impact assessment, which is to be undertaken for each fishery in order to identify its effects on marine species and habitats.

Catchment management regime that effectively manages marine sedimentation and pollution from land-based sources

Sedimentation and pollution of the marine area from land-based sources is managed under the RMA. Under the Act, regional councils and territorial authorities develop and enforce regional policy statements and plans and district plans respectively, and process resource consents for individual proposals.

Management of the coastal environment at a regional and local level is carried out within the framework established by the New Zealand Coastal Policy Statement (NZCPS). The NZCPS, which came into force in 1994, was prepared by DoC and sets out a number of policies to be applied to the coastal environment. These relate to protecting the natural character of the coastal environment, maintaining and enhancing coastal water quality, protecting the characteristics of the coastal environment of special value to tangata whenua, and maintaining public access on the coast.

The jurisdiction of regional councils is based on water catchments and extends over land and the marine area to the edge of the territorial sea. Regional councils are, therefore, potentially able to apply an integrated approach to the management of the land and the sea. This can, firstly, be achieved through the development of regional policy statements which are designed to provide an overview of resource management issues throughout the region and to identify policies and methods to achieve the integrated management of natural and physical resources (section 59).

Secondly, regional plans deal with specific resource management issues within the region and usually contain rules which directly control activities. They can address issues such as soil erosion, marine water quality and the management of marine ecosystems. Regional councils are required to prepare a regional coastal plan which applies to activities below mean high-water springs. However, other regional plans are optional. Several regional councils have prepared regional coastal environment plans which extend over the land and water portions of the coast. They typically contain objectives and policies which apply to the whole area but rules which apply only to the coastal marine area, leaving district councils to develop rules to control land-based development within the guidance provided by the regional plan (Peart, 2005).

Territorial authorities are charged with controlling the 'effects' of the use, development or protection of land (section 31(b)), including effects within the land portion of the coastal environment but not in the marine area. District plans prepared by territorial authorities are where the 'rubber hits the road' in terms of managing coastal development. As well as containing objectives and policies, district plans usually include rules which control activities such as subdivision and the building of houses and structures. Such rules determine the location, density and design of development which is likely to be permitted within the catchment and on the coast. Some district plans identify special coastal zones where greater control is exercised over development in recognition of the sensitivity of the particular environment (Peart, 2005).

The policy statements and plans prepared under the RMA have been of generally poor quality, particularly with respect to establishing a robust facts base on which to base the subsequent planning approaches (Ericksen et al., 2003:120 & 138). Because of the weakness of plans and of the types of planning approaches which have commonly been adopted, environmental management under the RMA has also been particularly weak in addressing the cumulative effects of activities. Instead, the focus has been on

assessing and managing the impacts of specific activities on an ad hoc basis (Peart, 2004:23–26).

Managing cumulative environmental impacts in the marine area is particularly complex because regional councils, which have management jurisdiction over the marine area, have little control over the actions of territorial authorities which authorise land use activities. Conversely, territorial authorities have no jurisdiction over the marine area and therefore less vested interest in its management even though their actions can have a major impact on these areas. Unitary councils, which undertake the functions of both a regional council and a territorial authority, overcome this split and five of these operate within New Zealand compared to 69 territorial authorities and 12 regional councils.

Some of today's problems have resulted from past infrastructure investment decisions and will be very expensive to remedy. For example, in the Auckland region it is estimated that between \$4 and \$11 billion needs to be spent over 20 years to improve the quality of beaches and waterways – much more than the \$1.9 billion which councils had planned to spend on such works over the same period (The Boston Consulting Group 2004). Other problems are integrally related to the form that urban development takes on the coast. For example, the high levels of pollution of some sediments in the Auckland region are largely related to growth in motor vehicle usage, which is in turn related to the dispersed nature of settlement in the region and the lack of an adequate public transport system.

There are some notable success stories, however. For example, water quality in Whangarei Harbour has improved in recent years because of stronger controls on sedimentation and other discharges into the marine area (Thompson, 2005:A13).

Environmental impact regime that ensures that structures and activities which are established in the marine area do not disrupt marine ecosystems

Within the territorial sea, structures and activities within the marine area are managed by regional councils. Regional plans can address the environmental impacts of such activities through establishing activity status and assessment criteria which apply to specific activities. All applications for resource consents under the RMA are required to include an assessment of effects on the environment (section 88(2)) and such effects are considered when determining resource consent applications (section 104(1)).

There are few explicit controls on the environmental impacts of activities outside the 12 nautical mile limit of the territorial sea. Under the Continental Shelf Act 1964, licences issued by the Minister of Energy are required in order to mine for petroleum and minerals on the continental shelf. In the case of petroleum, applications are dealt

with through procedures under the Crown Minerals Act 1991. Neither Act has any explicit environmental provisions.

The Territorial Sea, Contiguous Zone and Exclusive Economic Zone Act 1977 has a catch-all provision which enables regulations to be promulgated, 'where no other provision is for the time being made by any other enactment for any such purposes ...' (section 27) for a range of purposes including 'prescribing measures for the protection and preservation of the marine environment of the zone'. No such regulations have been made, but presumably could apply in cases such as proposals for offshore aquaculture or wind farms.

In the absence of environmental regulatory controls, voluntary initiatives are being developed. For example, OMV New Zealand Limited has voluntarily agreed to undertake a full environmental effects assessment of its proposal to extract oil from the offshore Maari field and to involve key stakeholders in the process (MfE, 2005b:23). The Ministry for the Environment is currently championing an initiative, involving central government agencies and the petroleum exploitation industry, to develop a voluntary accord on the management of environmental impacts outside territorial waters.

In respect of impacts on marine species, the Wildlife Act 1952 identifies protected species, and it is an offence to hunt or kill such species (section 63A). Marine species protected under the Act include black coral (all species in the order of *Antipatharia*), red coral (all species), the spotted black groper and marine reptiles such as turtles and sea snakes.

All species of seabirds are protected under the Wildlife Act except for the southern black-backed gull. Sooty shearwaters and grey-faced petrels may be killed subject to conditions set by the Minister of Conservation and their chicks can be legally harvested by iwi and manawhenua (land trustees) at identified sites (Taylor, 2000:18).

Marine mammals are managed by DoC under the Marine Mammals Protection Act 1978 and a permit is required to hold a marine mammal in captivity or to remove one from its natural habitat (section 4).

Marine pollution management regime that effectively manages the discharge of pollutants from ships and marine structures

The Maritime Safety Authority, under the Maritime Transport Act 1994, has developed Marine Protection Rules which govern:

- Marine pollution by ships
- Oil pollution and discharges from offshore installations engaged in seabed exploration, exploitation or processing of minerals throughout New Zealand's entire marine area including the territorial sea and EEZ

- The discharge of garbage and the dumping of waste or other matter at sea

The Authority is in the process of developing an accord with offshore operators on the management of environmental effects.

Effective bio-invasion prevention and response capability

The transport of invasive species by vessels is primarily controlled by Biosecurity New Zealand, a division of the Ministry of Agriculture and Forestry, under the Biosecurity Act 1993. Under this Act an Import Health Standard has been established, which requires all ships to exchange their ballast water with mid-ocean water. This was one of the first initiatives in the world to control the discharge of ballast water in national seas (Hewitt et al., 2004:432). The International Convention for the Control and Management of Ships' Ballast Water and Sediments, adopted in February 2004 but yet to come into force, will provide internationally consistent controls.

Ships and recreational yachts coming into New Zealand waters are not currently inspected to establish the extent of hull fouling. Biosecurity New Zealand is currently undertaking a research programme focused on investigating the impact of hull fouling on marine biosecurity risks. It also operates an educational programme targeted at boat owners to reduce hull-fouling risk and is developing standards for facilities to treat fouled vessels (Hewitt et al., 2004:432).

The following measures are in place in order to identify and respond to a marine bioinvasion if and when it occurs (Hewitt et al., 2004:433–434):

- Baseline surveys in New Zealand's highest-risk ports
- Targeted surveillance for high-risk exotic species in areas most at risk of invasion
- An incursion response protocol
- A vector management programme to reduce the spread of undaria

The management of existing pests can help strengthen the effort to control new organisms, through educating people about the risks posed by exotic organisms, establishing the infrastructure required to manage marine biosecurity issues at a regional and industry level, and helping characterize vector movements to sensitive locations (Cassidy, 2005).

Evaluation

Insufficient information is currently available to accurately assess the extent to which the management practices described above are sustainable in terms of ensuring that

human activities are not significantly disrupting marine ecosystems. However, at least four weaknesses are evident in marine management outside special management and highly protected areas:

- Management of fisheries activity has yet to seriously grapple with understanding and managing environmental impacts and, in particular, addressing the impacts of trawling and dredging on benthic habitats. This is of particular concern because, as noted in Chapter 6, a large proportion of New Zealand's benthic marine species are endemic and they are also the species about which least is known and of which the smallest percentage has been discovered or described. The development of the Strategy for Managing the Environmental Effects of Fishing will hopefully help focus efforts on this key issue.
- The management of land-based activities under the RMA has been weak in addressing the high levels of sedimentation and pollution entering marine areas. Such high levels can significantly impede the important ecosystem functions these areas perform, including providing nursery grounds for commercially important fish species. Dedicated efforts are urgently required to get on top of this problem before the problem rapidly escalates, as the numbers of people living on and visiting the coast, and the consequent development, continues to increase.
- For areas outside territorial waters, there is no consistent environmental regime with which to manage the environmental effects of activities such as the exploitation of petrochemicals and minerals, and the establishment of offshore wind and marine farms.
- Mechanisms currently in place to control the arrival of invasive marine species on the fouling of vessels' hulls appear weak. This is of concern because, as indicated in Chapter 8, it is estimated that about 70 per cent of exotic marine species accidentally introduced into New Zealand arrived on vessels' hulls.

Special Marine Management Areas

There are many types of special marine management areas that can be created under statute in New Zealand. These include marine mammal sanctuaries, areas closed to fisheries, mataitai reserves, taiapure-local fisheries, foreshore and seabed reserves, marine parks, areas established under special legislation, areas of conservation value, areas to be avoided by shipping, and cable exclusion areas. These are discussed in more detail below.

Marine mammal sanctuaries

Marine mammal sanctuaries can be established under the Marine Mammals Protection Act 1978 and are managed by DoC. These can be established only for the purposes of protecting, conserving and managing marine mammals. Only two sanctuaries have been established under the legislation, the first over an area generally extending seawards 4 nautical miles from Banks Peninsula. Controls on set netting apply to this area in order to protect Hector's dolphins. In 1993, a second sanctuary was established around the Auckland Islands to protect the New Zealand sea lion, and 10 years later the area was declared a marine reserve.

Areas closed to fishing activity

Areas can be closed to specified types of fishing activity by the Minister of Fisheries, under the sustainability provisions of the Fisheries Act 1996, after taking into account the effects of fishing on any stock and the aquatic environment. Before making a decision to close an area, the Minister must consult with stakeholders (section 12) and take into account any conservation or fisheries services and any relevant fisheries plan (section 11(2A)).

Numerous area-based restrictions have been placed on fishing activities, particularly in harbours and coastal areas, but with few outside the territorial sea (see Froude and Smith, 2004). Of specific significance to biodiversity protection are:

- The exclusion of trawling, Danish seine netting and scalloping in an area off Spirits Bay and Tom Bowling Bay at North Cape to protect a highly biologically diverse area of bryozoans

- The exclusion of trawling, Danish seine netting and dredging off Separation Point in Tasman Bay to protect an area of bryozoans
- The exclusion of trawling from 19 seamounts, covering an area of about 100,000 square kilometres, to protect representative seamount fauna from a variety of habitat types (Clark and O'Driscoll, 2003:452–453)

In addition to these statutory closures, members of the fishing industry have from time to time instituted a range of voluntary closures of areas to fishing activity. Examples include (Gibbs, 2004):

- The closure of some areas to trawling and others to purse seining in Tasman Bay, Golden Bay and the Marlborough Sounds to protect juvenile snapper and kahawai
- The closure of areas of the west coast of the North Island to set netting to protect the Maui's dolphin, prior to regulatory closures being imposed
- Closure of an area at Separation Point to protect bryozoan beds
- Closure of areas of the east coast of the South Island to protect elephant fish nursery grounds and the Hector's dolphin
- Closure of areas on the west coast of the South Island to improve access to paua by recreational fishers

Mataitai reserves

Mataitai reserves can be declared under the Kaimoana Customary Fishing Regulations 1998 on application by tangata whenua. These are areas which are recognised as being traditional fishing grounds and with which tangata whenua have a special relationship. Commercial fishing is generally prohibited in these areas and tangata whenua manage non-commercial fishing activity. Only two mataitai reserves have been declared, the first at Rapaki Bay in Lyttelton Harbour in 1999 and the second at Koukourarata, Banks Peninsula in 2000. The creation of a third in Paterson Inlet, Stewart Island, was announced in June 2004. There are currently a further eight outstanding applications.

Taiapure-local fisheries

Taiapure-local fisheries may be established under the Fisheries Act 1996 in an estuarine or littoral coastal area that is traditionally important to iwi or hapū. These are areas which are managed by a management committee representative of the local Māori community. Commercial fishing may occur in a taiapure and the committee may recommend to the Minister regulations to manage the area. Seven taiapure-local fishery areas, covering 328 square kilometres of ocean, have been established as follows:

- Whakapuaka Delaware Bay (25 square kilometres)
- Palliser Bay (3 square kilometres)
- Porangahau (67 square kilometres)
- Maketu (55 square kilometres)
- Kawhia Aotea (137 square kilometres)
- Waikare Inlet (18 square kilometres)
- East Otago (23 square kilometres)

There are currently three further outstanding applications.

Foreshore and seabed reserves

These reserves may be established under the recently enacted Foreshore and Seabed Act 2004 to acknowledge the exercise of kaitiakitanga (guardianship) over a specified foreshore and seabed area (section 40). The area is managed by a board comprised of members proposed by the group that holds territorial customary rights over the reserve. The board has the function of preparing a management plan for the reserve. Because the Act has only recently been passed by Parliament no foreshore and seabed reserves have yet been established.

Marine parks

Two marine parks were established under the now repealed Harbours Act 1950 and regulations under the Fisheries Act 1996. These are the Tawharanui Marine Park, which covers 3.5 square kilometres and was established in 1981, and the Mimiwhangata Marine Park, which covers 20 square kilometres and was established in 1983. The fisheries regulations which apply to the parks prohibit the taking of all marine life in the Tawharanui Marine Park and exclude all commercial and some recreational fishing within the Mimiwhangata Marine Park.

The marine park concept as applied in these areas has not been successful. Research has indicated that fish stocks have not recovered from depletion during the 1970s, probably owing to illegal fishing at Tawharanui and increasing recreational fishing pressure at Mimiwhangata. Proposals are currently being developed to convert both areas to marine reserves.

Areas protected by special legislation

Special legislation, the Sugar Loaf Islands Marine Protected Area Act 1991, was passed to establish the Sugar Loaf Islands Marine Protected Area. This covers an area of 8

square kilometres and is managed by DoC. Mining is prohibited in the Protected Area but there is no restriction on fishing activities.

The Hauraki Gulf Marine Park Act 2000 (HGMPA) established a Hauraki Gulf Marine Park, which primarily consists of reserve land, foreshore and seabed owned by the Crown and sea water (but not marine life). It provides a set of management objectives for the Gulf and adjacent catchments. These seek to promote the protection and enhancement of the life-supporting capacity of the environment, of the Gulf's natural, historic and physical resources and people's relationship to and enjoyment of these resources. They must be given 'effect to' by all regional policy statements, regional plans and district plans which apply within the HGMPA area. In addition, all agencies undertaking statutory functions within the Gulf must have particular regard to the management objectives (section 13).

As well as setting out management objectives, the HGMPA established a Hauraki Gulf Forum for the purposes of integrating management of the Gulf and facilitating coordination. Members of the Forum include representatives of the Ministers of Conservation, Fisheries and Māori Affairs; 12 local authorities; six representatives from the tangata whenua of the Hauraki Gulf and its islands, appointed by the Minister of Conservation; and two further representatives appointed by the Auckland Regional Council. Functions of the Forum include:

- Integrating the management of the Gulf
- Facilitating communication, cooperation and coordination of members' statutory functions exercised within the Gulf
- Preparing a list of strategic issues
- Determining priorities for action on each issue
- Publishing a report on the state of the environment every three years

Day-to-day management of the Gulf continues to be carried out by individual statutory agencies.

The Fiordland (Te Moana o Atawhenua) Marine Management Act 2005 establishes a Fiordland (Te Moana o Atawhenua) Marine Area and establishes a body called the Fiordland Marine Guardians which has an advisory and facilitative role, amongst other things. Members of the Guardians are appointed by the Minister for the Environment with a requirement that one member be nominated by Te Rūnanga o Ngāi Tahu, that at least five members are ordinarily resident in the Otago or Southland regions and that the membership reflects a balanced mix of knowledge and experience in relation to the Marine Area (section 15). Unlike the legislation applying to the Hauraki Gulf, this Act does not provide any common management objectives for the Fiordland Marine Area.

Areas of significant conservation value

Areas of significant conservation value are sometimes identified in regional coastal plans under the Resource Management Act 1991 and measures to ensure that these areas are not adversely affected are specified. These can include avoiding activities such as aquaculture, marinas and moorings, reclamations and building structures. The areas identified as having conservation value are usually small and located adjacent to the coast. There is some uncertainty as to how far regional councils can restrict fishing methods within these areas through their plans.

Areas to be avoided by shipping

Areas to be avoided by shipping can be established under the Maritime Transport Act 1994 and through the International Maritime Organisation. There are three categories of such areas: recommended areas to be avoided, mandatory areas to be avoided and particularly sensitive areas. New Zealand has a recommended area to be avoided around the Three Kings Islands and the world's first mandatory area to be avoided around the Poor Knights Islands.

Cable exclusion areas

Fishing activities and anchoring are excluded from cable areas under the Submarine Cables and Pipelines Protection Act 1996. This exclusion is not for environmental purposes, but it may have the indirect effect of restricting activities that could impact on biodiversity.

Marine Protected Areas Policy Statement and Implementation Plan

The Department of Conservation and MFish have been developing a Marine Protected Areas Policy Statement and Implementation Plan. This arose out of the New Zealand Biodiversity Strategy, which included an objective to (DoC 2000a: 67):

protect a full range of natural marine habitats and ecosystems to effectively conserve marine biodiversity, using a range of appropriate mechanisms, including legal protection.

Actions identified in the Strategy to achieve this objective included 'developing and implementing a strategy for establishing a network of areas that protect marine biodiversity ...'.

For several years, work has been under-way on developing the Policy Statement and Implementation Plan which has not yet been publicly released. It is expected to define a marine protected area as:

an area of the marine environment especially dedicated to, or achieving, through adequate protection, the maintenance and/or recovery of biological diversity at the habitat and ecosystem level in a healthy functioning state.

The definition makes no explicit distinction between the functions performed by highly protected areas (marine reserves) and other less fully protected areas.

The final policy statement and plan is expected to include a series of network design principles, which will guide the development of a representative network of marine protected areas, and a series of planning principles, which will guide the planning process to establish new protected areas. The network design principles are likely to include the establishment of national priorities for marine protected area planning, which will be set for a five-year period and reviewed annually.

The implementation of the policy is planned around a four stage framework consisting of preparation for implementation, strategic analysis, development of a marine protected areas network, and evaluation and monitoring.

The preparation for implementation is likely to include developing a nationally consistent approach to classification, which is likely to draw on work that has been undertaken by DoC and the Ministry for the Environment (MfE). The Department of Conservation system, called Nearshore Marine Classification and Inventory (NMCI), relates only to coastal areas and brings together all available published material and survey information on specific areas. The MfE system, termed Marine Environments Classification, covers the entire EEZ and is based on mainly physical data (Snelder et al., 2004).

A strategic analysis in stage two of the planned implementation is likely to involve the development of an inventory of current marine protected areas, the identification of current gaps in the network and the prioritisation of habitats and ecosystems for new protected areas.

The development of the network is likely to involve applying an integrated regional approach in nearshore areas. This may involve the establishment of Marine Protection Planning Forums to identify sites and potential tools for protection and to facilitate consensus on areas to be set aside. In offshore areas it is likely that Expert Offshore Panels, which have specific expertise and include representation of offshore interests, will be convened.

A proposed evaluation and monitoring programme will assess progress against priorities. This is likely to include the performance of the MPA network as a whole as well as the effectiveness of individual marine protected areas.

Evaluation

There are at least 10 different mechanisms providing, either directly or indirectly, some special management for marine areas. These include measures taken under nine pieces of legislation:

- Marine Mammals Protection Act 1978
- Sugar Loaf Islands Marine Protected Area Act 1991
- Resource Management Act 1991
- Maritime Transport Act 1994
- Submarine Cables and Pipelines Protection Act 1996
- Fisheries Act 1996
- Hauraki Gulf Marine Park Act 2000
- Foreshore and Seabed Act 2004
- Fiordland (Te Moana o Atawhenua) Marine Management Act 2005

The areas are administered by 10 different management bodies:

- Department of Conservation
- Ministry of Fisheries
- Ministry of Transport
- Maritime Safety Authority
- Regional councils
- Tangata tiaka
- Taiapure management committees
- Foreshore and seabed reserve boards
- Hauraki Gulf Forum
- Fiordland Marine Guardians

The legislative framework has only sought to apply an integrated management approach within two of the areas – the Hauraki Gulf Marine Park and the Fiordland (Te Moana o Atawhenua) Marine Area.

Until an adequate analysis is undertaken of the location of these special management areas, and their contribution to the management of the oceans, it is difficult to comment in a meaningful way on their adequacy. Such an analysis will hopefully be produced from the work to be undertaken under the Marine Protected Areas Implementation Plan. Some initial observations can, however, be made on current information:

- There appears to be little coordination between the management bodies in designating and managing these areas, and the overall system is highly

fragmented. The implementation of the Marine Protected Areas Policy Statement and Implementation Plan should help establish common priorities and a common information base for the establishment of a more cohesive network of marine protected areas. However, it will not serve to rationalise the plethora of legislation in this area. Ultimately each managing authority must operate within the jurisdiction of its own governing legislation, which has its own purposes and processes, irrespective of whether these are consistent with non-statutory government policy documents

- Although there are numerous areas closed to certain types of fishing activity within harbours and along some areas of the coast, there is little protection for the rest of the marine area, particularly that outside the territorial sea, with the notable exception of the recent closure to trawling of 19 seamounts. Hopefully the implementation of the Marine Protected Areas Policy Statement and Implementation Plan will help prioritise further protection of offshore areas.

Network of Highly Protected Marine Areas

Legislative provisions

Highly protected areas, called marine reserves, can be established in New Zealand under the Marine Reserves Act 1971. At present, these can be established only within the territorial sea, although proposed amendments to the Act, contained in a Marine Reserves Bill currently before Parliament, would extend its jurisdiction to include the EEZ.

The New Zealand Biodiversity Strategy contains a commitment to protecting 10 per cent of New Zealand's marine environment (which includes the territorial sea and EEZ) by 2010. However, this figure includes less rigorously protected marine areas as well as marine reserves and fails to distinguish between the different functions of these two types of areas.

The purpose of establishing marine reserves under the Marine Reserves Act focuses on scientific study rather than preserving biodiversity. The long title of the Act states: 'An Act to provide for the setting up and management of areas of the sea and foreshore as marine reserves for the purpose of preserving them in their natural state as the habitat of marine life for scientific study.'

The Marine Reserves Bill proposes to change this purpose to more explicitly refer to biodiversity considerations, as follows (clause 7):

The purpose of this Act is to conserve indigenous marine biodiversity in New Zealand's foreshore, internal waters, territorial sea and exclusive economic zone for current and future generations, by preserving and protecting within marine reserves –

- (a) representative examples of the full range of marine communities and ecosystems that are common or widespread; and*
- (b) outstanding, rare, distinctive, or internationally or nationally important marine communities and ecosystems; and*
- (c) natural features that are part of biological and physical processes of the marine communities and ecosystems referred to in paragraphs (a) and (b), in particular those nature features that are outstanding, rare, unique, beautiful or important.*

Marine reserves are managed by DoC with the aim of maintaining them as far as possible in their natural state. Although non-commercial fishing can be permitted in a marine reserve, it is only currently permitted within a small part of the Kapiti Marine Reserve and is restricted to whitebaiting (DoC, 2000b:26). Proposed amendments to the Marine Reserves Act in the Bill would prohibit any fishing in marine reserves. Mining activity is not necessarily excluded from marine reserves although it is not currently undertaken in any of them.

An application to establish a marine reserve can currently be made only by a limited category of entities including a university, a body administering a reserve adjacent to the coast, an organisation involved in the study of marine life and natural history, iwi or hapū with tangata whenua status over the area and the Director-General of Conservation (section 5(a)). The Bill proposes to amend these provisions to provide that any person can prepare a proposal for a marine reserve to be established (clause 47).

Currently, marine reserves must be approved by the Minister of Conservation, with the concurrence of the Ministers of Fisheries and Transport, before they can be established through Order in Council (section 5(9)). A Marine Reserves Protocol has been developed to guide the interactions between DoC and MFish when considering marine reserve applications (DoC and MFish 2003b). The Bill proposes to remove the concurrence roles of the Ministers of Fisheries and Transport, with the Minister of Conservation making the final decision on a marine reserve application (section 67).

A marine reserve may not be established if it would (section 5(8)):

- Unduly interfere with any interest in land in or adjoining the reserve
- Interfere unduly with any existing right of navigation
- Interfere unduly with commercial fishing
- Interfere unduly with or adversely affect any existing usage of the area for recreational purposes
- Otherwise be contrary to the public interest

Current marine reserves

There are currently 28 marine reserves in New Zealand (see Figure 12.1) covering 12,595 square kilometres, being around 7 per cent of New Zealand's territorial sea but only 0.3 per cent of the entire marine area under New Zealand's jurisdiction.

The first marine reserve, at Cape Rodney–Okakari Point, was established 30 years ago and was hailed as the first no-take reserve in the world. The number of marine reserves being established has increased significantly in the past 15 years. Although only one was established during the 1970s and one during the 1980s, 14 were established during the 1990s and 12 in the first half of the 2000s, including eight (totalling 94.3

square kilometres) recently established in Fiordland, under the Fiordland (Te Moana o Atawhenua) Marine Management Act 2005.

Name	Year established	Size (km ²)	Proposer
Cape Rodney – Okakari Point	1975	5.2	Auckland University
Poor Knights Islands	1981	24.0	Hauraki Gulf Maritime Park Board
Kermadec Islands	1990	7480.0	Department of Lands and Survey
Kapiti	1992	21.7	DoC
Tuhua – Mayor Island	1992	10.6	DoC and Ministry of Agriculture and Fisheries (proposed by Māori trustees of Tuhua)
Whanganui A Hei – Cathedral Cove	1992	8.4	DoC
Long Island – Kokomohua	1993	6.2	Marlborough Combined Underwater Clubs
Piopiotaahi (Milford Sound)	1993	6.9	New Zealand Federation of Commercial Fisherman
Te Awaatu (the Gut)	1993	0.9	New Zealand Federation of Commercial Fisherman
Tonga Island	1993	18.4	DoC
Westhaven – Te Tai Tapu	1994	5.4	DoC
Long Bay – Okura	1995	9.8	East Coast Bays Protection Society
Motu Manawa – Pollen Island	1995	5.0	Royal Forest and Bird Protection Society
Te Angiangi	1997	4.5	DoC
Pohatu	1999	2.2	Akaroa Harbour Recreational Fishing Club and Canterbury Marine Recreational Fishers Association

Name	Year established	Size (km ²)	Proposer
Te Tapuwae O Rongokako	1999	24.5	Ngati Konohi and DoC
Auckland Islands	2003	4840.0	DoC
Te Matuku Bay	2005	6.9	Royal Forest and Bird Protection Society
Paterson Inlet	Approved in 2004	10.1	Paterson Inlet Marine Protection Committee
Horoirangi (Glenduan – Atoata Point)	Approved in 2005	9.5	Royal Forest and Bird Protection Society
Te Hapua (Sutherland Sound)	2005 (FMMA)	4.5	Guardians of Fiordland
Hawea (Clio Rocks)	2005 (FMMA)	4.1	Guardians of Fiordland
Kahukura (Gold Arm)	2005 (FMMA)	4.6	Guardians of Fiordland
Kutu Parera (Gaer Arm)	2005 (FMMA)	4.3	Guardians of Fiordland
Taipari Roa (Elizabeth Island)	2005 (FMMA)	6.1	Guardians of Fiordland
Moana Uta (Wet Jacket Arm)	2005 (FMMA)	20.1	Guardians of Fiordland
Taumoana (Five Fingers Peninsula)	2005 (FMMA)	14.7	Guardians of Fiordland
Te Tapuwae o Hua (Long Sound)	2005 (FMMA)	36.7	Guardians of Fiordland
TOTAL		12,595.3	

Figure 12.1. Marine reserves in New Zealand

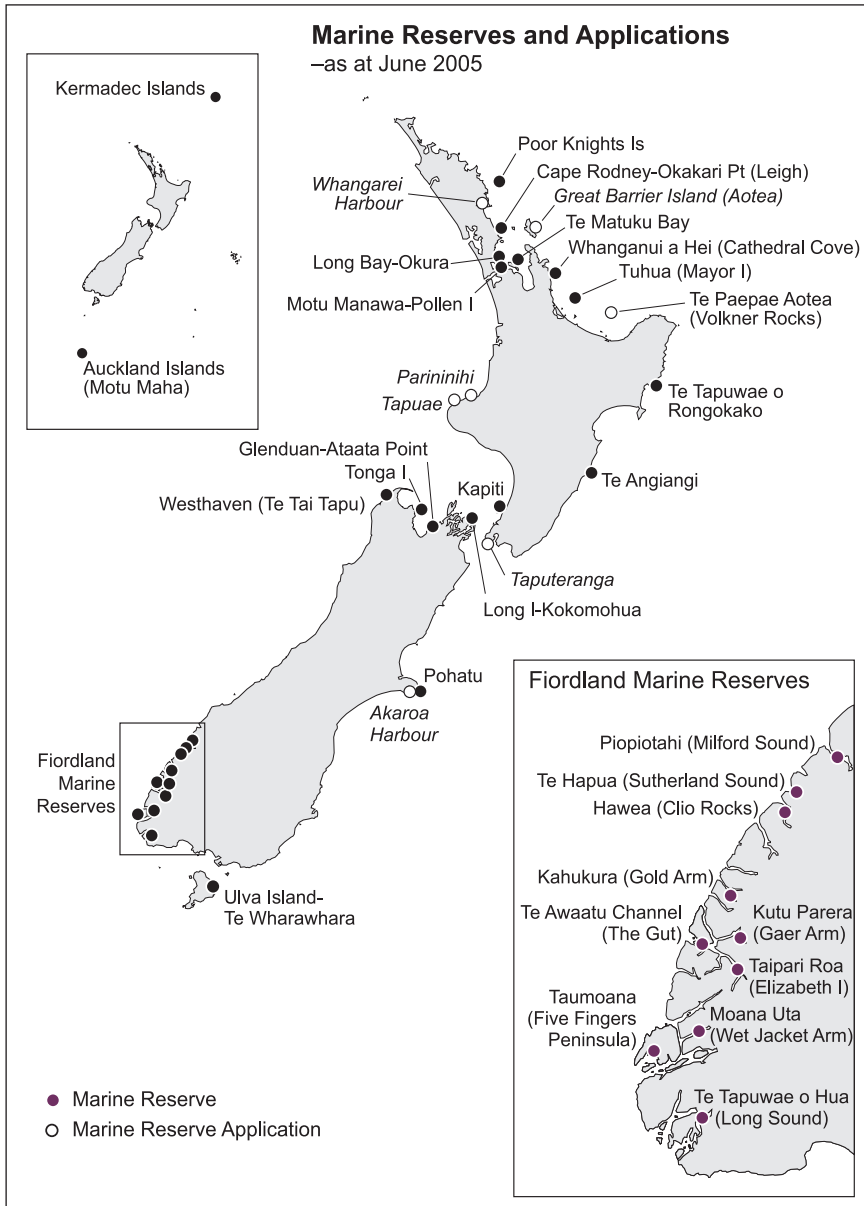


Figure 12.2. Location of marine reserves and applications

Provided courtesy of Chris Edkins, Department of Conservation

These marine reserves are concentrated around parts of the northern east coast of the North Island, the northern end of the South Island and Fiordland. Apart from two large reserves, comprising 98 per cent of the total marine reserve area and covering offshore island groups, most of the reserves extend over small marine areas (see Figure 12.2).

The small size of many existing marine reserves is illustrated by the fact that, although there are five marine reserves in the Hauraki Gulf, they protect less than 0.3 per cent of the Gulf's marine area (Hauraki Gulf Forum, 2005:118).

Establishment of marine reserves

The identification of potential marine reserve sites has been, to date, a largely ad hoc affair with applications being made by a large number of different parties including a university, environmental groups, fishing groups, community organisations and government departments, often in response to local concerns about marine exploitation.

The process to establish marine reserves can be long and conflictual. Strong opposition can be mounted by the potential short-term 'losers' as a result of areas being removed from fishing grounds. This is because, although there is strong evidence that the closure of areas to fishing leads to increases in abundance, body size, biomass and reproductive output of exploited species within reserves, there is less evidence to demonstrate that reserves enhance fisheries elsewhere except on their boundaries (Gell and Roberts, 2003:6). This does not mean that such effects do not occur, just that they are very difficult to prove, largely because the generally small area of reserves relative to large surrounding marine ecosystems makes it difficult to isolate the effects from the reserve.

There is some evidence that the views of fishers can change in favour of reserves once they have been established and that they can benefit from spill-over effects. A case study of the Cape Rodney–Okakari Point, Tonga Island and Pohatu Reserves, found that commercial fishers in these areas now actively support the reserves. Forty per cent of fishers at Leigh, adjacent to the Cape Rodney–Okakari Marine Reserve, considered catches to be higher outside the reserve than before its establishment. The most heavily fished area is on the boundaries of the reserve, probably owing to an increase in the size and abundance of fish (Taylor and Buckenham, 2003:31).

Fishers have also been applicants for marine reserves in Fiordland and Akaroa. In at least one case this was to head off a more substantive proposal in a similar area. The Pohatu Marine Reserve was proposed by local fishers to head off a proposal by the Akaroa Harbour Marine Protection Society for a reserve within Akaroa Harbour (Taylor and Buckenham, 2003:23).



Figure 12.3. There can be strong local opposition to marine reserve proposals – signs displayed at Nugget Point, the Catlins.

Hapū and iwi can be opposed to the marine reserve model in principle on the grounds that it prohibits their access to the natural resources and sites within their rohe moana (territorial waters). Absolute protection is not a feature of Māori resource management.

In some areas, however, iwi and hapū have been closely associated with proposals for marine reserves (Taylor and Buckenham, 2003:15).

Opposition to the establishment of marine reserves can be exacerbated by uncertainty as to their likely ultimate extent in any given area. Because of the ad hoc nature of the process to identify candidate areas for marine reserves, stakeholders have no certainty that the proposal they are being asked to consider is the full extent of protection sought rather than the 'thin end of the wedge'. In addition, the fragmented system for identifying marine areas for special protection means that it is difficult for stakeholders to negotiate offsetting benefits. For example, local iwi may be willing to agree to a marine reserve in their rohe moana if they are also able to obtain assurances about the establishment of other types of protected areas in an adjacent area, such as a mataitai reserve or taipure-local fishery, which will enable them to exercise kaitiakitanga (guardianship) over marine resources.

Opposition may be exacerbated through public misperceptions about the existing extent of marine protection and may not accurately reflect broader public opinion. A recent survey undertaken by WWF-New Zealand of the attitudes of 1001 New Zealanders to marine reserves, found that 90 per cent of those surveyed overestimated how much of New Zealand's oceans was already protected. The average estimate of the area protected given by respondents was just over 23 per cent of New Zealand's marine environment (WWF-New Zealand, 2005:7). This can be compared to the 7 per cent of the country's territorial sea and less than 1 per cent of the EEZ which is fully protected. Survey respondents on average indicated that they considered that 36 per cent of New Zealand's marine environment should be protected in marine reserves (WWF-New Zealand, 2005:9).

The high level of opposition to some marine reserve proposals means that it can take considerable time for applications to get through the process. Figure 12.4 indicates that the processing time has considerably shortened in recent years for some applications. The longest time taken for a proposal to be processed was 18 years, and not surprisingly this was the first one at Cape Rodney-Okakari Point when the concept was very new.

The subsequent four proposals to be developed between the early 1960s and 1980s also took a considerable length of time; two took 17 years, one took 15 years and one took 14 years. The time period subsequently shortened, with the seven applications which began to be developed between 1988 and 1993 taking much shorter time periods. One took six years, two took four years, three took three years and one took only two years from inception.

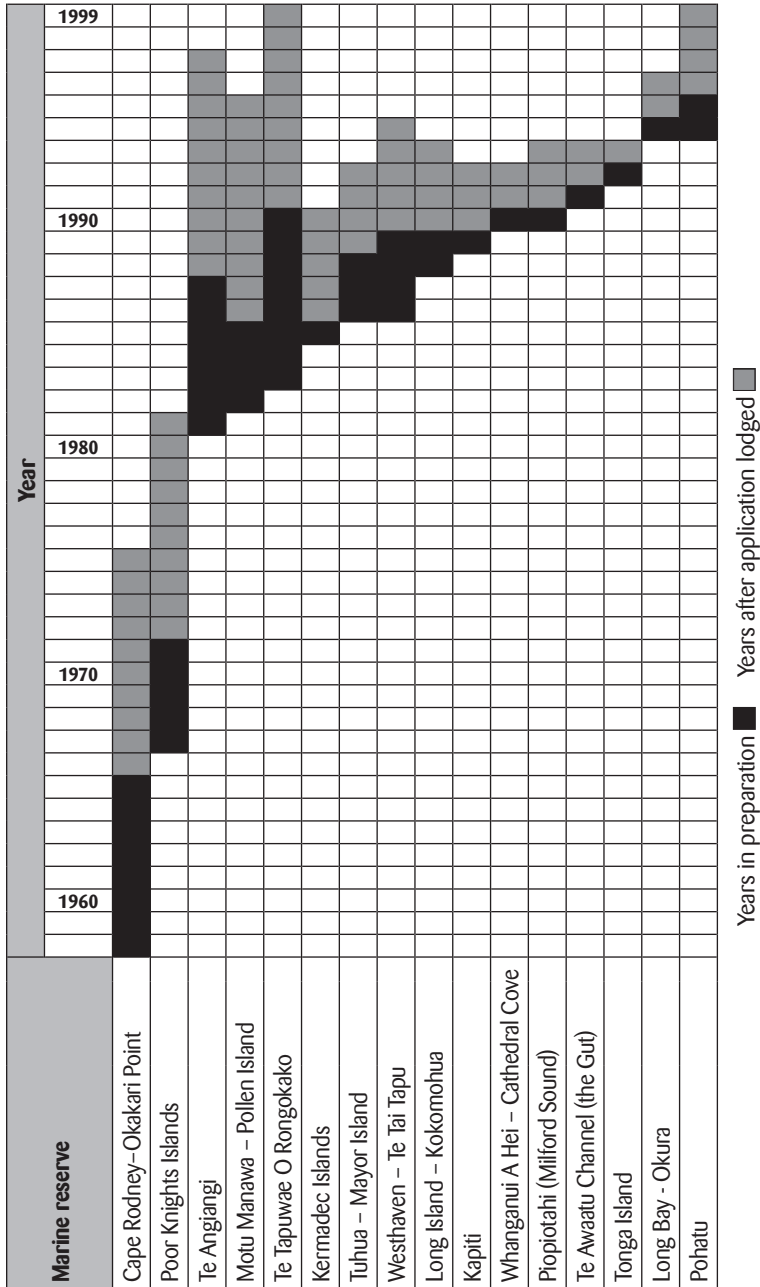


Figure 12.4: Time taken to process marine reserves established between 1971 and 1999

The more recent figures, however, can be misleading, as some applications that have not yet been approved have been languishing for years. For example, the processing of the application for Nugget Point, lodged in 1992, has recently been recommenced at the beginning of the process, and a final decision is some way off. The Fiordland (Te Moana o Atawhenua) Marine Management Act 2005 has managed to shortcut the process for the establishment of eight marine reserves in Fiordland through establishing them immediately through a statutory provision (section 7). However, the Act also prohibits the establishment of any additional marine reserves in the area for seven years or until a ministerial review is undertaken five years after the Act came into force (section 10).

Although a marine reserve application has yet to be declined outright, the area covered can be significantly reduced and/or the application can languish for years as attempts to overcome opposition are made.

Evaluation

In terms of the adequacy and representativeness of New Zealand's current marine reserve network, a detailed analysis has yet to be undertaken. Such an analysis is proposed as part of the Marine Protected Areas Policy Statement and Implementation Plan described in Chapter 11. It is abundantly clear, however, that New Zealand's marine reserves do not contain representative areas of all habitats or protect all the country's important marine biodiversity hot spots. Large areas of the coastline, including most of the west coast and all of the northern tip of the North Island, the east and south coasts of the South Island and the central portion of the west coast of the South Island have no marine reserves. In addition, there are no marine reserves outside the territorial sea, as the marine reserve legislation does not currently apply in this area. None of the area that is considered to be the New Zealand marine biodiversity hot spot, the Three Kings Islands and shelf, Spirits Bay and Pandora Bank (extending to the north of North Cape), is fully protected (Arnold, 2005:51).

Recent initiatives, including the Marine Reserves Bill and the Marine Protected Areas Policy Statement and Implementation Plan, should facilitate the establishment of additional marine reserves. The Marine Reserves Bill, if passed by Parliament in its current form, will enable marine reserves to be established within the EEZ, thereby removing the current legal impediment to the establishment of a representative highly protected network. The Bill should also streamline and shorten the statutory process for establishing marine reserves.

As already indicated, the proposed Marine Protected Areas Policy Statement and Implementation Plan is likely to apply an integrated regional approach to the planning and establishment of new marine protected areas including marine reserves. This should help address some stakeholder concerns about the inability to engage in the bigger issue of integrated marine protection. The focus on establishing national priorities and identifying gaps in the current network should also help ensure a more representative network.

Functioning of Governance System

As discussed in Chapter 9, an effectively functioning governance system is likely to have high connectivity, be adaptive and be supported by a strong constituency. The performance of the New Zealand governance system in these respects is discussed in the following sections.

Connectivity

There are several mechanisms which can be used to achieve connectivity throughout an oceans governance system, as described in Chapter 4. These include establishing a common purpose and principles for ocean management efforts, establishing integrative bodies and processes for oceans policy makers and managers and creating incentives to encourage integrative behaviour. The extent to which these mechanisms have been adopted within the New Zealand oceans governance system is explored in the sections below.

Integrative purpose and principles

There is currently no common purpose or set of principles for oceans governance. Oceans management is currently undertaken within the jurisdiction of many pieces of legislation, each with its own specific purpose. For example:

- The Resource Management Act 1991 has as its purpose '*... to promote the sustainable management of natural and physical resources*' (section 5(1))
- The Fisheries Act 1996 has as its purpose '*... to provide for the utilization of fisheries resources while ensuring sustainability*' (section 8(1))
- The Maritime Safety Act 1994 includes as its purpose '*... to protect the marine environment ...*' and '*... to continue, or enable, the implementation of the obligations on New Zealand under various international conventions relating to the pollution of the marine environment ...*' (long title)
- The Continental Shelf Act 1964 has as its purpose '*... to make provision as to the exploration and exploitation of the continental shelf of New Zealand and for matters connected with that purpose*' (long title)
- The Territorial Sea, Contiguous Zone, and Exclusive Economic Zone Act 1977 has as part of its purpose '*... to make provision for the exploration and exploitation, and conservation and management, of the resource of the zone ...*' (long title)

Integrative bodies and processes

Apart from those established for the development of the national oceans policy itself, as will be described in Chapter 14, there are currently no fully integrative bodies or processes for oceans governance in New Zealand. However, there have been some positive initiatives to better integrate management efforts, many of which have been mentioned in previous sections.

At a central government level, there have been several joint initiatives between DoC and MFish on issues of common concern. These include the development of:

- A National Plan of Action to Reduce the Incidental Catch of Seabirds in New Zealand Fisheries
- A joint approach to the protection of the Maui's dolphin (DoC and MFish, 2003a)
- A Marine Protected Areas Policy Statement and Implementation Plan

In addition, as already mentioned, MfE is spearheading a collaborative effort amongst a wide range of central government agencies and industry to develop an environmental accord to apply to the management of activities outside territorial waters.

Generally, linkages between central government and local government oceans management agencies have been weak. In a survey of 13 local authorities, no agency reported a close working relationship with MFish and few indicated a strong working relationship with DoC (Willis, 2003:9). There have been several initiatives on a sub-national level to integrate oceans management across different levels of government. These apply an integrated approach to planning and management and, in some cases, set up an integrative management body. They include:

- The initiative to provide integrated management of the Hauraki Gulf, under the Hauraki Gulf Marine Park Act 2000, through providing a set of common management objectives and establishing an integrative body, the Hauraki Gulf Forum, which has representatives from central government agencies, local authorities and tangata whenua
- The initiative to provide an integrated management solution for Fiordland under the Fiordland (Te Moana o Atawhenua) Marine Management Act 2005, which includes the establishment of an advisory and facilitation body, the Fiordland Marine Guardians, but not a set of common management objectives
- The proposed Waitakere Ranges Heritage Bill, which puts forward a set of common objectives for the management of the Ranges, coastal villages and foothills, but does not extend to the marine area and does not establish an integrative body, leaving management to existing statutory bodies

Integrative incentives

The only incentive identified during the research undertaken for this publication, for integrated action on oceans management, is the dedicated funding provided to support the implementation of the oceans elements of the Biodiversity Strategy. This funded the development of a web-based national aquatic biodiversity information system (NABIS) and directed research on the diversity of selected marine communities within the EEZ and in the Ross dependency. The research topics included research on the biodiversity of seamounts, bryozoan communities around Separation Point, coastal soft-sediment communities and crustose coralline algae (MFish, 2002:2 & 8).

Adaptive management

An adaptive governance system could be expected to incorporate ongoing monitoring of the health of the oceans, regular reviews of the effectiveness of management actions and flexible management tools which can be readily adjusted in light of changing information and conditions.

State of the oceans monitoring

There is currently no comprehensive programme to monitor the state of the oceans. The New Zealand State of the Environment Report, which was released in 1997, failed to address the oceans as a system, reporting separately on the state of water, mammals, birds, fish, plants and invertebrates.

An integrated and comprehensive report on the state of the oceans at a sub-national level has been undertaken for the Hauraki Gulf. The Hauraki Gulf State of the Environment Report 2004 (Hauraki Gulf Forum, 2005) describes the underlying pressures on the Gulf environment and current environmental management efforts. It then reports on six key issues: water quality, biological diversity, natural character and landscape values, access, cultural heritage and coastal hazards.

The Report notes in its introduction that the challenges in presenting a picture of the Gulf were considerable. This was because monitoring of different aspects of the Gulf is carried out by multiple agencies for a whole host of purposes. The issue-, site- and purpose-specific information collected by a host of agencies did not lend itself to collation into a state of the environment report for a particular environment such as the Gulf (Hauraki Gulf Forum, 2005:14).

A State of the Environment Report has also recently been prepared for the Southland Coastal Marine Environment through the collaboration of five agencies, Environment Southland, DoC, Invercargill City Council, Southland District Council and Te Ao

Marama Incorporated – the resource management agency for the four Murihaka rūnanga of the Ngāi Tahu iwi (Environment Southland and others, 2005).

The Ministry for the Environment devoted considerable effort into developing a set of environmental performance indicators for the marine environment (MfE, 2001), but most of them are still not associated with monitoring regimes and have therefore not been implemented (Oceans Policy Secretariat, 2003b:3).

Monitoring of some aspects of the marine environment is being undertaken by various statutory agencies. The Department of Conservation undertakes some monitoring of marine reserves, which commonly includes monitoring of the habitat and selected key species but not of the effects of human activity on the reserve (McCrone, 2001:16). The Ministry of Fisheries monitors the state of commercially important fisheries stocks. Monitoring by regional councils varies between councils, but is primarily focused on issues such as beach water quality, sedimentation, heavy metal contamination and coastal erosion.

Reviews of management actions

There is currently no system to ensure regular independent reviews of the effectiveness of oceans management. Despite this, several of these have been undertaken during recent years.

The Parliamentary Commissioner for the Environment (PCE) undertook a review of oceans governance in the late 1990s and released his report titled *Setting Course for a Sustainable Future: The Management of New Zealand's Marine Environment* in December 1999. The review identified a number of problems with the current governance system including (PCE, 1999:96–98):

- Lack of an overarching framework or strategy to guide the many stakeholders towards sustainable marine management and to resolve incompatible goals between the various pieces of legislation
- Failure of current management structures to reflect the complexities and interconnections within the ecological, cultural and economic systems being managed
- Failure of the current fisheries management rights regime to deliver sustainable management of fish stocks or the marine ecosystems they inhabit
- Very inadequate investment in ecosystem research, with a marked decrease in government investment in such research over the past six to seven years. This was in favour of a high-risk strategy of devolving research to the industry exploiting the resource

- Lack of knowledge about marine species and their role in maintaining the integrity and resilience of marine ecosystems, including lack of basic information on the status of commercial fish stocks
- Lack of communication and a grave lack of trust among stakeholders

A key recommendation of the report was to establish a Coastal and Oceans Task Force to develop a strategy comprising, firstly, goals and principles and, secondly, actions and policies for the future sustainable management of New Zealand's marine environment (PCE, 1999:99). This recommendation helped prompt the government to embark on the preparation of a national oceans policy.

An independent review of the New Zealand Coastal Policy Statement was undertaken by Dr Johanna Rosier, with the results set out in the report titled *Independent Review of the New Zealand Coastal Policy Statement* released in May 2004 (Rosier, 2004). The review made a number of findings including:

- The preparation and implementation of the NZCPS had effectively generated debate about New Zealand's national priorities for coastal management and had encouraged local government to change the way in which coastal issues were considered in local planning frameworks
- NZCPS policies had been effectively implemented through regional policy statements and regional coastal plans
- The NZCPS had been only partially effective in influencing district plans and subsequent land-use planning decisions within the coastal environment
- The area of poorest implementation had been the monitoring of environmental outcomes and assessment of the degree to which plans and policy statements had influenced environmental results

The reviewer recommended that the Policy Statement be formally reviewed under the procedures set out in the Resource Management Act 1991, and amended to both revoke policies now obsolete because of other policy developments and provide additional policy guidance required by sub-national levels of planning (Rosier, 2004:11). The Minister of Conservation has committed to undertake such a review and initial work on the review has commenced.

In 1999, the Controller and Auditor-General undertook an investigation into information requirements for the sustainable management of fisheries (Controller and Auditor-General, 1999). The report contains a range of findings and recommendations including:

- Owing to a lack of adequate information, MFish manages most fish stocks without being sure that their management regime leads to sustainable fisheries

- The Ministry of Fisheries has been slow to commit resources to meet the environmental principles of the Fisheries Act 1996 and needs to invest in significantly more research to fulfil these environmental principles

A formal review process has been incorporated in the Fiordland (Te Moana o Atawhenua) Marine Management Act 2005. The Minister for the Environment is required to initiate a review to determine the effectiveness of the management of the Fiordland (Te Moana o Atawhenua) Marine Area five years after the commencement of the Act (section 25).

Flexible management tools

Different oceans management tools have varying levels of flexibility built into them. In the fisheries sectors, the provision for regular reviews of the total allowable catch (TAC) enables frequent adjustments to be made in light of new information about the state of fish stocks. This has, in some cases, allowed substantial reductions to be made to the TAC such as the recent reduction in hoki fisheries from 180,000 tonnes to 100,000 tonnes (Benson-Pope, 2004). In some other cases, however, reductions can be strongly opposed by members of the fishing industry, making it politically difficult to adapt management in light of new information.

Regional coastal plans prepared under the Resource Management Act 1991 are an example of an oceans management tool which has proved very inflexible. There are currently only seven operative regional coastal plans prepared by regional councils (out of 12 regions) 13 years after the Act came into force, even though the preparation of such plans is mandatory.

The lengthy time taken to prepare the plans and to go through the public submission process, involving council and Environment Court hearings as well as mediation (see Figure 13.1), is costly and time consuming and means that councils can be reluctant to change them. Plans can take over a decade to become operative, they can be out of date by the time they come into force, and are in place for another 10 years before review. Although regional councils can introduce changes to the plan in the interim, this is a costly and lengthy process.

A recent survey of 12 local authorities and three DoC conservancies identified a common opinion that the oceans policy should not require the reworking of regional coastal plans. This was despite the view expressed by seven respondents that regional coastal plans were only partially successful (Willis, 2003:29) and indicates a lack of enthusiasm to change despite misgivings about effectiveness.

Regional Council	Year plan notified	Year plan operative
Manawatu-Wanganui		1997
Taranaki	1994	1997
Hawkes Bay	1994	1999
Wellington	1994	2000
Otago	1994	2001
West Coast	1997	2001
Bay of Plenty	1995	2003
Canterbury	1994	Not operative
Southland	1997	Not operative
Waikato	1994	Not operative
Northland	1994	Not operative
Auckland	1995	Not operative

Figure 13.1. Processing times for regional coastal plans prepared by regional councils

Source: www.mfe.govt.nz/issues/resource/plans/plans.php

Strong constituency

A strong constituency for oceans governance can be developed through a range of mechanisms including disseminating high-quality and accessible information to stakeholders and facilitating networks and forums through which oceans issues can be discussed and information shared.

The research undertaken for this study revealed that there is a paucity of accessible and informative data about the state of the oceans and the effectiveness of current management efforts. Information is fragmented, often highly technical and situation specific and difficult to integrate into a broader picture.

The New Zealand Coastal Society, inaugurated in 1992 as a technical group of the Institution of Professional Engineers New Zealand (IPENZ), plays an important role in promoting networking and information dissemination on the management of the coastal environment amongst professionals. The Society currently has over 300 members drawn from the coastal science, engineering and planning disciplines. It hosts

a national annual conference, holds other regional meetings and produces issues of the publication *Coastal News* three times a year (New Zealand Coastal Society, 2005).

There are few broader fora, however, for a wide range of stakeholders to exchange information and viewpoints on oceans issues.

Evaluation

The current governance system appears weak on some attributes which could indicate effective functioning. Areas of weakness include:

- The lack of a common purpose or set of principles which applies to the entire governance system
- Few integrative bodies or processes, although there have been some innovative initiatives at a regional level
- Few integrative incentives
- No comprehensive programme to monitor the state of the oceans and no system for regularly reviewing management performance
- Some inflexible management tools in key areas, especially regional and district planning
- Poor public availability of accessible information on oceans

What Should New Zealand's Oceans Policy Contain?

This chapter addresses the key issue of what a world-leading oceans governance system for New Zealand might look like and how an oceans policy might help New Zealand get there.

What might a world-leading oceans governance system look like?

Summarising the material in previous chapters, one might expect a world-leading oceans governance system in New Zealand to have the following features:

- *Sustainable management practices* applied to all human activities which impact on the marine environment to ensure that such activities do not undermine marine ecosystems. This might have several key components including:
 - A *fisheries management regime* that identifies and effectively manages the impacts of fishing activity on marine ecosystems
 - A *catchment management regime* that effectively manages marine sedimentation and pollution from land-based sources
 - An *environmental impact assessment regime* that ensures that structures and activities which are established in the marine area – such as reclamations, wharves, minerals exploitation, aquaculture and power generation facilities – do not disrupt marine ecosystems
 - A *marine pollution management regime* that effectively manages the discharge of pollutants into the marine area from ships and marine structures
 - An effective *bio-invasion prevention and response capability* to ensure that ecologically and/or economically damaging species are not introduced and established in New Zealand
- A network of *special marine management areas* incorporating sensitive and important marine environments where additional management effort is deployed
- A representative network of *highly protected areas* where natural processes within the marine environment are able to operate undisturbed

In addition, the oceans governance system could be expected to exhibit the following characteristics:

- *High levels of connectivity* between the different elements of the system so that they work together as a cohesive whole to achieve a common purpose
- *Strong adaptive capability* to respond to changes in information and circumstances
- *Strong stakeholder support* for ecosystems-based and integrative oceans governance to strengthen ongoing management actions

How might an oceans policy help New Zealand get there?

Drawing on the elements of a national oceans policy described in Chapter 1, some proposed elements of a New Zealand oceans policy are set out below.

Common purpose

- Provide a clear statement of the purpose of oceans management in New Zealand and establish a set of common principles to guide oceans management. If clear and well articulated, these can provide an inspirational focus for ocean management efforts.
- Make a commitment to progressively apply ecosystems-based approaches to oceans management. An oceans governance system for the twenty-first century needs to be sophisticated enough to treat oceans as ecological systems rather than focusing on discrete parts or organisms.
- Make a commitment to protecting representative areas of all New Zealand marine habitats, and all special and unique areas, in a viable network of highly protected areas. This is an important element of any internationally credible and world-leading oceans governance system.

Integrative mechanisms

- Establish a permanent *Oceans Ministerial Group* to provide high-level policy direction. This would continue the work of the ad hoc group of ministers set up to oversee the oceans policy development process. It would help ensure coordination at the highest political level.
- Establish an *Oceans Management Group*, consisting of high-level officials of relevant central, regional and local government departments and agencies to coordinate activities, share information and undertake joint projects. This would

continue the work of the Oceans Policy Steering Group established to oversee the development of the oceans policy.

- Establish an *Oceans Science Group* to provide independent and relevant scientific advice to the Oceans Ministerial Group and the public. The credibility of scientific advice, when the vast expanse of the sea is away from the public eye, is vital. An independent Science Group would help ensure that the latest scientific knowledge is incorporated into policy making at a high level, as well as provide the public with a source of information independent of government or vested stakeholders.
- Establish *Oceans Stakeholder Advisory Groups*, including a Māori Advisory Group, to provide stakeholder input into oceans policy priorities and initiatives.

Strategic actions

- Support the passage of the Marine Reserves Bill and provide substantial dedicated funding to speed up the identification and establishment of marine reserves to ensure a viable and fully representative network. The passage of the Bill needs to be supported to revise the dated provisions in the Marine Reserves Act 1971. The passage of this Bill needs to be supported. Marine reserves could possibly be renamed 'Marine National Parks' to more closely align them with the existing land-based national park system and to avoid any negative connotations which may have become associated with the marine reserve concept. New Zealand led the world in 1975 when it established the first no-take marine reserve and it is time to retake a leadership position in this area.
- Undertake legislative reform to rationalise the statutory provisions for the establishment of *special marine management areas*. There is currently a myriad of pieces of legislation, which have been increasing in number, for the management and protection of particular spatial areas of the oceans. This field of law is ripe for review and rationalisation. Legislative reform is required to provide for the establishment of special management areas within the marine environment within a clearly defined framework based on current international best practice.
- Undertake a national investigation into the impacts of trawling and dredging on benthic habitats and identify effective mitigation measures to be adopted, incorporating the precautionary principle. Little action is being taken to manage the impacts of such fishing activity and this area needs urgent action to ensure irreversible damage is avoided. New Zealand became an international leader in fisheries management when it established the Individual Transferable Quota system. The management of the impacts of trawling and dredging, an issue which many other countries are also grappling with, provides an excellent

opportunity for New Zealand to cement its position as an innovator in fisheries management.

- Prepare a *Guide to Sustainable Coastal Development* and give key provisions statutory force through incorporation into the NZCPS. Many land-based impacts on marine areas are caused by the way coastal settlements and related infrastructure are designed and located. A national guide could usefully define what constitutes sustainable coastal development and indicate how it could be achieved at local and regional levels.
- Undertake an investigation into coastal planning under the RMA, at both regional and territorial authority levels, and identify reforms to increase both the effectiveness of planning in managing coasts and its ability to respond quickly to changing information and circumstances. Current planning efforts are failing to sustainably manage impacts on the marine area, are very costly to undertake and lack flexibility. New Zealand was an innovator in establishing regional councils with jurisdiction over both water catchments and marine areas. Both review and reform of current planning practices are urgently required to maximise the advantages of these regional institutional structures and to re-establish New Zealand's international position in respect of resource management.
- Develop a *National Agenda for Sustainable Estuaries Management* that scopes the problem, develops solutions and targets resources to improve the management of estuaries and harbours of national ecological importance. Regional councils are struggling to manage the ongoing sedimentation and pollution of estuaries and seem unlikely to get on top of the issue in the near future. This is an area requiring centralised government support, perhaps drawing from the experience of the Lake Taupo water quality initiative.
- Develop and resource the implementation of a *Marine Biosecurity Strategy* that identifies the key sources of marine bio-invasion risks and effective actions to manage those risks.
- Develop an environmental regime for the management of activities outside the territorial sea. New Zealand currently lacks any such coherent environmental system, so this provides an excellent opportunity to develop a world-leading regime drawing on experiences in other countries and current international thinking.
- Establish a contestable *Oceans Constituency Fund*. The constituency for the implementation of an effective oceans governance system is currently weak and dedicated efforts are needed to strengthen it. Non-governmental organisations can be more effective at reaching a wide group of stakeholders than government agencies. A contestable fund could usefully support such initiatives. Potential projects could include establishing stakeholder networks, preparing and

disseminating information on oceans issues and convening workshops, seminars and conferences.

Performance management systems

- Develop a performance management system for oceans management, drawing on international best practice. The National Oceans Office in Australia has put considerable effort into developing such a system, and learning from that initiative could help inform the development of an appropriate system for New Zealand.
- Develop a national *Oceans Information Strategy*, building on the Ocean Survey 20/20 Project, to address the current fragmentation, inconsistency and gaps in oceans management information.
- Prepare a *State of the Oceans Report* and regularly update it (at least five-yearly). This would provide a touchstone for how well the oceans governance system is doing in an accessible manner for both stakeholders and the public more generally.
- Task the Parliamentary Commissioner for the Environment with undertaking regular (three-yearly) independent reviews of progress in implementing the oceans policy. The PCE is an appropriate person to undertake such reviews, being independent of government through reporting directly to Parliament, and having already undertaken an investigation into oceans governance.

Implementation

- Continue the role of *Oceans Minister* to provide oversight of the implementation of the initiative. As discussed in Chapter 5, the review of comparator countries indicated that high-level political support, driven by a strong government cabinet minister, is important for successful implementation.
- Establish an *Oceans Agency* to champion the further development and implementation of the policy. The comparator country review also indicated the importance of establishing an agency which can focus solely on implementation, rather than giving the role to an existing department which has other competing priorities.
- Provide sufficient *dedicated funding* to enable successful implementation. The review of the development of oceans policies in Australia, Canada and the USA highlighted the importance of providing significant dedicated funding to implement oceans policy, if rapid progress is to be made.

Proposed Elements of a New Zealand's Oceans Policy

Common purpose

- Clear statement of purpose and principles
- Commitment to progressively apply ecosystems-based management
- Commitment to establishing a representative and viable network of highly protected areas

Integrative mechanisms

- Oceans Ministerial Group
- Oceans Management Group
- Oceans Science Group
- Oceans Stakeholder Advisory Groups

Strategic actions

- Support the passage of the Marine Reserves Bill
- Develop a revised statutory framework for the establishment and management of special marine management areas
- Undertake a national investigation into the impacts of trawling and dredging on benthic habitats and mitigation measures
- Prepare a Guide to Sustainable Coastal Development and incorporate it into the NZCPS
- Undertake a national investigation into coastal planning and identify reforms
- Develop and resource the implementation of a National Agenda for Sustainable Estuaries Management
- Develop an environmental regime for activities outside the territorial sea
- Develop and resource the implementation of a Marine Biosecurity Strategy
- Establish a contestable Oceans Constituency Fund

Performance management systems

- Develop a performance management system for oceans governance drawing on international best practice
- Develop an Oceans Information Strategy
- Prepare a State of the Oceans Report and regularly update it
- Task the Parliamentary Commissioner for the Environment with undertaking three-yearly independent reviews of the implementation of oceans policy

Implementation

- Oceans Minister
- Oceans Agency
- Dedicated funding

Part Three Conclusions

An oceans governance framework which incorporates ecosystems-based and integrated approaches is likely to consist of three key elements: sustainable management practices applied to activities impacting on the marine area, a network of special management marine areas whereby management is integrated and/or focused on key threats and a representative network of highly protected areas.

There are at least four notable gaps in sustainable marine management in New Zealand. These include managing the environmental impacts of fishing and in particular the impacts of trawling and dredging on benthic habitats, reducing the levels of land-based sedimentation and pollution entering the marine area to a sustainable level, managing the environmental impacts of activities outside territorial waters and controlling the arrival of invasive marine species on the hulls of ships.

The legislation and management framework for marine protected areas is highly fragmented and, with the exception of 19 seamounts, provides little protection for areas outside the territorial sea. New Zealand currently falls far short of adequate provision for areas that need to be highly protected (through marine reserves), with inadequate coverage of representative areas of all habitats and important marine biodiversity hot spots.

The current governance system is weak on attributes which would indicate effective functioning. There is a lack of a common purpose, no fully integrative bodies, processes or incentives, no comprehensive monitoring and review system, some inflexible management tools and poor accessibility to information.

There are many elements, which the oceans policy could contain, that would contribute to improved oceans governance in New Zealand. A number of these are proposed with the aim of helping promote constructive debate on this critical issue.

Conclusions

Oceans cover nearly three-quarters of the earth's surface, regulate the climate and are rich in biodiversity. The establishment of EEZs extending 200 nautical miles seawards from coastlines has resulted in more than one-third of all oceans being brought under the jurisdiction of coastal countries. The challenge of effectively managing these vast ocean areas, along with their increasing economic importance and a greater understanding of the fragility of ocean ecosystems, has led to a growing interest in the development of fully integrated national oceans policies.

The effective governance of oceans within national jurisdictions has proved to be enormously difficult, as has the development of fully integrated national oceans policies. Although interest in developing comprehensive and integrated policies began in the 1960s, surprisingly little progress has been made. Countries have yet to apply the same level of effort to managing the oceans as has been devoted to addressing sustainability issues on land. Yet the oceans will play an increasingly important role in future economic, social and cultural welfare.

New Zealand governs an expansive area of the world's oceans. It is an area rich in biodiversity as well as oil, gas and minerals. It supports major export earners for the country, including commercial fishing, aquaculture and marine tourism. The ocean also provides significant economic, social and cultural opportunities as well as governance obligations.

The current governance of New Zealand's oceans is not up to the task in several key areas, including providing adequate protection for marine biodiversity, effectively managing the environmental effects of fishing, reducing sedimentation and pollution of the marine area, managing the environmental impacts of activities outside territorial waters and preventing damaging bio-invasion.

A national oceans policy could provide significant assistance in addressing these problems, and others, through providing an improved organisational, legal and policy framework and focusing coordinated efforts and resources on key areas.

New Zealand is in a unique position to 'get it right'. It is a relatively small country where there is close interaction between policy makers and stakeholders. It has a unicameral parliamentary system which avoids the complexities of the federal and state constitutional arrangements encountered in many other countries.

At the same time, the country governs a large area of ocean and is experiencing many of the same marine management issues that larger states are grappling with. The oceans policy initiative provides New Zealand with the opportunity not only to improve outcomes for its own oceans, but also to become an exemplar of best practice oceans governance for the rest of the world.

Preparing a national oceans policy provides an opportunity to strengthen New Zealand's international reputation as an innovator in natural resource governance. Setting a new benchmark for international best practice in oceans governance will enhance New Zealand's reputation as an environmental leader. This will, in turn, reinforce the country's 'clean and green' brand which helps sell our goods overseas and attract tourists. It is an opportunity not to be missed.

Appendices

Comparator Country Case Studies

A

Australia's Oceans Policy

Context

Australia has jurisdiction over an EEZ consisting of 8.2 million square kilometres of ocean, excluding the area around Antarctica which increases the area to 10.2 million square kilometres.

Under the *Offshore Constitutional Settlement*, which entered into force in 1983, states and territories within Australia have title to water landwards of the 3 nautical mile limit (with some areas specifically excluded), which at the time was the extent of the territorial sea. States were given the same power to legislate over this ocean area as they had over their land territory.

In 1990, the Commonwealth government extended Australia's territorial sea from 3 to 12 nautical miles and in 1994 passed legislation asserting jurisdiction over a contiguous zone extending seawards 24 nautical miles and a 200 nautical mile EEZ (Rothwell, 1996:61–61). Consequently, the Commonwealth government has jurisdiction over an area of oceans generally extending between the 3 nautical mile and 200 nautical mile limits.

Agreements between Commonwealth, state and territorial governments provide for specific roles and responsibilities in terms of different marine sectors and these may extend beyond the established jurisdictional boundaries. State governments effectively control management of the coastal zone and oversee local government authorities which regulate land-use planning and development control over coastal areas (Haward, 1996:21).

Over 85 per cent of Australians live in the coastal zone, which extends from the tropics to temperate zones (Zann, 1996:64). The ocean supports significant economic activity. In 1994, Australian marine industries contributed around A\$30 billion to the national economy, or 8 per cent of the country's gross domestic product, and provided 7 per cent (A\$6.6 billion) of total exports (Commonwealth of Australia, 1998a:8).

Key marine economic sectors are (Commonwealth of Australia, 1998b):

- Commercial fisheries – which generates around A\$1.8 billion per annum
- Aquaculture – predicted to generate A\$1.4 billion per annum by 2005
- Offshore petroleum – which generates A\$8 billion per annum
- Marine tourism – which generates a significant proportion of the around A\$16.1 billion in foreign exchange produced by international tourism

Australia's ocean areas support rich marine habitats and biodiversity. The country has the largest area of coral reefs, the largest area and diversity of temperate seagrass, the greatest development and diversity of temperate macroalgae (seaweed) and amongst the largest areas of tropical seagrass and mangroves in the world. The south has a very high proportion of endemic marine species (around 90 per cent in most groups) and the north has some of the world's last secure populations of threatened species such as dugongs, turtles and giant clams (Zann, 1996:72).

A State of the Marine Environment Report (SOMER) completed in 1995 found that the health of Australia's marine environment was, on average, generally good although not as good near urban coastal settlements. Over-harvesting of fish and other marine life, coastal development and conflicting resource use were identified as serious issues around Australia. In particular, estuarine environments in much of eastern and southern Australia are declining because of eutrophication and sedimentation, acid soil run-off, coastal developments, loss of habitat and overfishing (Zann, 1996:73–74).

Declining water quality and sedimentation were found to be probably the most serious issues affecting Australia's marine environments. Land erosion is a major source of nutrients in rural areas, and sewage and urban run-off the major source in urban areas. Because of elevated nutrient levels, blooms of blue-green algae are frequent in southern Australian estuaries and bays. Eutrophication is a serious threat to Australia's estuaries, temperate seagrass beds and tropical corals. Oil is also entering the marine environment mainly through urban run-off (Zann, 1996:73).

SOMER reported widespread concern about the environmental effects of trawling and scallop dredging on the sea-floor community, on juvenile fish, on by-catch species and on marine food webs. The effects of fishing on the ecosystem are little known but are thought likely to be significant. Serious overfishing of southern bluefin tuna, southern sharks, gemfish, rock lobsters and other species has occurred. Of the 100 most important fisheries, nine are considered to be overfished, 23 are fully or heavily fished, nine are underfished and 59 are of unknown status (Zann, 1996:75).

The introduction of exotic pests and diseases via ships' ballast water was also identified as potentially a very serious problem, with around 55 species known to have been introduced this way (Zann, 1996:75).

Origin of national oceans policy

The Commonwealth government's Ocean Rescue 2000 initiative, which commenced in 1991, laid important foundations for the subsequent national oceans policy. The prime objective of the initiative was the conservation and sustainable use of Australia's marine environment. One component of the initiative was the establishment of the Marine and Coastal Community Network. This network, which was established in 1993 with funding provided by the Commonwealth government, made a significant

contribution to increasing interest in oceans policy development. Ocean Rescue 2000 also included the preparation of SOMER, which was completed in 1995. This provided the first comprehensive description of Australia's marine environment and highlighted key issues and threats. Other components of the initiative included the development of an Australian Marine Conservation Strategy and a national representative system of marine protected areas.

In 1994, the Australian government ratified UNCLOS and in the same year asserted jurisdiction over a very extensive and largely unstudied EEZ. The need to exercise sovereign rights over this area and to ensure the ecologically sustainable development of its resources, both for wealth creation and environmental protection, was the prime motivation for developing a national oceans policy (Westcott, 2000:862).

In August 1994, a seminar hosted jointly by the Centre for Maritime Policy at the University of Wollongong and the Royal Australian Navy directly addressed the need for an oceans management policy and developed a policy framework based strongly around the development of marine science, technology and industry (Westcott, 2000:859).

During the 1996 Commonwealth government election campaign, both major political parties supported the development of a comprehensive and integrated oceans policy, and work to prepare the policy commenced shortly after the election in May 1996.

Development of policy¹²

Shortly after the 1996 Commonwealth government election, the Department of Environment, Sports and Territories (Environment Australia) was tasked with the role of overseeing the development of the national oceans policy. The selection of an environmental agency, rather than an economic development one, to play this pivotal role was significant in emphasising environmental considerations associated with oceans management. In mid-1996, Environment Australia established an inter-departmental committee of major Commonwealth government agencies involved in marine affairs to assist with the policy's preparation.

The Prime Minister launched a consultation paper titled *Australian Oceans New Horizons* in March 1997. This document proposed a draft vision, goals, a list of objectives and a series of major issues for Australia's oceans. Comment was invited on the scope of the policy before the end of April 1997. The issues identified in the paper included (Commonwealth of Australia, 1997):

- Conservation and management of Australia's marine biodiversity
- Marine pollution
- Ocean industries

12 Unless otherwise referenced, the material in this section has been sourced from Westcott 2000.

- Employment and economic opportunities
- Scientific understanding
- Indigenous issues
- Surveillance and enforcement
- International considerations
- Community participation
- Public awareness and understanding

The government asked the Marine and Coastal Community Network (of just under 7000 participants) to use its resources and participants to inform the community about the development of the oceans policy. The Network published progress reports on the policy development process, distributed information supplied by the government, and also distributed a questionnaire. In addition, Environment Australia organised several workshops and undertook face-to-face interviews in respect of the policy proposals. Only 63 submissions were received on the discussion document by the closing date. These were primarily from academics and representatives of non-governmental organisations, indicating the limited involvement of the general public in the process.

In September 1997, the Minister for the Environment announced that he would establish a Ministerial Advisory Group on Ocean Policy (MAGOP) comprised of people from a range of sectors, but with no formal representation from the state governments. In March 1998, the Advisory Group presented the Minister with a comprehensive report with 99 specific recommendations. Although MAGOP was comprised of a range of people from diverse interest groups and sectors, the committee achieved remarkable consensus on the scope of the policy and the wish to tie it to environmentally sustainable development principles. However, disagreement did occur on whether an ecosystems-based management system approach, combined with an integrated regional planning and management system, should be adopted. Conservative views, held mainly by commercial development industries, favoured the retention of the current sectorally-based regulatory arrangements. Other sectors wished to see a truly integrated planning and management system implemented.

In late 1997, Environment Australia prepared a series of discussion papers around the issues identified in the consultation paper. A two-day forum for 133 invited participants was organised in December 1997, using the discussion and issues papers as a basis for further policy development.

In May 1998 *Australia's Oceans Policy – an Issues Paper* was released for public comment. Submissions were invited by July 1998. The document addressed a range of relevant issues but did not include any details on implementation or recommend future institutional arrangements. Approximately 660 submissions were received on the document, the vast majority coming from individuals, indicating that the policy development process was achieving better traction amongst the broader community.

Although the Commonwealth government initiated consultation with state and territory governments during the policy development process, no agreement was reached on the involvement of state governments in the implementation of the policy. This was because of the unwillingness of the Commonwealth government to provide state government representation on the proposed National Oceans Ministerial Board and to provide funding to assist state governments with the implementation of the policy (Foster et al., 2005). The Commonwealth government, wishing to launch the policy during the international Year of the Oceans, released the final policy document in December 1998 without the endorsement of state and territorial governments. This failure to obtain buy-in from the level of government with jurisdiction over coastal areas was to prove a substantial hurdle in achieving the subsequent implementation of the policy.

The Commonwealth government committed A\$50 million over a three-year period to implement the initiatives identified within the policy (Commonwealth of Australia, 1998a:5).

Content of policy

The Oceans Policy released in late 1998 consisted of two volumes. The first volume set out the overall vision and goals for the policy, described the concept of ecosystems-based oceans planning and management, which was proposed to be applied to oceans management, set out the institutional arrangements for implementation and identified some key initial implementing actions to be carried out by the Commonwealth government. The second volume of the policy identified specific actions, which were proposed to be taken in respect of specific economic sectors and other areas, including marine science and defence.

Vision and policy goals

The policy set out a very broad vision for Australia's oceans as follows:

Healthy oceans; cared for; understood and used wisely for the benefit of all, now and in the future.

It then described nine policy goals:

- To exercise and protect Australia's rights and jurisdiction over offshore areas, including offshore resources
- To meet Australia's international obligations under UNCLOS and other international treaties

- To understand and protect Australia's marine biological diversity and the ocean environment and its resources, and ensure ocean uses are ecologically sustainable
- To promote ecologically sustainable economic development and job creation
- To establish integrated oceans planning and management arrangements
- To accommodate community needs and aspirations
- To improve our expertise and capabilities in ocean-related management, science, technology and engineering
- To identify and protect our natural and cultural marine heritage
- To promote public awareness and understanding

These goals were further explicated as policy objectives which included managing actions to:

- Ensure continuing marine ecosystem health
- Safeguard marine biological diversity
- Promote diverse, strong and sustainable marine industries
- Provide increased certainty and long-term security for all marine users
- Ensure the establishment of a representative system of marine protected areas

Underlying the policy was the intention to implement integrated and ecosystems-based oceans planning and management. This was to be primarily through the preparation and implementation of regional marine plans based on large marine ecosystems.

Institutions

Four institutions were put in place to promote an integrated management approach. Significantly, none of them included formal representation from state or territory governments, reflecting the failure to effectively engage this level of government in the policy development process.

A *National Oceans Ministerial Board of Commonwealth Ministers*, chaired by the Minister of Environment and Heritage, was established in 1999 to oversee and approve regional marine plans, coordinate cross-sectoral policy issues and develop priorities for government funding and research. Members included the Minister for Environment and Heritage, the Minister for Industry, Science and Resources, the Minister for Sport and Tourism, the Minister for Transport and Regional Services and the Minister for Fisheries.

The Board brought together Ministers across a range of government policy areas in a formal decision-making structure. In practice, the Board proved useful in promoting cross-sectoral thinking at a high level on issues affecting oceans. However, it did not

initially operate as intended, becoming bogged down with numerous operational matters rather than focusing on key strategic issues (Norton, 2002:16). This problem has been more recently addressed with the establishment in 2003 of an *Oceans Board of Management*, comprising heads of relevant Commonwealth departments, to deal with operational issues and oversee the work of the National Oceans Office.

A *National Ocean Advisory Group* (NOAG) was also established in 1999 with members from industry, science, conservation, and community and government stakeholders. It essentially carried on the role of MAGOP and was given the task of advising the Ministerial Board on cross-sectoral and cross-jurisdictional ocean issues, on the effectiveness of regional planning efforts, and on the views of a wide range of non-governmental interests.

In practice, the role of NOAG was somewhat unclear. Many members represented sectoral interests rather than a broader approach to oceans management. NOAG has acted both as an advisory body, bringing a range of expertise to bear on strategic issues, and as a sounding board where the Ministerial Board can test the reaction of major stakeholders to proposals. Trying to deal with both roles in the one entity has reduced its effectiveness, led it to adopt a reactive rather than proactive role, and resulted in confusion and frustration amongst some members (Norton, 2002:16).

The *National Oceans Office* was established in Hobart during December 1999 as an independent executive agency reporting to the Oceans Ministerial Board. A second, smaller office was subsequently established in Canberra and a third in Cairns to oversee the development of the Northern Regional Marine Plan. The National Oceans Office was a small agency with around 40 staff members. It spent approximately A\$15 million during its first three years of operation. The efforts of NOO were initially focused primarily on preparing the South-East Regional Marine Plan (SERMP), which was released in May 2004. With the preparation of SERMP completed, the major work focus is now shifting to the Northern Regional Marine Plan and the South-West Regional Marine Plan.

In practice, NOO has shown both significant strengths and weaknesses. On the positive side, the office has managed to recruit a group of talented, energetic and results-oriented staff, is perceived as being independent of the sectoral-focused departments in Canberra, has developed good links with a broad range of external stakeholders and has carried out ground-breaking work on planning for large marine ecosystems. On the negative side, the office has lacked a clear direction, has been dominated by technical and scientific viewpoints rather than policy thinking, has been weak on the social sciences and has failed to engage effectively with other Commonwealth departments (Norton, 2002:18).

There was some uncertainty over the role of NOO, which resulted in considerable time being spent on debating its role with stakeholders rather than getting on with the job. The initial focus on the preparation of one regional marine plan also meant that

NOO failed to engage with ocean managers and stakeholders in many other parts of the country.

Being a small executive agency with no statutory basis or large budget allocation, NOO was reliant on relationship building to achieve change. This proved to be a slow and uncertain process. In practice, NOO had little actual power to overcome institutional inertia or the predominance of sectoral approaches to ocean management, particularly when powerful and well-resourced sectoral interests mobilised to protect their 'patch'. In November 2004, NOO lost its independence and became a branch of the Department of Environment and Heritage.

The fourth type of institution provided for in the Oceans Policy was *Regional Marine Plan Steering Committees*. These are established by the Ministerial Board and are comprised of non-governmental and governmental regional stakeholders who are to work closely with the Ministerial Board to oversee the development of regional marine plans. In practice, the Steering Committee, which was established to oversee the development of the SERMP, has suffered from confusion about its role and there has been tension between the role of providing technical advice and the role of representing stakeholder interests (Norton, 2002:16).

More recently, during 2003, an *Oceans Policy Science Advisory Group* was established to provide scientific advice and support to the Oceans Board of Management. The Advisory Group consists of senior representatives of government and non-government agencies with a strong interest in marine science. It provides a forum for setting priorities for marine science and sharing information.

Actions

The Oceans Policy contained a large number of 'key initial actions' which addressed integrated oceans planning and management; conservation of marine biological diversity; oceans uses and impacts including marine industries, marine pollution, and marine tourism; community participation; understanding the oceans; protecting national interests; and assessing performance. The key mechanism identified for implementing an integrated and ecosystems-based approach to oceans management was regional marine planning.

This planning was designed to provide a structured and orderly process for the ecosystems-based allocation of resource access and use across and within sectors. It aimed to determine the conservation requirements of each region, including the establishment of marine protected areas. It also aimed to establish complementary management regimes in both state and Commonwealth waters. Regional marine plans were to (Commonwealth of Australia, 1998a:13):

- Identify ocean resources and economic and other opportunities

- Identify current and emerging threats to ecosystem health and determine planning and management responses to those threats
- Within the region, set out what is known of ecosystem characteristics and establish a broad set of objectives for those systems
- Identify the requirements and priorities for environmental baselines and basic biological inventories and other surveys in the development of Regional Marine Plans
- Identify priorities and put in place measures to meet conservation requirements and determine those areas that should be assessed for marine protected area declaration
- Identify community and sectoral interests, including the interests of Aboriginal and Torres Strait Islander communities
- Identify priorities for industry and economic development of the region
- Put in place a planning regime to prevent conflict between different sectors over resource access and allocation
- Provide a framework within which there is increased certainty and long-term security for marine-based industries
- Establish indicators of sustainability and requirements for monitoring, reporting and performance assessment

Although it was clear what tasks regional marine planning was to undertake, the Oceans Policy failed to provide a detailed framework for the implementation of such planning or a clear idea of what a regional marine plan might actually look like and what effect it might have in practice. This failure to clearly think through the implementation phase when the policy was developed proved to be a major obstacle to its subsequent successful implementation.

Implementation

As already indicated, the key mechanism for implementing the Oceans Policy was regional marine planning based on large marine ecosystems. The ocean areas of Australia were divided into 13 large marine domains.

The first plan prepared was SERMP, which covered an area of around 2 million square kilometres, at the junction of three major oceanic systems and bordering the coastlines of four states. The scale of the planning exercise was unprecedented anywhere in the world (S & J Woodley Pty Ltd, 2002:11). Around A\$9.3 million was spent on the planning process.

In April 2000, the National Oceans Advisory Group convened a National Oceans Forum to launch the SERMP initiative. The plan preparation process included the following three broad stages (Sullivan, 2002):

- *Scoping phase*, in which the objectives for the planning process were defined, as well as the nature of the assessments
- *Assessment phase*, which involved six assessment streams: biological and physical, indigenous, management and institutional arrangements, impacts on the natural system, uses, and community and cultural values.
- *Options development*, which included consultation with sectoral interests as well as key stakeholders, including indigenous communities and state governments

In November 2000, a Regional Marine Plan Committee for the South-East Region was established consisting of nine members from a broad range of disciplines but not including any representatives from the Commonwealth or state governments (Vince, 2005:4).

In practice, the development of SERMP proved far more difficult and complicated than anticipated. Much of the difficulty was owing to the lack of a clear framework, at the beginning of the process, indicating what the relationship between integrated planning and sectoral implementation would be. As a result, the planning process met with some scepticism in its early stages, resulting from a fear of the unknown and concerns that SERMP would simply offer another layer of bureaucracy (Foster and Haward, 2003:556). Even in 2002, when the planning process was some way down the track, there was still uncertainty amongst stakeholders as to what would be delivered, how it would work in practice and whether it would add value (Norton, 2002:3).

The initial stages of the planning process were dominated by a significant effort to generate and analyse data. During 2000, a project to map the seabed and assess the marine life in the deep waters of the region was initiated. The A\$2.7 million project was jointly funded by NOO, the CSIRO and the Australian Geological Survey (Vince, 2005:5). One interviewee observed that the data-gathering phase was essentially a 'data mining' exercise rather than being guided by policy or management objectives. However, NOO was grappling with how to apply an ecosystems-based approach to oceans management when it had only a very sketchy understanding of marine ecosystems, especially in the deep ocean areas. Considerable effort was put into bioregionalisation – characterising areas of the ocean into different habitat types. This was to be used as a basis for delineating planning units and developing a better understanding of ecosystem functions and linkages. Significant effort also went into mapping present uses of the oceans.

Despite the large effort put into this analysis, the data remained poor, leading a reviewer of the planning, scientific and technical support components of the Oceans Policy to note that the information base for undertaking bioregionalisation, and for identifying economic values and attributes of existing and future marine industries, needed to be treated with care and should be updated as new data was collected (S & J Woodley Pty Ltd, 2002:3).

Extensive public consultation was carried out during the development of SERMP, but without a clear guiding framework, leading to some confusion and frustration amongst stakeholders. Much of the consultation was carried out on detailed issues, without an overall context within which to place the information (Norton, 2002:3).

As the state governments were not on board before the Oceans Policy was launched, considerable effort was undertaken to re-engage them through the SERMP process. Because of the political impasse, state governments were not formally involved in the planning process, although government officials participated in a consultative committee, technical committees and information dissemination. State governments also provided much data which was used in the planning process. While the SERMP was launched as a Commonwealth government initiative, state governments have agreed to participate in its implementation where this requires cooperative action and addresses matters of mutual concern.

It is unclear how committed key Commonwealth agencies and other stakeholders are to implementing the Plan. A 2002 review notes 'an urgent need to improve the involvement and ownership of the planning and management processes (and model) by State and Commonwealth agencies and industries that are likely to have significant roles in implementation or be affected by it' (Norton, 2002:7). The NOO is attempting to address this problem in the Northern Regional Marine Plan process through developing a memorandum of understanding with the Northern Territory and Queensland state governments, describing the roles each will play in the planning process.

In March 2003, a group of Australian conservation organisations released a report on the implementation of Australia's Oceans Policy and ecosystems-based regional marine planning titled *Oceans Eleven*. It contained six core recommendations to strengthen the oceans policy initiative, which were (Smyth et al., 2003:6):

1. That ecosystems-based management is reinforced as being the heart of Oceans Policy and regional marine planning
2. That a National Oceans Act and a sufficiently empowered National Oceans Authority be created
3. That a National Marine Research Council is formed to review and fund an expansion of marine research
4. That the National Representative System of Marine Protected Areas (NRSMPA) is established with a core network of marine national parks that are of sufficient number and size to protect the range of marine ecosystems in Australia's care
5. That the Commonwealth and state governments make a commitment to community capacity building, education and engagement in regional marine planning
6. That the 11-step process for ecosystems-based regional marine planning (set out in the report) be adopted

The SERMP, which was released in May 2004, consisted of (National Oceans Office, 2004):

- A description of environmental, economic and social characteristics of the region
- A set of nine very broad objectives for the region which do little more than restate the objectives set out for regional marine planning in the oceans policy
- A description of current ocean uses, management of those uses and proposed future management actions
- A description of the conservation and protection of ecosystems, cooperative management and implementation, and review actions
- A description of current initiatives being undertaken to meet the objectives
- An action plan consisting of a tabulated list of actions, with a lead agency identified for the implementation of each
- A series of status reports outlining in more detail progress and future work on some of the actions. These include reports on adaptive management and performance assessment, a multiple-use management case study, pilot sea country plans, a decision support toolkit for estuarine managers, risk assessment, a regulatory efficiency review, and the establishment of Commonwealth marine protected areas.

This appears to be somewhat short of the original objectives for regional planning as identified in the Oceans Policy document. These included determining areas which should be assessed for marine protected area declaration, identifying priorities for industry and economic development of the region, and putting in place a planning regime to prevent conflict between different sectors over resource access and allocation. It may also fall short of the expectation of some stakeholders that expected SERMP, at the very least, to show zonings for some primary uses and to identify definitive boundaries for marine protected areas (Westcott, 2004). This divergence is indicative of a shift which occurred, during the development of SERMP, from developing a tangible 'plan' to developing an ongoing 'process' (Foster et al., 2005).

The main tangible outcome is in respect of Commonwealth marine protected areas. Eleven broad areas of interest have been identified throughout the region and the boundaries of two proposed protected areas are shown in the SERMP.

There were mixed reactions to SERMP from stakeholders. The World Wide Fund for Nature saw it as 'a useful first step', the Australian Conservation Foundation described it as 'a plan to create a plan' and the Australian Seafood Industry indicated that there was still a high level of uncertainty across the seafood industry as many of the actions within SERMP were still unknown (Vince, 2005:8).

The SERMP process has been comprehensive and has had many positive outcomes:

- It has mobilised considerable scientific effort to develop a better understanding of the regional marine environment
- It has enabled the identification and better recognition of existing use and access rights
- The National Oceans Office and a range of stakeholders now have a vastly improved knowledge of the ecosystems and their uses within the region
- There is considerably greater awareness amongst stakeholders of ocean management issues
- Stakeholders from a range of sectors and perspectives have begun talking to each other about ocean management issues and have a better understanding of each other's point of view
- The action plan, which specifies many actions which can be taken now, provides for tangible outcomes and early 'runs on the board'
- Broad areas of interest for marine protected areas have been identified as a basis for ongoing stakeholder engagement in identifying specific sites
- Methodologies to support oceans managers are being developed and tested
- The reflective nature of the initiative, with NOO involved in regular performance reviews, has facilitated much learning throughout the process, and this is being applied to the next regional marine plan being developed in the Northern Region
- A cadre of officials with expertise in integrated oceans management has been developed

Recognising the ground-breaking nature of SERMP, and the need to develop it within an environment of uncertainty, NOO intends to apply an adaptive management model to its implementation. This will include adopting a comprehensive performance assessment system; monitoring and reporting on ocean ecological, economic and social outcomes; and adopting mechanisms to continuously improve management as new information becomes available. The NOO is also developing a generic planning model and identifying success factors, drawn from its experience with the SERMP process, which can be applied to future marine planning exercises.

In 1993, in an attempt to map out a methodology for integrating sectoral management responsibilities, the Oceans Office released proposals for an *Integrated Oceans Process* (National Oceans Office, 2003). The process will apply to issues identified through strategic ocean planning, regional marine planning, feedback from monitoring, stakeholders' input or government priorities. The process consists of three steps. Step one involves applying a relevance test to determine whether an issue needs

to be dealt with through the integrated process. In step two a lead agency is identified, the information base and expert advice needed for decision making defined, and risks analysed. Step three involves strategy development.

The National Oceans Office has indicated that it will be developing *Ocean Guidelines* to provide practical assistance to oceans managers and users. The Guidelines are to address a range of issues including (National Oceans Office, 2003:4):

- The application of ecosystems-based management and the precautionary principle
- Implementation of user-pays principles
- The definition and use of indicators of ecologically sustainable development
- The assessment of cultural and socioeconomic values
- The application of information to decision making
- The evaluation of cumulative impacts

The National Oceans Office is also working cooperatively with state governments to develop a national approach to Integrated Oceans Management which will apply to agreed cross-jurisdictional areas. The Natural Resource Management Ministerial Council, which has Commonwealth and state government representation, is directing this initiative. It commenced in 2002 through the establishment of an Integrated Oceans Management Working Group as part of the Ministerial Council's Marine and Coastal Committee. It is focusing on developing a framework and institutional arrangements to address oceans-related issues as they arise (Foster et al., 2005). This initiative represents a valuable attempt to bypass the roadblock of state government non-participation in the Oceans Policy. It starts to bring oceans issues within the purview of a joint federal and state government decision-making body, rather than under the control of the federal government's Oceans Ministerial Board.

Work is also continuing on the preparing of the Northern Regional Marine Plan and a substantive amount of research has been undertaken to develop a better understanding of that region. Preparation has also begun for a South-West Regional Marine Plan.

Evaluation

Australia's Oceans Policy, as described in the 1998 document, represented an ambitious and comprehensive attempt to map out the path towards more effective management of Australia's oceans. It sets out a powerful vision for a new approach to oceans governance, based on applying an integrated and ecosystems-based approach to planning and management. The regional marine planning undertaken is world leading and has generated considerable learning. However, several problems with the

development and content of the policy created significant obstacles to its successful implementation.

State and territory governments were not successfully engaged in the policy development process and there was a failure to secure their endorsement of the policy and their agreement to implement it. This is widely acknowledged as being the biggest impediment to achieving the aims of the policy (Norton, 2002:8).

The document represents 'all things to all people' (Norton, 2002:8) and postpones the more difficult matters of integration, the resolution of conflicts between sectoral interests and the role of state and territorial government in the implementation phase (Westcott, 2000:873).

The Oceans Policy document proposes an integrated management approach and also proposes to maintain existing sectoral and jurisdictional management arrangements. It does not, however, identify how these two aims are to be achieved (Norton, 2002:8). In particular, it is not clear how conflicts between different interests will be resolved. The document points to regional marine planning as being the mechanism to resolve such conflicts, but this was not achieved in SERMP. One interviewee observed that, five years on, there is no clarification on what integrated management means for sectoral management and the resolution of sectoral conflicts. 'This is a difficult issue but goes to the very heart of the policy' (Westcott, 2004). Vince (2005) observes that 'it is informally acknowledged, particularly outside the NOO, that the aim of the Oceans Policy has changed from 'integration' to 'cohesion' of process across sectors and jurisdictions'.

The Oceans Policy postponed a definition of what regional marine plans would be, how they would achieve their objectives, and how they would be implemented. This made it difficult to develop the first regional marine plan within tight time frames. Preparing this first plan turned out to be difficult but it was part of a 'learning-by-doing' approach that, in the absence of statutory backing, may have been unavoidable.

On the positive side, considerable progress has been made on implementing the concept of integrated and ecosystems-based management at a regional level. The open and reflexive nature of the process has enabled high levels of learning to take place, which can now be applied to future regional marine planning processes. While it is still not clear whether regional marine planning actually works in practice, a much stronger understanding of the constraints and how to address them will be carried over into future planning.

Ultimately, the success of the policy will be dependent on 'whether the initiative stays with the existing powerful sectoral interests (industry, government departments, central government bureaucrats) or whether a constituency for integrated non-sectoral implementation of a comprehensive policy continues to develop and is sufficiently influential to offset pre-existing power elites and to establish new integrated institutional arrangements' (Westcott, 2000:868). This transition appears to have yet to happen.

B

Canada's Oceans Policy

Context

Canada has jurisdiction over about 4.7 million square kilometres of ocean and has the longest coastline of any country in the world, extending 244,000 kilometres and bordering three oceans: the Arctic, Atlantic and Pacific (Petrachenko, undated:1).

The Federal government has jurisdiction over the ocean in respect of such matters as marine navigation and shipping, international affairs, defence, environmental protection, the conservation and protection of migratory birds and commercial fisheries resources. Provincial governments generally manage non-renewable resources and own and manage the seabed in a significant proportion of coastal waters (Petrachenko, undated:2).

In 1970, Canada expanded its territorial sea limit from 3 to 12 nautical miles, and in 1977 put in place a 200 nautical mile fisheries zone (Dorman, 1996:118 & 120). This formally became an EEZ when the Oceans Act was passed in 1996. This Act also put in place a contiguous zone extending seawards 24 nautical miles. Canada was slow in ratifying UNCLOS and only did so in 2003.

Unlike populations in the other countries studied, Canadians are not predominantly coastal dwellers. Less than one-quarter of Canadians live in the coastal zone. However, the marine area is critical for trade and Can\$85 billion worth of goods and commodities were moved through Canada's national marine transportation system in 1998 (Petrachenko, undated:2). Oceans industries contribute well over Can\$20 billion a year to the Canadian economy. Major industries include commercial fisheries (Can\$2.1 billion), aquaculture (Can\$639 million), oil and gas production (estimated at Can\$2.3 billion in 1998), shipping, shipbuilding and tourism (Government of Canada, 2002a:14; Fisheries and Oceans Canada, 2003a:1).

Canada's marine ecosystems are relatively healthy by world standards. However, spoiled beaches, destroyed habitats, shellfish bed closures and contaminated sites can be found all around Canadian coasts (Fisheries and Oceans Canada, 2003b). Canada has also suffered from major fisheries collapses in the Atlantic and Pacific oceans.

Origins of national oceans policy

The Minister for Science and Technology announced Canada's first national ocean policy in 1973. This was a result of concern that Canada needed to develop the capability

to exploit offshore resources and to assert sovereignty in Arctic waters (Herriman et al., 1997). The Department of Fisheries and Oceans (DFO), a federal government department with a specific mandate to take charge of the oceans jurisdiction, was created in 1979 (Hildebrand and Chircop, 2000:3).

In 1986, the Federal government sponsored an oceans forum at the Institute of Ocean Sciences 'to discuss opportunities and strategies for developing the nation's oceans economy'. Representatives from industry, universities and government attended and resolved that ocean activities in Canada should develop in accordance with a 'comprehensive oceans strategy' (Herriman et al., 1997).

In 1987, DFO released a publication titled *Oceans Policy for Canada: A Strategy to Meet the Challenges and Opportunities on the Oceans Frontier*. This focused on the economic and technological opportunities that Canada's ocean space presented, together with the need to develop proper management for resource exploitation. It concentrated on specific sectors, particularly fisheries, contained no reference to the principles of sustainable development, and failed to provide for integrated coastal and ocean management. The document indicated that consideration would be given to an Oceans Act to consolidate and clarify the legislative basis of the new oceans strategy (Day, 1995:251; Hildebrand and Chircop, 2000:3).

In the late 1980s, a number of institutional arrangements were established to improve oceans management coordination. These included (Herriman et al., 1997; Hildebrand and Chircop, 2000:4):

- The establishment of an ad hoc *Ministers Oceans Group* to advise on the best mechanisms for ongoing industry consultation
- The establishment of an *Interdepartmental Committee on Oceans*, chaired by the Deputy Minister of Fisheries and Oceans, to improve interdepartmental coordination and raise the profile of the government's ocean-related efforts. A marine environmental quality initiative, operating under the auspices of this committee, developed a federal framework and action plan for marine environmental quality. Although this was approved interdepartmentally at the Deputy Prime Minister level in 1992, it did not proceed to a national strategy
- The creation of a *National Marine Council* with membership drawn from major sectors and constituencies of the Canadian ocean economy. Its purpose was to provide a forum for communicating and harmonising views and informing and advising the Minister
- The appointment by the Prime Minister of an *Ambassador of Marine Conservation*
- The drafting of an *Oceans Act*

These initiatives subsequently faltered. The National Marine Council ceased to meet in 1990 and was disbanded in the 1993 federal budget; the Interdepartmental Committee

on Oceans stopped meeting regularly after 1990; the marine environmental quality initiative declined during the 1990s; and the Oceans Act was not passed for another 10 years.

The 1990s has been referred to as a 'decade of disarray and fire-fighting in Canadian oceans' (Hildebrand and Chircop, 2000:2). Drastic depletion of fish stocks led to massive unemployment among the fishing population of the east coast. Between 1990 and 1994 Atlantic fisheries landings declined by nearly 50 per cent in volume. The northern cod fishery collapsed and was closed in 1992, affecting around 25,000 fishermen and plant workers in Newfoundland and prompting the government to introduce a Can\$484 million income replacement programme (Mitchell, 1998:403).

Canada has also experienced difficulties managing its fisheries on the Pacific west coast. Between 1990 and 1994 the volume of landings declined by 38.3 per cent, largely owing to a reduction in salmon landings caused by both overfishing and environmental factors. In March 1996, the Minister of Fisheries announced a Can\$80 million government buy-back programme to reduce the number of vessels in the salmon fleet by 50 per cent (Mitchell, 1998:405).

Some of these problems arose because of lack of coordination between federal and state programmes. The federal government used licensing to restrict entry and curtail fishing effort whereas, at the same time, provincial governments were subsidising vessel construction and processing plants to increase capacity (Day, 1995:252). In addition, scientific assessment of stocks was unsound and underestimated the effect of improved technology on commercial catches. Catch expansion was thought to be a result of healthier fish stocks rather than improved ways of catching fish (Day, 1995:253).

In addition to addressing problems with fisheries management, DFO was facing threats of illegal fishing by foreign vessels and an increase in illicit movement of goods and people. The need to address these issues, along with changes in key personnel at the DFO and substantial cuts in the department's size, resulted in the demise of the 1987 oceans policy initiative and a return within DFO to its traditional preoccupation with fisheries. After a brief four years of attempting to coordinate ocean policy making between 14 federal departments, the federal government returned to reactionary policy making (Day, 1995:251).

Interest in oceans policy development did, however, continue in other forums. In May 1994, the Committee on Oceans and Coasts of the National Advisory Board on Science and Technology recommended a comprehensive legal framework for Canadian oceans in its report *Opportunities from Our Oceans*. This was to include extending Canadian sovereignty over the EEZ and contiguous zones, extending environmental regulation to the EEZ and promoting international scientific research in support of a global resource management regulatory regime (Hildebrand and Chircop, 2000:4; Foster et al., 2005). In November 1994, the Minister for DFO released a document titled *A Vision for Oceans Management* and work started on preparing an Oceans Act.

The Act was finally passed by Parliament in December 1996 and came into force in January 1997.

Development of policy

Under the Oceans Act, the Minister of Fisheries and Oceans was given the task of leading and facilitating the development and implementation of a *national oceans management strategy* (Oceans Act 1996, section 29). The scope of the strategy was to be the management of 'estuarine, coastal and marine ecosystems in waters over which Canada has sovereign rights' and did not explicitly include land areas impacting on the ocean. The national strategy was to be based on the principles of sustainable development, integrated management and the precautionary approach (section 30). The preamble to the Act also refers to an ecosystems-based approach as being of fundamental importance to maintaining biological diversity and productivity in the marine environment.

The Minister was also tasked with leading and facilitating the development and implementation of *integrated management plans* (section 31) and a *national system of marine protected areas* (section 35) and was enabled to establish *marine environmental quality guidelines*, objectives and criteria (section 32(d)).

The Act strengthened the DFO through a merger with the Canadian Coast Guard. Its primary activities and organisational structure, however, have continued to be fisheries oriented (Mitchell, 1998:408). The Department was not allocated any additional funding for implementing the Oceans Act despite its enlarged mandate under the Act, and was forced to reallocate funds internally. Over eight years later, additional funds have still yet to be provided.

A discussion paper titled '*Towards Canada's Oceans Strategy*' was released in 1997 but failed to gain significant political momentum. It was another five years before a revised oceans policy was released.

Content of policy

Canada's Oceans Strategy: Our Oceans, Our Future was released in July 2002 and sets out the vision, principles and policy objectives for the future management of Canada's oceans (Government of Canada, 2002a). It was launched in a tentative way as a 'discussion paper' (Government of Canada, 2002b:1).

The strategy fleshes out the three principles in the Act:

- *Sustainable development* – which is to be implemented through integrated management planning

- *Integrated management* – which is to be implemented through the development of large-scale and local integrated management plans for all Canada's oceans, starting with priority areas. It also embodies integrated data collection, inclusive and collaborative governance structures and processes, flexible and adaptive management techniques to deal with uncertainty, and planning on the basis of natural and economic systems rather than political or administrative boundaries
- *The precautionary approach* – which includes the promotion of an ecosystems-based approach to management and the application of conservation measures, including the establishment of marine protected areas

Three policy objectives are identified in the document:

- *Understanding and protecting the marine environment.* Understanding involves integrating diverse and complex information, supplementing it with new research and sharing that knowledge with the Canadian public. Protecting includes the creation of a national network of marine protected areas and the establishment of marine environmental quality guidelines.
- *Supporting sustainable economic opportunities,* including the conservation and sustainable use of fisheries resources and the development of aquaculture
- *International leadership,* which is about advancing Canadian and global ocean-related interests broadly and proactively

Ocean governance was to be undertaken through working collaboratively within federal government and among various levels of government, sharing responsibility for achieving common objectives, and engaging Canadians in oceans-related decisions in which they have a stake. This was to be achieved through the establishment of intergovernmental committees and management boards, sharing information and undertaking integrated management planning.

Implementation

In 2002, the strategy discussion document was accompanied by a *Policy and Operational Framework for Integrated Management of Estuarine, Coastal and Marine Environments in Canada* (Government of Canada, 2002b). This was presented as a paper to explain how DFO is addressing its responsibilities for integrated management and to 'foster discussion' with other federal, provincial and territorial governments, aboriginal authorities and the general public. It defines an integrated management process for oceans and identifies as its principles ecosystems-based management, sustainable

development, a precautionary approach, conservation, duty in shared responsibility, flexibility and inclusiveness.

The document indicated DFO's long-term intention to establish a system of Large Ocean Management Areas and smaller Coastal Management Areas, which will eventually cover all the marine waters within Canadian jurisdiction. Initial efforts are to focus on areas currently under pressure or soon to come under pressure from human activities. Integrated management plans of various scales may be nested within a Large Ocean Management Area. The DFO indicated an intention to play a lead role in the process and to coordinate the development of Integrated Management Plans. The process of development is to include:

- Assessment of ecosystem characteristics that must be maintained to ensure the maintenance of the natural functions of the ecosystem
- Identification of ecologically-sensitive habitats and marine species and features in need of protection, including areas of interest for marine protected areas
- Assessment of existing open space uses and future economic opportunities.

As management plans are developed, integrated management bodies are to be established with a role in information, consultation and advice, but not decision making. They will be composed of both governmental and non-governmental representatives. Their main aim will be to formulate a plan of action for the management of activities that are in, or affect, a specific geographic area. All parties are expected to agree to carry out their respective responsibilities in accordance with the plan. Once plans are approved by mandated authorities, the management bodies may shift to an overseer role.

In its performance report for the period ending 31 March 2003, DFO reported allocating Can\$15 million to the implementation of the Oceans Strategy (Fisheries and Oceans Canada, 2003b). Progress in implementing the policy has included the following actions:

- The appointment of two *Ministerial Oceans Ambassadors* (2000)
- The creation of a *Ministerial Advisory Council on Oceans*, comprised of seven distinguished Canadians with oceans expertise and experience, to provide independent advice on oceans policy matters (2000). The Council meets at least quarterly with senior officials of DFO and/or Ministers.
- The launch of the *Oceans Management Research Network*, which is comprised of over 200 academic researchers (2001)
- The establishment of a federal-provincial-territorial *Oceans Task Group*, under the Canadian Council of Fisheries and Aquaculture Ministers, to provide an intergovernmental forum for discussing oceans issues of national significance (2001)

- The initiation of planning processes for four Large Oceans Management Areas and 21 smaller Coastal Management Areas (Fisheries and Oceans Canada, 2003b)
- In March 2003, the first marine protected area to be declared under the Oceans Act was announced: the Endeavour Hydrothermal Vents. In May 2004, a second area, the Gully Marine Protected Area, was designated, covering 2364 square kilometres. Progress had also been made on the establishment of at least 11 additional marine protected areas. The department of Fisheries and Oceans has also initiated the development of a federal strategy to achieve a national system of marine protected areas.
- In February 2004, the development of a federal Oceans Action Plan was announced in the Speech from the Throne (Foster et al., 2005)

Because of the lack of staff and financial resources, implementation of integrated management plans has been slow. One of the first pilot projects undertaken by DFO to apply the integrated management approach was the Eastern Scotian Shelf Integrated Management (ESSIM) Project, which was announced in December 1998. This project initially covered the entire Eastern Scotian Shelf, an offshore ocean management area of approximately 325,000 square kilometres, but subsequently incorporated coastal areas. It is a large marine system which provides important habitat for a number of species at risk, including the northern bottlenose whale, the leatherback turtle and the North Atlantic right whale. The area is also used for oil and gas extraction, commercial fishing, traditional fishing, shipping and defence. There are about 78,000 square kilometres of oil and gas lease areas and the rapid development of this industry was one of the main triggers for the ESSIM (National Round Table on the Environment and the Economy, undated).

The vision for the project is to '... have an effective, collaborative process that provides integrated and adaptive management plans, strategies and actions for social, economic, environmental, and institutional sustainability.' The four overarching objectives for the project are to (Rutherford et al., 2005):

- Integrate the management of all measures and activities in or affecting the area
- Manage for conservation, sustainability and responsible use of ocean space and marine resources
- Restore and maintain natural biological diversity and productivity
- Provide opportunities for economic diversification and sustainable wealth generation to foster social well-being for coastal communities and stakeholders

To date, major effort has concentrated on the collection and mapping of data from different sectors, with a focus on natural science data. This work has included assessing

contaminants, mapping benthic communities, undertaking zooplankton and nutrient studies, carrying out an inventory of existing marine environmental quality standards and guidelines, identifying ecosystem objectives and indicators, mapping key ocean activities and assessing noise levels and potential impacts on whales.

Supporting work for the eventual identification of marine protected areas has also been undertaken, including developing a systematic approach to mapping, classifying and assessing marine ecosystems and defining sensitive and important areas in terms of ecological parameters and the nature of human impacts.

Stakeholder engagement has been primarily through DFO holding informal, bilateral information and discussion sessions with different stakeholders, rather than through multi-stakeholder dialogue. These discussions focused initially on the Oceans Act and the key principles and approaches contained within it. Discussions have also focused on the scope and objectives of ESSIM, priority issues to be addressed, and proposed collaborative structures and institutional design. A discussion paper produced in November 2001 titled *Issues, Challenges and Opportunities* sought to bring together internal work carried out by DFO and the information gained during stakeholder dialogue, in order to identify the range of issues and challenges to be addressed (Rutherford et al., 2005).

In February 2002, the first *ESSIM Forum Workshop* was held to provide input into the initiative. Around 150 people attended. A second workshop was held in February 2003 at which discussion focused on the elements of the proposed ESSIM Ocean Management Plan and Forum.

Proposals to establish an *ESSIM Forum* are currently being considered (see Figure B.1). The Forum is expected to function as a networked structure for engaging and linking federal and provincial departments and agencies, First Nations, local authorities, user groups, community interests, non-governmental organisations, scientists and the general public. The Forum is likely to have the following elements (Rutherford et al., 2005):

- A multi-stakeholder *Oceans Management and Planning Group*, which will form the core of the Forum and provide opportunities for information sharing and advice
- A *Regional Committee on Ocean Management* comprised of federal and provincial government representatives, which will serve as a forum for information exchange
- A federal-provincial *ESSIM Working Group*, which will focus at an operational level on policy, management and regulatory coordination. This Group was established in January 2001 and comprises representatives from 20 oceans-related departments, agencies and boards. When the Forum is established, the Working

Group is likely to operate as a sub-committee of the Oceans Management and Planning Group.

- An ESSIM Forum *Secretariat*, which would support the activities of the Oceans Management and Planning Group. The Secretariat is currently operating within DFO to support the ESSIM initiative and provides support for the ESSIM Working Group.
- A number of issues-based *Working Groups*

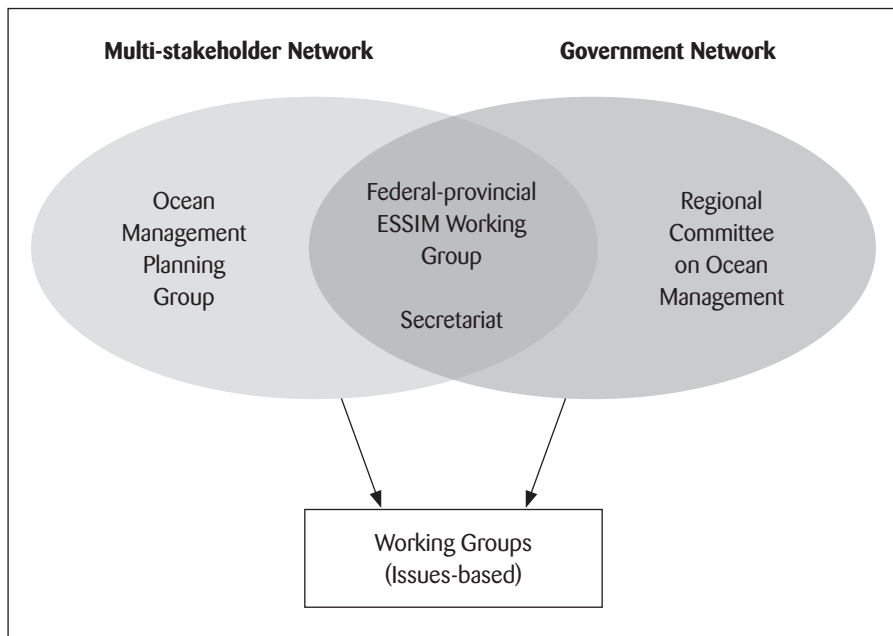


Figure B.1: Structure of proposed ESSIM Forum in Canada

Source: Rutherford et al., 2005

Work has been undertaken to scope the preparation of an ESSIM Oceans Management Plan. This is intended to be a five-year strategic plan aimed at integrating all activities impacting on the area. It is likely to include the following elements (Rutherford et al., 2005):

- A definition of the ocean management area covered by the Plan
- A definition of the purpose, scope and legislative basis of the Plan
- A management vision, goals, principles and approaches

- High-level management objectives to achieve environmental, social and cultural, economic and institutional sustainability
- Operational objectives, indicators and management strategies
- Ocean use planning measures such as zoning to address multiple ocean use
- An institutional planning framework of collaborative processes for the development and implementation of the plan
- Ongoing monitoring, performance measurement and adaptive management

It is proposed that the plan be approved by the Ocean Management Planning Group and the Regional Committee on Ocean Management. Implementation is to be facilitated through the preparation of annual action plans which will identify and prioritise issues. Individual agencies will be encouraged to undertake the required actions.

The ESSIM Project has had a number of positive outcomes (National Round Table on the Environment and the Economy, undated; Rutherford et al., 2005):

- Improved understanding about the Eastern Scotian Shelf and improved data sharing amongst government departments and other organisations
- The establishment of a multi-stakeholder forum and process for discussing and addressing conservation issues
- A more integrative and collaborative management approach being adopted by sectoral agencies
- The development of positive expectations that ESSIM will lead to more efficient and timely management and regulation as a result of the clarification of management roles and jurisdictions

Difficulties with the ESSIM Project have included (National Round Table on the Environment and the Economy, undated; Rutherford et al., 2005)):

- Difficulty in overcoming entrenched positions and mistrust among stakeholders
- Limited capacity of stakeholder groups to effectively participate in the initiative
- Difficulty in effectively coordinating government agencies, with ESSIM being regarded as a DFO initiative, and other agencies facing dwindling budgets and being concerned with operating under their separate legal mandates
- Lack of clear policy targets for the initiative, which may lead to the entrenchment of the status quo, and lack of clarity as to what the resultant plan will look like or do

- Slow process in designating marine protected areas and lack of any regulations under the Oceans Act defining what they should look like, how big they should be, or what restrictions should be put in place
- Difficulty adapting to the speed at which oil and gas exploration and development and other industrial pressures are proceeding, and establishing marine protected areas before conservation opportunities are foreclosed

The ESSIM Project was commenced prior to DFO developing a national framework for integrated oceans management, and the lessons learnt from the regional initiative have helped better define the national approach to regional marine planning, illustrating a bottom up 'learning by doing' approach.

Evaluation

The Oceans Act is cautious in attempting institutional or practical change. It focuses on the development of a national strategy, but fails to reduce the number of agencies and departments involved in the marine policy area, allowing the continuation of fragmentation and jurisdictional conflicts. The Minister of Fisheries and Oceans is a coordinator and facilitator rather than a supervisor of other Canadian departments and agencies (Juda, 2003a:173) and carrying out this role effectively has proven problematic. The lack of any additional funding being provided to DFO for implementing the Act has also been a significant hurdle to it being put into effect. A regional marine plan has yet to be prepared under the Act.

In October 2001, the Standing Committee on Fisheries and Oceans released its report on the Oceans Act. It concluded that the Act was fundamentally sound and did not recommend major amendments. However, it did express concerns about the implementation of some of its provisions. The report indicated that (Standing Committee on Fisheries and Oceans, 2001):

- There was a lack of tangible evidence of implementation of the Act
- There was slow progress in the establishment of marine protected areas, and confusion about the responsibilities of various agencies, as there were three different federal programmes for marine protected area establishment
- Responsibility for the management of Canada's oceans was becoming increasingly fragmented between different ministers, and DFO was struggling to perform its function as a lead agency for oceans matters. For example, new legislation was establishing other ministers as being responsible for issues such as oil exploration and marine conservation areas. This eroded the authority of the Minister of Fisheries and Oceans and undermined that minister's ability to lead a coordinated and comprehensive oceans management strategy.

C

The USA's Oceans Policy

Context

The USA has an EEZ of nearly 11.6 million square kilometres, an area just over 20 per cent larger than the nation's land area. It also has jurisdiction over a lengthy coastline, 133,312 kilometres long (Jacques and Smith, 2003:161).

State governments have jurisdiction over oceans inside the 3 nautical mile limit and over coastal land areas. The ocean area between the 3 nautical mile limit and the seaward boundaries of the 200 nautical mile EEZ is under federal government jurisdiction.

In 1976, the USA established a 200 nautical mile fishery conservation zone, which was converted to an EEZ in 1983. This was followed, in 1988, by the declaration of a territorial sea 12 miles offshore (Cicin-Sain and Knecht, 2000:3 & 78). Although the USA has adopted the oceans jurisdictions provided for under UNCLOS, it has yet to ratify the convention.

More than half the population lives in coastal areas which constitute only 17 per cent of the land area. This concentration of population on the coast is rapidly increasing. Over 40 per cent of new residential and commercial development is along the coast and the coastal population increases by around 3600 people each day. It is predicted that there could be an additional 27 million coastal residents by 2015 and by the year 2025 about 75 per cent of Americans may live in coastal areas. Half the country's gross domestic product, over US\$4.5 trillion, is generated in coastal watershed counties (US Senate, 2000:2; US Commission on Ocean Policy, 2002:3; 2004:2).

The ocean economy contributed US\$117 billion to the country's gross domestic product and supported well over 2 million jobs during 2000. The commercial fishing industry is valued at more than US\$28 billion annually. Significant oil and gas resources in the outer continental shelf contribute 30 per cent of domestic oil and 25 per cent of natural gas supply. Marine gas hydrates, which could be a potential long-term energy resource, have been identified for further evaluation. Coastal tourism, which generates around 180 million visitors annually, accounts for 85 per cent of tourism-related revenues (US Senate, 2000:3; US Commission on Ocean Policy, 2004:2–3).

Many marine and coastal habitats are threatened by human activities. Around 20,000 acres of coastal wetlands are disappearing each year. Current projections indicate that 50 to 60 per cent of coral reefs may be lost during the next 30 years. The Gulf of Mexico now suffers from a 7700 square mile dead zone, thought to be a result

of pollution flowing from the Mississippi River. There are growing numbers of red and brown tides, especially in mid-Atlantic estuaries bordered by intensive agricultural production (Cicin-Bain and Knecht, 2000:203; US Senate, 2000:4).

Around US\$200 million is lost annually in reduced fish catches owing to ongoing habitat loss. Only 23 per cent of the managed fish stock have been fully assessed, and over 40 per cent of these are considered depleted or are being fished beyond sustainable levels. Catches of groundfish off New England have sunk to levels one-third of what they were a decade ago, resulting in the loss of an estimated 20,000 jobs and US\$349 million. Decreasing salmon populations in the north-west have cost 72,000 jobs and more than US\$500 million (Cicin-Bain and Knecht, 2000:203; US Senate, 2000:4; US Commission on Ocean Policy, 2004:11).

The legal and bureaucratic framework related to the oceans has become increasingly complex. There is a patchwork of confusing and sometimes contradictory federal and state authorities and regulations, with no mechanism for establishing a common vision or set of objectives. Ocean conflicts have increased and are not easily resolved by existing policies (US Senate, 2000:3).

Origins of national oceans policy

The USA has a long history of concern about management of the ocean. In 1966, legislation (the Marine Resources and Engineering Development Act) was passed to enable the establishment of the Commission on Marine Science, Engineering, and Resources. The Commission, commonly known as the 'Stratton Commission' after its chair Julius Stratton, embarked on an unprecedented and, until 2001, unrepeated investigation of the nation's relationship with the oceans (US Senate, 2000:2).

The Commission issued its influential report, *Our Nation and the Sea*, in 1969. This report identified a number of basic defects in governmental efforts to manage the ocean, including responsibility for ocean activities being spread among a number of departments and agencies with overlapping jurisdictions, and important ocean-related programmes being placed in departments where they were seen to be of only marginal importance. The Commission recommended that ocean and coastal areas be managed in terms of 'systems', in which the interaction and interdependencies between different parts were recognised. It also recommended the creation of a new independent agency to provide a central and integrated focus for ocean-related programmes (Juda, 2003a:165–166).

The National Oceanic and Atmospheric Administration (NOAA) was subsequently established in 1970, not as an independent body as recommended by the Commission, but as part of the Department of Commerce. Although it consolidated some oceans-related activities, it was not the recipient of all relevant government ocean programmes. In 1971, the National Advisory Committee on Oceans and Atmosphere was created to

provide analyses and recommendations on oceans issues. It ceased meeting in the late 1980s primarily owing to lack of political support. In 1972, the Stratton Commission report also led to the passage of the Coastal Management Act, which encouraged states to adopt comprehensive coastal zone management plans (Juda, 2003a:166–167), laid the groundwork for the Fishery Conservation and Management Act in 1976 and established priorities for federal ocean activities that guided the nation for the next 30 years (US Senate, 2000:2; US Commission on Ocean Policy, 2004:50).

During the 1990s, various expert groups and panels articulated concern that a second review of ocean policy was required. In late 1997 and early 1998, a dozen teams, comprised largely of staff from federal agencies, prepared papers describing important oceans and coastal issues and the programmes in existence to deal with them. These provided background material for a series of workshops in mid-1998, organised by the Year of the Ocean Steering Group for the John H. Heinz Center for Science, Economics and the Environment. A report was issued out of this work, titled *Our Ocean Future: Themes and Issues Concerning the Nation's Stake in the Oceans* (Cicin-Bain and Knecht, 2000:204).

At the same time, work was undertaken on developing an Oceans Act. This Act, which was modelled on the legislation which set up the Stratton Commission, was passed by the Senate in November 1997 and by the House of Representatives in 1998. However, it was not considered by Congress before the October 1998 adjournment, was finally passed in 2000 and came into effect in January 2001 (Cicin-Bain and Knecht, 2000:205).

In 2000, prior to the Act coming into effect, the non-governmental Pew Oceans Commission was established to support the passage of the Bill, to promote public awareness of ocean issues and to develop recommendations for ocean governance reform and action.

Development of policy

The Oceans Act 2000 provides for the establishment of a new commission to make recommendations for a 'coordinated and comprehensive national ocean policy'. The commission is required to establish a multidisciplinary science advisory panel of experts, in the sciences of living and non-living marine resources, to assist with preparing the Commission's report (section 3(c)(3)). The Commission was to have submitted a final report to congress and the President by June 2003 (section 3(f)(1)). This time frame slipped, with the final report being adopted by the Commission in July 2004.

The Commission's report is required to include (section 3(f)(2)):

- An assessment of facilities (people, vessels, computers, satellites)
- A review of federal activities

- A review of cumulative effects of federal laws
- A review of the supply and demand for ocean and coastal resources
- A review of the relationships between federal, state and local governments, and the private sector
- A review of the opportunities for investment in new products and technologies
- Recommendations for modifications to the laws and/or the structure of federal agencies
- A review of the effectiveness of existing federal interagency policy coordination

Within 90 days of receiving and considering the report, the President was required to submit to congress a statement of proposal to implement or respond to the Commission's recommendations (section 4(a)).

The inaugural meeting of the Commission was held in September 2001. Four committees were established: governance; stewardship; research, education and marine operations; and investment and implementation. The Commission commenced with a fact-finding phase in November 2001, which included nine regional meetings in coastal areas. A year later, the Commission started exploring various policy options and held public meetings in January 2003 and April 2003 to debate proposed options. It released a preliminary report in April 2004, which was open to submissions. The final report was adopted three months later in July 2004. In December 2004, President Bush submitted his response to the recommendations of the Commission in the form of the *U.S. Ocean Action Plan*.

In a parallel initiative in May 2003, the Pew Oceans Commission released a report titled *America's Living Oceans: Charting a Course of Sea Change*, which was framed as 'A Report to the Nation'. The Commission's 18 members spent two and a half years travelling around the country to meet with various interested parties and hosted a series of workshops. The focus of the Pew Commission was narrower than the US Commission on Ocean Policy, being concentrated on environmental issues rather than broader economic and social concerns.

Content of policy

The US Commission on Ocean Policy developed guiding principles for the development and implementation of the oceans policy. These include sustainability, stewardship, ocean-land-atmosphere connections, ecosystems-based management, multiple-use management, preservation of marine biodiversity, best available science and information, adaptive management, understandable laws and clear decisions, participatory governance, timeliness, accountability and international responsibility.

Ecosystems-based management was to be implemented at a regional level. Regional management areas would incorporate large marine areas broadly based on

eight currently delineated for regional fishery management councils. The areas were proposed to extend from land-based water catchments to the outer limits of the EEZ.

Governance recommendations proposed by the Commission in its report included (US Commission on Ocean Policy, 2004:Chapter 4):

- A designated *Assistant to the President* – to provide leadership and support for national ocean policy. The Assistant would chair the National Ocean Council and co-chair the Presidential Council of Advisors on Ocean Policy.
- An *Office of Ocean Policy* – to be located within the Executive Office of the President. The Office would provide support to the Assistant, the National Ocean Council and the Presidential Council of Advisors.
- A *National Ocean Council* – which would be chaired by the designated Assistant to the President and be composed of cabinet secretaries of government ocean agencies and directors of independent ocean agencies. It would provide high-level attention to ocean issues, develop and guide the implementation of appropriate national policies, and coordinate the many federal departments and agencies with ocean and coastal responsibilities.
- A *Presidential Council of Advisors on Ocean Policy* – made up of coastal governors and other appropriate state, local and tribal government representatives, as well as the private sector, the research community and non-governmental and watershed organisations. It would be co-chaired by the Assistant and a nonfederal member.
- A *Committee on Ocean Science, Education, Technology and Operations* – which would report to the National Ocean Council and coordinate ocean-related education and research
- A *Committee on Ocean Resource Management* – which would include undersecretaries and assistant secretaries of departments and agencies which are members of the National Ocean Council. The purpose of the Committee is to better integrate the resource management activities of ocean-related agencies.

Regional Ocean Councils were proposed to be primarily coordinating bodies with authoritative action remaining the responsibility of official agencies under existing statutes. These voluntary and flexible bodies were designed to represent regional, state, territorial, tribal and local interests. The Councils were to be charged with bringing the goals of ecosystems-based management and improved coordination to bear on decision making, while lessening bureaucratic roadblocks. Possible roles for the Councils included (US Commission on Ocean Policy, 2004:58):

- Designating ad hoc sub-committees to examine issues of regional concern
- Mediating and resolving disputes among entities within the region

- Developing more formal mechanisms for implementing decisions such as interagency agreements
- Monitoring and evaluating the state of the region and the effectiveness of management efforts
- Building public awareness about regional ocean and coastal issues
- Actively soliciting stakeholder participation and leading the design and implementation of marine protected areas
- Facilitating the required government approvals or permitting processes which involve several agencies

The Commission's recommendations did not explicitly include the preparation of regional marine plans, but did include the establishment of regional ocean information programmes administered by regional boards. These programmes were designed to:

- Help fill gaps in current research
- Oversee the operation of regional ocean observation systems
- Translate ongoing regional research and data collection into useable products
- Educate decision makers
- Undertake regional ecosystem assessments

The Commission's report also contained specific recommendations for improving the coordination of federal oceans management efforts. These included identifying a federal lead agency for each current and foreseeable use of federal waters; establishing a coordinated, ecosystems-based offshore management regime; and developing national goals and guidelines, leading to a uniform process for the effective design and implementation of marine protected areas.

Implementation of the Commission's recommendations was not expected to be inexpensive, being estimated to require US\$1.5 billion in the first year of implementation and increasing to US\$3.9 billion per year in ongoing costs after full implementation.

These recommendations were similar to, but somewhat weaker than, those proposed by the Pew Oceans Commission, which included (Pew Oceans Commission, 2003:33–34):

- Enacting a *National Ocean Policy Act* requiring all agencies to 'protect, maintain, and restore marine and coastal ecosystems'. It should also provide clear and measurable goals and standards to govern activities affecting the oceans, establish mechanisms to ensure compliance with national policy and establish national and regional institutions capable of carrying out the policy.

- Establishing *Regional Oceans Ecosystem Councils* consisting of federal, state and tribal representatives. The councils would be charged with developing and overseeing the implementation of enforceable regional ocean governance plans. These plans must include performance goals and indicators, be binding on parties and meet standards established under the proposed Act. Each council should establish permanent advisory committees. It is envisaged that the plans would incorporate ocean zoning.
- Establishing a *national system of marine reserves*, with the federal government protecting areas of national significance and the councils creating marine reserves in their area of jurisdiction
- Establishing an independent *National Oceans Agency*, outside the Ministry of Commerce, which consolidates federal oceans programmes under one roof
- Establishing a permanent interagency *Oceans Council*, within the Executive Office of the President, chaired by the head of the National Oceans Agency. Membership would include heads of federal departments or agencies whose activities have a significant effect on oceans. Duties would include coordinating the implementation of the National Ocean Policy Act, resolving interagency disputes and coordinating agency ocean budgets. A position of National Oceans Adviser should also be established in the Executive Office of the President.

The *U.S. Ocean Action Plan*, developed by the President in response to the Commission's report, includes immediate action on around 40 of the report's 212 recommendations. The Plan established a new oceans governance structure, which includes (Tibbetts, 2005; Bush Administration, 2005; see Figure C.1):

- A cabinet-level *Committee on Ocean Policy*, which will advise the President and agency heads on the establishment or implementation of policies concerning ocean-related matters. The Committee is to facilitate the development and implementation of common principles and goals for governmental ocean activities, the use of science in ocean policy and the collection and use of information on ocean-related matters. The Committee has also been charged with developing an 18-month work plan to address some of the Commission's recommendations.
- An *Interagency Committee on Ocean Science and Resource Management Integration* within the Committee on Ocean Policy, which will consist of Under- and Assistant Secretaries from relevant executive branch agencies and departments. The functions of the Committee include to coordinate and integrate activities of ocean-related federal agencies and develop strategies to resolve conflicts; to address gaps and new emerging ocean issues; to ensure the effective use of science in ocean policy; to periodically assess the state of the oceans; and to

make recommendations to the Committee on Ocean Policy on developing and carrying out national ocean policy.

- *Sub-committee on Integrated Management of Ocean Resources* consisting of representatives from relevant executive branch agencies and departments. The Sub-committee will focus on the management of living and non-living marine resources.
- *National Science and Technology Council (NSTC) Joint Sub-committee on Ocean Science and Technology* consisting of representatives from relevant executive branch agencies and departments. The Joint Subcommittee was established in 2003 and focuses on the development and application of ocean science.
- *Ocean Research Advisory Panel*, which expands the existing Panel and will consist of 10 to 18 members representing universities, research institutions, ocean policy organisations, state governments, ocean industries, educators and others. Its purpose is to provide independent advice and guidance to the Interagency Committee on Ocean Science and Resource Management Integration.
- *National Security Council Policy Coordinating Committee on the Global Environment*, which was established in 2001 and has a Sub-committee on Oceans Policy which acts as a forum for interagency coordination on international oceans issues.

While this governance structure does not match the one proposed by the Commission, it effectively implements many of elements. What is most noticeably missing is a designated Assistant to the President and an Office of Ocean Policy located within the Executive Office of the President.

A wide range of actions are listed in the Action Plan, including promoting greater use of individual fishing quotas, building a global earth observation network, developing an ocean research priorities plan and implementation strategy, implementing coral reef local action strategies and supporting a regional partnership in the Gulf of Mexico. Significantly, the Action Plan indicates the Bush Administration's support for USA accession to UNCLOS and for the National Oceanic and Atmospheric Administration Organic Act, which seeks to provide a legislative basis for NOAA which currently exists under an Executive Order.

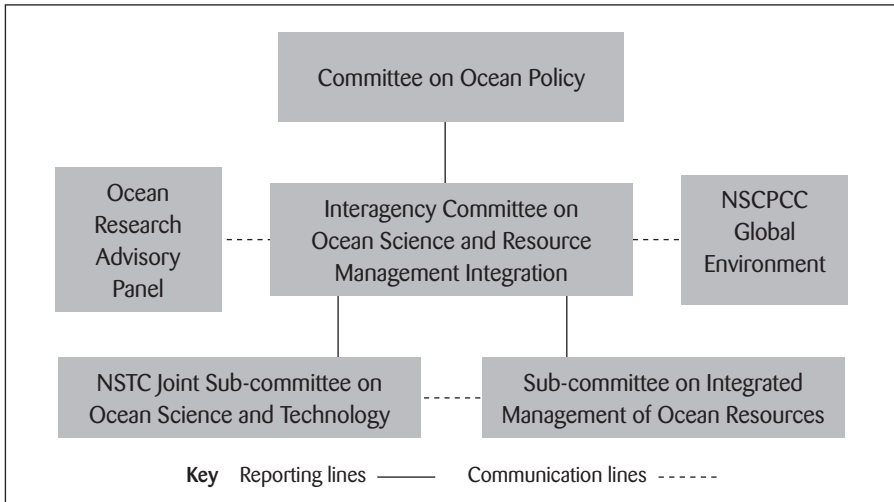


Figure C.1. New USA ocean governance structure

Source: Bush Administration, 2005:10

Implementation

As this publication goes to press, it is very early days for the implementation of the U.S. Ocean Action Plan, which was released less than a year ago. However, there are indications that the Bush Administration has not placed high priority on funding the Plan's implementation. The only new funding announced when the Plan was released was US\$2.7 million to be requested in the fiscal year 2006 budget for coral reef improvements (Tibbetts 2005). However, counteracting this small funding increase is a proposed cut of more than US\$391 million to the funding of NOAA's ocean and coastal programmes in the 2006 budget. This can be contrasted with the estimation of the US Commission on Ocean Policy that US\$1.5 billion funding would be required to implement the oceans policy during its first year (Rahall, 2005).

Evaluation

Initial reactions to the U.S. Ocean Action Plan have been mixed. Many participants at a stakeholder roundtable meeting, convened by NOAA in January 2005, saw the Action Plan as a positive first step, with the establishment of the cabinet-level Committee on Ocean Policy playing an important role in creating greater visibility for oceans issues,

coordinating federal agency activities and targeting budget needs. However, a number of criticisms of the Plan were expressed at the meeting, including (NOAA, 2005):

- The lack of a clear vision statement
- Failure to provide new funding for ocean-related activities
- Failure to recognise the need to work with Congress to set a national policy agenda
- The Plan being a repackaging of existing initiatives and low-cost actions rather than an effort to move towards real change

There is some uncertainty as to whether a sufficiently strong constituency exists in the USA to support the successful implementation of an oceans policy. The US Commission on Ocean Policy expressed concern in its mid-term report as to whether there was an adequate sense of national urgency to implement a coordinated and comprehensive national policy and to address the challenges (US Commission on Ocean Policy 2002: 2). This hesitancy has been borne out by the Bush Administration's response to the Commission's recommendations, which has been to adopt an incremental approach to implementation. The Action Plan identifies discrete changes which can be implemented relatively easily in the short term and sets up a process which can be used to tackle the harder problems (Tibbetts 2005).

D

New Zealand's Oceans Policy

Origins of national oceans policy

A key turning point in the government's involvement in marine affairs was the commissioning of a study in 1992 in response to the decommissioning of the marine research vessel *Rapuhia*. The resulting report, titled *Our Ocean: A Wealth of Opportunities – Research Vessel Needs for the 21st Century*, drew attention to the potential wealth of New Zealand's relatively unexplored oceans. In late 1993, after considering the report, Cabinet directed a group of officials to explore the issues raised, and work was carried out between 1994 and 1996. Topics investigated included Crown-owned research vessels, hydrography, seabed cadastral responsibilities, marine scientific research, Law of the Sea issues and general marine policy matters (Helm, 1998:251).

In February 1998, during the United Nations Year of the Ocean, the Environment and Conservation Organisations of New Zealand (ECO) hosted the *Seaviews – Marine Ecosystems Management Obligations and Opportunities* conference to address issues of oceans management. The conference addressed a range of issues related to applying an ecosystems approach to the management of human impacts on the marine environment (Wallace et al., 1998).

Just over a year later, in March 1999, the Ministers for the Environment, Conservation and Biosecurity directed officials to investigate current arrangements for the management of New Zealand's marine environment. The issues identified required a whole-of-government approach and the Department of the Prime Minister and Cabinet was given the responsibility of managing the ongoing project (Risk, 2002a:2). Later that year, in October 1999, the University of Canterbury-based Centre for Advanced Engineering hosted the *Our Oceans* conference, which further considered ocean management and governance issues (Risk, 2002a:2).

Shortly thereafter, in December 1999, the Parliamentary Commissioner for the Environment (PCE) presented to Parliament the report titled *Setting Course for a Sustainable Future: the Management of New Zealand's Marine Environment*. As already indicated in Chapter 13, this report identified a number of problems with the current system for managing New Zealand's oceans and recommended the establishment of a Coastal and Oceans Task Force to develop a strategy comprising, firstly, goals and principles and, secondly, actions and policies for the future sustainable management of New Zealand's marine environment (PCE, 1999:96–99).

On reading the Parliamentary Commissioner's report, the Prime Minister was reportedly moved to take action to prepare an oceans policy (Hodgson, 2000:6). In March 2000, she appointed the then Minister of Fisheries, Energy and Research, Science and Technology, the Honourable Pete Hodgson, to take over responsibility for the development of an oceans policy for New Zealand. In July 2000, Cabinet endorsed the proposal to prepare the policy (Risk, 2002a:2).

Development of policy

The development of the oceans policy was formally initiated in July 2000 when cabinet established an ad hoc group of six Ministers to oversee the policy development process. The group included the Ministers of Fisheries, Foreign Affairs and Trade, Conservation, Māori Affairs, Commerce and Environment. Cabinet directed that the project was to focus on managing the marine environment within New Zealand's jurisdiction and on the interaction between land management and the status and quality of the marine environment and the intertidal zone. It was to identify clear goals and principles and provide an integrated framework for managing the oceans (Risk, 2002a:2).

The policy development process was initially designed to have three stages. The first stage involved developing a vision. Stage two focused on designing policies to achieve this vision. Stage three was to deliver the policies, processes and tools identified in stage two as being necessary to achieve the vision (see Figure D.1).

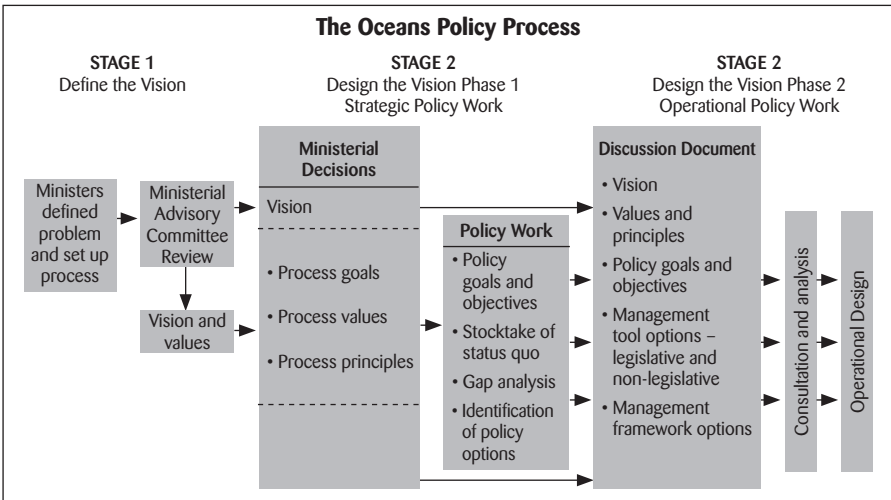


Figure D.1. Initial process to develop New Zealand's oceans policy

Source: www.oceans.govt.nz

A Ministerial Advisory Committee was appointed by Cabinet in March 2001, with the task of assisting 'Ministers to Define the Vision by managing and leading the process of identifying the shared vision, goals and objectives of New Zealanders for managing New Zealand's oceans' (Ministerial Advisory Committee on Oceans Policy (MACOP), 2001:42). Between June and August 2001 the committee undertook an extensive consultation process throughout New Zealand, including 47 public meetings and 24 hui (traditional consultative meeting with Māori) attended by around 2000 people. The committee also received 1160 written submissions (MACOP, 2001:3).

In September 2001, the committee produced a report titled *Healthy Sea: Healthy Society; Towards an Oceans Policy For New Zealand*. It indicated a high degree of interest in the idea of preparing a comprehensive national policy which would define a vision and develop an overarching management framework (MACOP, 2001:3).

A cross-departmental officials group was established which was led out of the office of the Minister in charge of the oceans policy initiative (the Honourable Pete Hodgson who was then the Minister of Fisheries) but which had members still physically located within their individual departments. This group worked on developing a vision and establishing process goals, values and principles for the development of the oceans policy (Risk, 2002b:7).

The Cabinet adopted a vision, goals and principles to guide the policy development process. The original vision statement (which was subsequently amended) was:

Healthy Oceans: New Zealanders understand marine life and marine processes and accordingly take responsibility for wisely managing the health of the ocean and its contribution to the present and future social, cultural, environmental and economic wellbeing of New Zealand.

This was supported by four goals:

- New Zealanders having confidence in, supporting and participating in the wise management, stewardship and sustainability of New Zealand's oceans
- Ecological integrity and abundant biodiversity for New Zealand society now and in the future
- New Zealand's oceans providing the best value for New Zealand society now and in the future
- A framework identifying the nature and extent of the rights and responsibilities of each Treaty partner in relation to the marine environment

Requirements identified for the policy framework included that it must:

- Clearly identify goals in relation to social, economic, environmental and cultural aspirations
- Clearly define roles and responsibilities
- Have clearly defined accountabilities
- Promote transparency
- Promote collaboration and constructive relationships between participants
- Require consideration of the best way to achieve the desired outcome whilst minimising cost
- Take account of relevant government policy and international obligations
- Decide where and how decisions are made and implemented
- Establish connections with other related policy frameworks
- Recognise the various sources of knowledge and information
- Be responsive to the changing context in which marine management issues arise – both domestically and internationally
- Promote a growing and sustainable economy that contributes to the social, cultural and economic well-being of New Zealanders
- Promote a healthy marine environment with strong natural character
- Take account of existing rights, interests and policies relevant to managing the marine environment and identify the implications and costs of any change to those rights or policies by which such rights are managed
- Assist in increasing the general understanding of the marine environment
- Encourage effective generation of, access to and use of information about the marine environment

These goals and requirements have since been reformulated into three conditions, three obligations and 12 operating principles.

To drive the initiative forward, a small dedicated Oceans Policy Secretariat was established within MfE. Although many of the Secretariat's staff had experience in policy development, they had little or no background in ocean science or management issues, although one staff member had been involved in fisheries policy. Several had been involved in the development of the government's climate change policy, which also involved a whole-of-government approach.

The work of the Secretariat was overseen by an Ocean Policy Steering Group of officials chaired by the Deputy Chief Executive of MfE, a Chief Executives Group comprising the heads of key agencies and, in turn, the ad hoc Ministerial Group, which was tasked with providing overall leadership and direction for the project. In March 2003, the Secretariat consisted of 10 staff seconded from the Ministries of Environment, Fisheries, Conservation, Economic Development and Te Puni Kokiri. Of these, seven were policy analysts with the other three acting in support roles in relation

to communication, business administration and project management (Oceans Policy Secretariat, 2003c:7).

The Oceans Secretariat was given the directive to deliver a draft policy within nine months with a budget of around NZ\$1 million (Risk, 2002c:9). The tight time frame appeared to have resulted from ministerial frustration at the slow pace of policy development during the previous two years and the wish to get something done. As well as focusing on delivering the policy within the stipulated time frame, the Secretariat also aimed to obtain the support of key stakeholders for the policy and to deliver a policy which was tangible and practical, so that people could clearly understand what impact it would have. The focus of the Secretariat was on identifying weaknesses in the oceans management system overall, and in processes within that system, rather than on identifying problems within specific sectors or drawing together information on key threats to the marine environment or ocean uses (Sheppard, 2004).

The first piece of work commissioned by the Secretariat was a stocktake of legislation, strategies and international instruments impacting on oceans. This was completed in December 2002 and identified the following weaknesses in the overall oceans management system (Willis, Gunn and Hill, 2002:1–2):

- Absence of an overriding goal
- Inconsistent decision-making structures and opportunities for participation
- Inconsistent management of 'like' activities (and potential effects), particularly beyond the territorial sea
- Uncertainty about Māori rights in relation to the oceans
- Ecologically arbitrary spatial management units and a general lack of integrated management
- Lack of information

The Secretariat subsequently sought to identify the issues that the oceans policy would need to address. It opted for a 'qualitative' rather than 'evidence-based' approach to the issues identification. This was owing to significant gaps in the existing information base on the state of the marine environment and activities and uses occurring in it, and a lack of time and resources to fill these gaps. The process therefore heavily relied on input from stakeholders (central government, local government, industry and other groups). This input was summarised by Secretariat staff in 11 working papers, which explored specific issues that had been raised during the stage one consultative process. The style and content of the working papers, which were descriptive, heavily summarised and contained little quantification, reflected the nature of the preparation process.

During March 2003, a series of three workshops and one hui were held with key stakeholders to discuss the material in the working papers. These papers were also

made available for stakeholder comment during March and April 2003, and feedback was received from 31 stakeholders.

As a result of this process, five key issues which the policy solutions would focus on addressing were identified. These paid particular attention to the overall management system as follows:

- The need for more effective integration of activities, policies, agencies and legislation
- The need for improved coastal and marine environmental management and performance
- The need to reconcile competing uses of marine space and resources, and to provide for new uses
- The need for more effective coordination and generation of marine information
- The need to ensure that obligations under the Treaty of Waitangi are met

Although different stakeholders had different views about what was most important and what priority each of these issues should be given, there appeared to be general consensus that these, collectively, comprised the key issues that needed to be addressed (Sheppard, 2004).

The Secretariat then undertook a series of brainstorming sessions to develop potential policy solutions to address these issues. This process was informed by considering different theoretical approaches to dealing with the problem of integration, and the Canadian and Australian experiences of developing oceans policy, which indicated that both a strong legislative basis and a sound planning process were required to deliver the desired outcomes (Sheppard, 2004).

In April 2003, a Māori Working Group of eight people was formed to assist the Secretariat in the development of the policy package and to provide advice on engagement with Māori. The Secretariat found this group to be very helpful (Sheppard, 2004).

Two further pieces of work were commissioned by the Secretariat to inform the policy development process. A report on oceans issues facing local government and staff in DoC's conservancies was released in May 2003 and a report on economic opportunities for New Zealand's oceans in June of the same year.

The first report described the findings of a survey of 13 local authorities and identified that they did not have a strong strategic focus on oceans, that ocean issues were seldom given high priority by local authorities and that, in general, relationships with relevant central government authorities were not strong. The report also identified the key barriers to better environmental management as being a lack of strategic focus and the inherent difficulties in providing for use and development in an environment

in which the accommodation of one objective inevitably comes at the expense of another (Willis, 2003:1–2).

The Secretariat prepared a discussion document for approval by Cabinet, outlining the policy options and proposed solutions, and planned to carry out broad public consultation on the preferred policy options during September and October 2003. However, before the paper could go to Cabinet, the policy process came to an abrupt halt. In June 2003, the Court of Appeal released a decision finding that eight iwi in Marlborough could have their claim to a stretch of foreshore and seabed heard by the Māori Land Court. This created the potential for customary rights over areas of the foreshore and seabed to be recognised through private title. The possibility that private owners might control access to and use of the foreshore and seabed was contrary to the commonly held view that these areas were governed by the Crown for the benefit of all. The day after the Court decision, the Government announced that it would pass legislation clearly vesting ownership of the foreshore and seabed in the Crown. This created significant Māori opposition and the partial resolution of the issue took some time.

Consultation on the oceans policy was suspended until the seabed and foreshore issue had been resolved. Legislation in the form of the Foreshore and Seabed Act 2004 was passed by Parliament in November 2004 and clarified that the foreshore and seabed, which were not subject to a freehold interest, were owned by the Crown. Now that the ownership issue has been resolved, the oceans policy is likely to be reconvened in early 2006. Somewhat ironically, the oceans policy process has been derailed, at least temporarily, by one of the issues it identified as needing attention: uncertainty about Māori rights in relation to the ocean.

In the interim, the cross-governmental Oceans Policy Secretariat has been disbanded and oceans issues are being dealt with by MfE. Two projects have been undertaken in the interregnum stage: a project to develop an information-based framework for setting priorities for oceans management and research, and a project to integrate processes for regulating activities beyond the territorial sea (including environmental impact assessment requirements). Reports setting out the results of these investigations were released by MfE in June 2005 (MfE, 2005a and 2005b).

Possible content of policy

No proposals for the content of the New Zealand oceans policy have been approved by Government as yet. However, public presentations given by the Secretariat have indicated that a proposed solution may include the following elements:

- The statement of a *vision* for the oceans and the establishment of a set of overarching objectives to govern management of the oceans. These are likely to

consist of two types: those which are non-negotiable and establish a 'bottom line' and those which can be traded off or balanced against each other. The 'non-negotiable' objectives are likely to include sustaining ecosystem health, meeting obligations under the Treaty of Waitangi and meeting international obligations. 'Balancing' objectives are likely to relate to providing for public use, access and enjoyment, economic return and enhanced environmental protection.

- The establishment of a set of overarching *decision-making principles* which relate to achieving transparent and participatory decision making; managing within the context of uncertainty; respecting existing rights, interests, values and future options; and avoiding, remedying or mitigating adverse environmental effects
- The development of a *National Oceans Plan* which sets national priorities, baseline ecological standards and processes and tools for reconciling competing uses

Other specific measures to be adopted may include:

- An information strategy
- A revised approval regime for beyond the territorial sea
- A register of marine rights and interests
- Capacity building for local government, Māori and the community
- An economic opportunities programme
- Public awareness work
- Improved devolution tools for Māori
- Improved support for existing work programmes relating to oceans
- Improved surveillance and enforcement
- Improved protection of marine cultural heritage

These elements could be implemented through:

- An *Oceans Act*
- Appointment of an *Oceans Minister*
- Establishment of an *Oceans Agency*, which could prepare and administer the Oceans Act and National Oceans Plan, monitor and review Act and Plan implementation, coordinate key cross-government projects, carry out a 'future watch' role, coordinate information and advice to ocean managers and users, and align international and domestic policy
- Establishment of *Māori advisory groups* to inform the development of Oceans Policy tools and to hold regular hui to connect to the 'flax roots' level

Policy and delivery for specific activities would remain the responsibility of existing agencies.

Evaluation

The New Zealand exercise to develop a national oceans policy put a large amount of effort into upfront public consultation in the absence of any policy framework. Although this was effective in developing some consensus over what was important, it raised expectations and gave little direction as to the way forward.

The policy development process was carried out at a very high and qualitative level but was light on scientific information and factual analysis. No quantitative assessment was made of the severity of oceans management problems or the identification of 'hot spot' areas which might require priority action. This reflected the lack of readily available quantitative information and the shortage of time and resources to collect it.

This lack of high-quality information means that the conclusions reached during the policy analysis process are only a rough 'best guess' and will need to be verified and adjusted as new information becomes available. It has also limited the extent to which stakeholders have become better informed about the issues and possible ways forward as a result of the process to date. It is proposed to develop a framework for adaptive management based on prioritised, shared and accessible data gathering and analysis (Sheppard, 2004), but details of how such an approach might be implemented are not yet available.

There is still some way to go in the policy development process and it is unclear what the policy will contain, how it will be implemented and how effective integration will be achieved across the ocean management system.

References

- Anderson O F, 2004a, 'Fish discards and non-target catch in the fisheries for southern blue whiting and oreos', *New Zealand Fisheries Assessment Report 2004/9*.
- Anderson O F, 2004b, 'Fish discards and non-target catch in the trawl fisheries for arrow squid, jack mackerel, and scampi in New Zealand waters', *New Zealand Fisheries Assessment Report 2004/10*.
- Anderson O F, D J Gilbert and M R Clark, 2001, 'Fish discards and non-target catch in the trawl fisheries for orange roughy and hoki in New Zealand waters for the fishing years 1990–91 to 1998–99', *New Zealand Fisheries Assessment Report 2001/16*.
- Andrew N and M Francis, 2003, *The living reef: The ecology of New Zealand's rocky reefs*, Nelson, Craig Potton Publishing.
- Annala J H, K J Sullivan, N W McL Smith, M H Griffiths, P R Todd, P M Mace and A M Connell, 2004, *Report from the fishery assessment plenary, May 2004: Stock assessments and yield estimates*, Wellington, Ministry of Fisheries.
- Aquaculture Steering Group, 2003, *He maara mataitai*, Wellington, Te Ohu Kai Moana.
- Arnold A (ed), 2005, *Shining a spotlight on the biodiversity of New Zealand's marine ecoregion: Experts workshop on marine biodiversity 27–28 May 2003*, Wellington, New Zealand, Wellington, WWF-New Zealand.
- Auckland Regional Council, 2003, *Auckland regional land transport strategy 2003*, Auckland, Auckland Regional Council.
- Auckland Regional Council, 2004, 'A study of roof runoff quality in Auckland, New Zealand: Implications for stormwater management', *ARC Technical Publication 213*.
- Baird S, 2004, Chief, Oceans Policy – Pacific Region, Fisheries and Oceans Canada, telephonic communication, 29 April.
- Baird S J, 2004, 'Incidental capture of seabird species in commercial fisheries in New Zealand waters, 2000–01', *New Zealand Fisheries Assessment Report 2004/58*.
- Ballantine B, 2004, 'A marine reserve manifesto', *New Zealand Geographic*, 69, 10–12.
- Bathgate A, 2004, 'Relief at trawling reprieve', *The Dominion Post*, 29 September, C2.

- Batson P, 2003, *Deep New Zealand: Blue water, black abyss*, Christchurch, Canterbury University Press.
- Batson P B and P K Probert, 2000, 'Bryozoan thickets off Otago Peninsula, New Zealand', *Fisheries Assessment Report 2000/46*.
- Bell R G, T M Hume and D M Hicks, 2001, *Planning for climate change: Effects on coastal margins*, Wellington, Ministry for the Environment.
- Benson-Pope D, 2004, 'Review of sustainability measures and other management controls for the 2004–05 fishing year', Minister's decision letter, Wellington, Ministry of Fisheries.
- Benson-Pope D, 2005, 'Minister calls commercial squid fishers back to port to save sea birds', Press Release, 6 May, available at www.beehive.govt.nz/ViewDocument.aspx?DocumentID=22919 (accessed 5 August 2005).
- Beston A, 2005, 'Fleet ordered into port over seabird slaughter', *The New Zealand Herald*, 6 May, 1.
- Boyd R O and J L Reilly, 2002, *1999/2000 national marine recreational fishing survey: Harvest estimates*, Draft New Zealand Fisheries Assessment Report, Wellington, Ministry of Fisheries.
- Brownell B, 2004, 'Mangroves: The cornerstone of a dynamic coastal environment', in B Brownell (ed), *Muddy feet; Firth of Thames Ramsar site update 2004*, Kaiaua, Ecoquest Education Foundation.
- Bush Administration, 2005, *U.S. Ocean Action Plan: The Bush administration's response to the U.S. Commission on Ocean Policy*, Washington.
- Cannon J, 2004, Conservation International, email communication, 28 April.
- Carbines G, W Jiang and M P Beentjes, 2004, 'The impact of oyster dredging on the growth of blue cod, *Parapercis colias*, in Fouveaux Strait, New Zealand', *Aquatic Conservation: Marine and Freshwater Ecosystems*, 14, 491–504.
- Cassidy M, 2005, Biosecurity New Zealand, email peer review note, 13 June.
- Centre for Advanced Engineering, 2003, *Economic opportunities in New Zealand's oceans: Informing the development of oceans policy*, Christchurch, Centre for Advanced Engineering.

- Chang F H, 2004, Scientist, National Institute of Water and Atmospheric Research Limited, personal communication, 27 July.
- Church D, 1998, 'State of the marine environment and marine ecosystem indicators', in C Wallace, B Weeber and S Buchanan (eds) *Seaviews marine ecosystem management: Obligations and opportunities*, Wellington, Environment and Conservation Organisations of New Zealand.
- Cicin-Sain B and R W Knecht, 1998, *Integrated coastal and ocean management: Concepts and practices*, Washington D.C., Island Press.
- Cicin-Sain B and R W Knecht, 2000, *The future of U.S. ocean policy: Choices for the new century*, Washington D.C., Island Press.
- Clark M and R O'Driscoll, 2003, 'Deepwater fisheries and aspects of their impact on seamount habitat in New Zealand', *Journal of Northwest Atlantic Fishery Science*, 31, 441–458.
- Collins S, 2004, 'Water fails clean, green test', *The New Zealand Herald*, July 3–4, A5.
- Commonwealth of Australia, 1997, *Australian oceans new horizons*, at www.oceans.gov.au/aust_oceans_new_horizons/default.jsp (accessed 17 May 2005).
- Commonwealth of Australia, 1998a, *Australia's oceans policy*, Canberra, Environment Australia.
- Commonwealth of Australia, 1998b, *Australia's oceans policy – specific sectoral measures*, Canberra, Environment Australia.
- Controller and Auditor-General, 1999, *Report of the Controller and Auditor-General: Fifth report for 1999*, Wellington, Parliament.
- Costanza R, R d'Arge, R de Groot, S Farber, M Grasso, B Hannon, K Limburg, S Naeem, R V O'Neill, J Paruelo, R G Raskin, P Sutton and M van den Belt, 1997, 'The value of the world's ecosystem services and natural capital', *Nature*, 387, 253–260.
- Cranfield H J, K P Michael and I J Doonan, 1999, 'Changes in the distribution of epifaunal reefs and oysters during 130 years of dredging for oysters in Fouveaux Strait, southern New Zealand', *Aquatic Conservation: Marine and Freshwater Ecosystems*, 9, 461–483.
- Cryer M, B Hartill and S O'Shea, 2002, 'Modification of marine benthos by trawling: Toward a generalization for the deep ocean?', *Ecological Applications*, 12(6), 1824–1839.

- Day D, 1995, 'Public policy and ocean management in Canada', *Marine Policy*, 19(4), 251–256.
- Department of Conservation, 2000a, *The New Zealand biodiversity strategy*, Wellington, Department of Conservation.
- Department of Conservation, 2000b, *Tapui taimoana: Reviewing the Marine Reserves Act 1971*, Wellington, Department of Conservation.
- Department of Conservation and Ministry of Fisheries, 2003a, 'Protection of critically endangered Maui's dolphin', brochure, Wellington, Department of Conservation and Ministry of Fisheries.
- Department of Conservation and Ministry of Fisheries, 2003b, *Marine reserves protocol*, Wellington, Department of Conservation and Ministry of Fisheries.
- Donoghue M, 1998, 'Management of the by-catch of protected species in New Zealand: A government agency perspective', in C Wallace, B Weeber and S Buchanan (eds) *Seaviews marine ecosystem management: Obligations and opportunities*, Wellington, Environment and Conservation Organisations of New Zealand.
- Dorman T L, 1996, 'Canada's ocean limits and boundaries: An overview', in L K Kriwoken, M Haward, D Vanderzwaag and B Davis (eds) *Oceans law and policy in the post-UNCED era: Australian and Canadian perspectives*, London, Kluwer Law International.
- Ecologic Foundation, 2001, *Towards an environmental management strategy for fisheries: A report to the Ministry of Fisheries*, Nelson, Ecologic Foundation.
- Environment Southland, Department of Conservation – Southland Conservancy, Invercargill City Council, Southland District Council and Te Ao Marama Incorporated, 2005, *The state of Southland's coastal marine environment*, Environment Southland, Invercargill.
- Eriksen N J, P R Berke, J L Crawford and J E Dixon, 2003, *Planning for sustainability; New Zealand under the RMA*, Hamilton, University of Waikato.
- Fisheries and Oceans Canada, 2003a, 'The overall state of Canada's fishery in 2002', at www.dfo-mpo.gc.ca (accessed 3 March 2004).
- Fisheries and Oceans Canada 2003b, *Performance report for the period ending March 31 2003*, Ottawa, Fisheries and Oceans Canada.

- Foster E G and M Haward, 2003, 'Integrated management councils: A conceptual model for ocean policy conflict management in Australia', *Ocean & Coastal Management*, 46, 547–563.
- Foster E G, M Haward and S Coffen-Smout, 2005, 'Implementing integrated oceans management: Australia's South East Regional Marine Plan (SERMP) and Canada's Eastern Scotian Shelf Integrated Management (ESSIM) initiative', *Marine Policy*, 29, 5, 391–405.
- Froude V, 1998, *Environmental performance indicators: An analysis of potential indicators for fishing impacts*, Technical Paper No 43 Marine Environment, Wellington, Ministry for the Environment.
- Froude V A and Smith R, 2004, *Area-based restrictions in the New Zealand marine environment*, Wellington, Department of Conservation.
- Gell F R and C M Roberts, 2003, *The fishery effects of marine reserves and fishery closures*, Washington, WWF-US.
- Gibbs N, 2004, email communication, 14 December.
- Goñi R, 1998, 'Ecosystem effects of marine fisheries: An overview', *Ocean & Coastal Management*, 40, 37–64.
- Gordon D, 2000, 'The Pacific ocean and global OBIS: A New Zealand perspective', *Oceanography*, 13(3), 41–47.
- Gordon D, 2004, unpublished notes, Wellington, National Institute of Water and Atmosphere.
- Government of Canada, 2002a, *Canada's oceans strategy: Our oceans, our future*, Ottawa, Fisheries and Oceans Canada.
- Government of Canada, 2002b, *Policy and operational framework for integrated management of estuarine, coastal and marine environments in Canada*, Ottawa, Fisheries and Oceans Canada.
- Government of Canada, 2002c, *Government response to the fourth report of the Standing Committee on Fisheries and Oceans 'Report on the Oceans Act'*, Ottawa, Fisheries and Oceans Canada.
- Hancock N and J Hewitt, 2004, 'Modelling the effects of muddy waters on shellfish', *Water & Atmosphere*, 12(2), 16–17.

- Hauraki Gulf Forum, 2005, *The Hauraki Gulf state of the environment report*, Auckland, Auckland Regional Council.
- Haward M, 1996, 'Institutional framework for Australian ocean and coastal management', *Ocean & Coastal Management*, 33, 19–39.
- Hayward B W, 1997, 'Introduced marine organisms in New Zealand and their impact in the Waitemata Harbour, Auckland', *Tane*, 36.
- Helm P, 1998, 'New Zealand's ocean future opportunities and responsibilities', in C Wallace, B Weeber and S Buchanan (eds) *Seaviews marine ecosystem management: obligations and opportunities*, Wellington, Environment and Conservation Organisations of New Zealand.
- Hembry O, 2005, 'Fishers launch research venture', *New Zealand Herald*, 21 February, C4.
- Herriman M, M Tsamenyi, J Ramli and S Bateman, 1997, *Review of international agreements, conventions, obligations and other instruments influencing use and management of Australia's marine environment*, Background Paper 2, Australia's Ocean Policy, Canberra, Environment Australia.
- Hewitt C L, J Willing, A Bauckham, A M Cassidy, C M S Cox, L Jones and D M Wotton, 2004, 'New Zealand marine biosecurity: delivering outcomes in a fluid environment', *New Zealand Journal of Marine and Freshwater Research*, 38, 429–438.
- Hicks M, U Shankar and A McKerchar, 2003, Sediment yield estimates: A GIS tool, *Water & Atmosphere* 11(4), 26–27.
- Hildebrand L and A Chircop, 2000, 'Integrated coastal and ocean management in Canada', paper presented at the first workshop phase 2 of the Australian-Canadian Oceans Research Network, Vancouver, December 10–11.
- Hodgson P, 2000, 'The oceans policy: Managing NZ's marine environment', speech delivered to the Energy and Natural Resources Law Association, 12 October.
- Hodgson P, 2003, 'Minding our oceans', *Oceans Blueprint*, 1, 3.
- Honeywill C, R Ford and R Babcock, 2002, *Long Bay-Okura marine reserve: Ecological baseline summary report*, Auckland, North Shore City Council.
- Hume T, 2003, 'Estuaries and tidal inlets', in J R Goff, S L Nichol and H L Rouse (eds), *The New Zealand coast: Te tai o Aotearoa*, Palmerston North, Dunmore Press.

- Hume T and A Swale, 2003, 'How estuaries grow old', *Water & Atmosphere*, 11(1), 14–15.
- Industry New Zealand, 2001, *An initial survey of the biotechnology industry in New Zealand: A discussion document*, Wellington, Industry New Zealand.
- Jacques P and Z A Smith, 2003, *Ocean politics and policy: A reference handbook*, Santa Barbara, ABC-CLIO.
- Juda L, 1999, 'Considerations in developing a functional approach to the governance of large marine ecosystems', *Ocean Development & International Law*, 30, 89–125.
- Juda L, 2003a, 'Changing national approaches to ocean governance: the United States, Canada and Australia', *Ocean Development & International Law*, 34, 161–187.
- Juda L, 2003b, 'Obstacles to ecosystem-based management', paper presented at the Global Conference on Oceans, Coasts, and Islands, November 12–14, Paris.
- Kaiser M J, J S Collie, S J Hall, S Jennings and I R Poiner, 2002, 'Modification of marine habitats by trawling activities: Prognosis and solutions', *Fish and Fisheries*, 3, 114–136.
- Land Information New Zealand (2005) at http://www.linz.govt.nz/rcs/linz/pub/web/root/core/Hydrography/CurrentProjects/continentalsheffolder/cont_shelf_ausnz/index.jsp (accessed 19 May 2005).
- Livingston M E, M R Clark and S – J Baird, 2003, 'Trends in incidental catch of major fisheries on the Chatham Rise for fishing years 1989–90 to 1998–99', *New Zealand Fisheries Assessment Report 2003/52*.
- Maritime Safety Authority, 2003, *Statement of intent 2003/04, 2004/05, 2005/06*, Wellington, Maritime Safety Authority.
- Markowitz T M, A D Harlin, B Würsig and C J McFadden, 2004, 'Dusky dolphin foraging habitat: Overlap with aquaculture in New Zealand', *Aquatic Conservation: Marine and Freshwater Ecosystems*, 14(2), 133–149.
- McCarthy D, 2002, *Mapping constraints to future marine farming in the Auckland and Waikato regions: Stage one*, Auckland, Auckland Regional Council.
- McCrone A, 2001, *National overview of biological monitoring in New Zealand's marine protected areas*, Wellington, Department of Conservation.

- McDermott Fairgray Group Limited, 2000, *Economic impact assessment for New Zealand regions*, Wellington, New Zealand Seafood Industry Council.
- Minister of Fisheries and Minister for the Environment, 2001, *Aquaculture reforms paper A: Overview of proposed aquaculture reforms*, paper presented to the Cabinet Finance, Infrastructure and Economic Committee.
- Ministerial Advisory Committee on Oceans Policy, 2001, *Healthy sea: Healthy society: Towards an oceans policy for New Zealand*, report on consultation undertaken by the Ministerial Committee on Oceans Policy, 30 September, Wellington, Oceans Policy Secretariat.
- Ministry for the Environment, 2001, *Environmental performance indicators: Confirmed indicators for the marine environment*, Wellington, Ministry for the Environment.
- Ministry for the Environment, 2005a, *Getting our priorities right: The role of information in setting priorities for management of New Zealand's ocean*, Wellington, Ministry for the Environment.
- Ministry for the Environment, 2005b, *Offshore options: Managing environmental effects in New Zealand's exclusive economic zone*, Wellington, Ministry for the Environment.
- Ministry of Economic Development, 2004, 'Government provides incentives to generate increased exploration activity', media statement, 14 June, Wellington, Ministry of Economic Development.
- Ministry of Fisheries, 2002, *Marine biodiversity: Medium-term research plan 2002/03 to 2005/06*, draft July, Wellington, Ministry of Fisheries.
- Ministry of Fisheries, 2004, *Briefing for the Minister of Fisheries*, Wellington, Ministry of Fisheries.
- Ministry of Fisheries, 2005, *Strategy for managing the environmental effects of fishing*, Wellington, Ministry of Fisheries.
- Ministry of Fisheries and Department of Conservation, 2004a, *National plan of action to reduce the incidental catch of seabirds in New Zealand fisheries*, Wellington, Ministry of Fisheries and Department of Conservation.
- Mitchell C L, 1998, 'Sustainable oceans development: the Canadian approach', *Marine Policy*, 22(4-5), 393-412.
- Morgan J, 2004, 'Slashing of hoki quota hailed', *The Dominion Post*, 24 September, C1.

- Morrison M, M Francis, D Parkinson, E Skipworth, M Timperley and K Webster, 2004, 'Quantifying the relative contributions of different estuaries to coastal stocks', powerpoint presentation, Wellington, National Institute of Water and Atmospheric Research.
- National Oceanic and Atmospheric Administration, 2005, 'NOS's contribution to NOAA's role in executing the U.S. Ocean Action Plan' at www.oceanservice.noaa.gov/v05roundtables/rt1/welcome.html (accessed 17 May 2005).
- National Oceans Office, 2002, *Impacts – identifying disturbances*, South-East Regional Marine Plan Assessment Reports, Hobart, National Oceans Office.
- National Oceans Office, 2003, *Oceans policy: Principles and processes*, Hobart, National Oceans Office.
- National Oceans Office, 2004, *South-East Regional Marine Plan: Implementing Australia's oceans policy in the south-east marine region*, Hobart, National Oceans Office.
- National Round Table on the Environment and the Economy, undated, *Eastern Scotian Shelf integrated management (ESSIM) project*, Ottawa, National Round Table on the Environment and the Economy.
- New Zealand Coastal Society, 2005, *NZ Coastal Society strategic plan (2004–2007)*, at www.coastalsociety.org.nz (accessed 23 August 2005).
- New Zealand Press Association, 2005, 'Prospector confident of rich sea deposits', *The New Zealand Herald*, 16 August, C3.
- Nicholls P and J Ellis, 2002, 'Fringing habitats in estuaries: The sediment-mangrove connection', *Water & Atmosphere*, 10(4), 24–25.
- Norton D, 2002, *Review of the implementation of oceans policy: Final report*, TFG International Pty Limited.
- Oceans Policy Secretariat, 2003a, *Marine biosecurity*, Working Paper Six, Wellington, Ministry for the Environment.
- Oceans Policy Secretariat, 2003b, *Information issues*, Working Paper One, Wellington, Ministry for the Environment.
- Oceans Policy Secretariat, 2003c, 'The team', *Oceans Blueprint*, 1, 7.

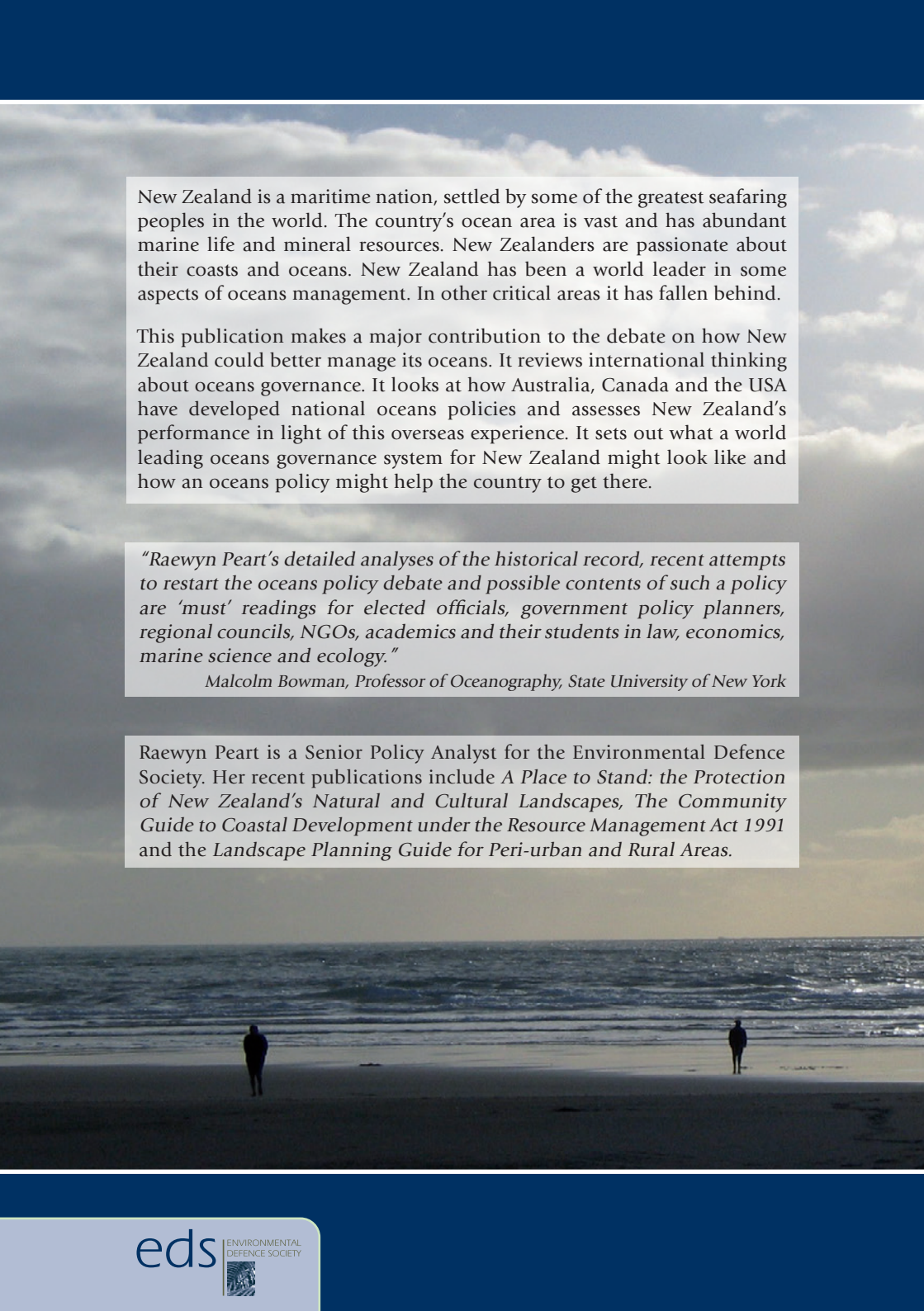
- Parliamentary Commissioner for the Environment, 1997, *Management of the environmental effects associated with the tourism sector*, Wellington, Parliamentary Commissioner for the Environment.
- Parliamentary Commissioner for the Environment, 1999, *Setting course for a sustainable future: The management of New Zealand's marine environment*, Wellington: Parliamentary Commissioner for the Environment.
- Parliamentary Commissioner for the Environment, 2003, *Just cruising? Environmental effects of cruise ships*, Wellington, Parliamentary Commissioner for the Environment.
- Peart R, 2004, *A place to stand: The protection of New Zealand's natural and cultural landscapes*, Auckland, Environmental Defence Society.
- Peart R, 2005, *The community guide to coastal development under the Resource Management Act 1991*, Auckland, Environmental Defence Society.
- Petrachenko D M, undated, *Practicalities on oceans development in Canada*, Vancouver, Fisheries and Oceans Canada.
- Pew Oceans Commission, 2003, *America's living oceans: Charting a course for sea change*, Arlington, Pew Oceans Commission.
- Probert P K, 1999, 'Seamounts, sanctuaries and sustainability: Moving towards deep-sea conservation', *Aquatic Conservation: Marine and Freshwater Ecosystems*, 6, 601–605.
- Rahall N, 2005, 'Bush budget torpedoes his own Oceans Commission', *News from the Democrats*, 8 March at www.house.gov/apps/list/press/ii00_democrats/oceanfunding.html (accessed 3 May 2005).
- Read G and S Handley, 2004, 'New alien mudworm now becoming a pest in longline mussels', *Water & Atmosphere*, 12(3), 30–31.
- Risk C M, 2002a, 'An oceans policy for New Zealand: Why, what, how?', speech, Wellington, Office of the Honourable Pete Hodgson.
- Risk C M, 2002b, *Oceans policy project: Structure and methodology for stage two*, Ad Hoc Ministerial Group on Oceans Policy, Wellington, Office of the Honourable Pete Hodgson.
- Risk C M, 2002c, *Oceans policy: Stage two work programme*, Ad Hoc Ministerial Group on Oceans Policy, Wellington, Office of the Honourable Pete Hodgson.

- Rosier J, 2004, *Independent review of the New Zealand coastal policy statement*, Palmerston North, Massey University.
- Rothwell D R, 1996, 'Australia and the law of the sea: Recent developments and post-UNCED challenges' in L K Kriwoken, M Haward, D Vanderzwaag and B Davis (eds) *Oceans law and policy in the post-UNCED era: Australian and Canadian perspectives*, London, Kluwer Law International.
- Rutherford R J, G J Herbert and S S Coffen-Smout, 2005, 'Integrated ocean management and the collaborative planning process: The Eastern Scotian Shelf integrated management (ESSIM) initiative', *Marine Policy*, 29, 1, 75–83.
- S & J Woodley Pty Limited, 2002, *Review of implementation of ocean's policy: Report on planning and scientific and technical support components*, Arcadia, S & J Woodley Pty Ltd.
- Seafood Industry Council, undated, *Sustaining the environment*, pamphlet, Wellington, Seafood Industry Council.
- Secretariat of the Convention on Biological Diversity, 2004, *Technical advice on the establishment and management of a national system of marine and coastal protected areas*, Montreal, Secretariat of the Convention on Biological Diversity.
- Sheppard L, 2004, Manager, New Zealand Oceans Policy Secretariat, personal communication, 2 April.
- Smith P, L Hauser and G Adcock, 2003, 'Overfishing leads to loss of genetic diversity in Tasman Bay', *Water & Atmosphere*, (11(1), 7.
- Snelder T, J Leathwick, M Weatherhead and M Wild, 2004, *Pilot marine environment classification of New Zealand's exclusive economic zone and the Hauraki Gulf*, Christchurch, National Institute of Water and Atmospheric Research.
- Southern Seabird Solutions, undated, *Conservation through co-operation*, flyer, Wellington, Department of Conservation.
- Standing Committee on Fisheries and Oceans, 2001, *Report on the Oceans Act*, October, Ottawa, House of Commons.
- Statistics New Zealand, 2005a, *Tourism satellite account 2001–2004*, Wellington, Statistics New Zealand.
- Statistics New Zealand, 2005b, *New Zealand external trade statistics: December 2004*, Wellington, Statistics New Zealand.

- Suisted R and D Neale, 2004, *Department of Conservation marine mammal action plan for 2005–2010*, Wellington, Department of Conservation.
- Sullivan S, 2002, *Beyond our coasts – Australia's ocean policy*, paper presented at Coast to Coast 2002.
- Taylor G A, 2000, 'Action plan for seabird conservation in New Zealand, part A: Threatened seabirds', *Threatened Species Publication No 16*.
- Taylor N and B Buckenham, 2003, 'Social impacts of marine reserves in New Zealand', *Science for Conservation* 217.
- Taylor R and I Smith, 1997, *The state of New Zealand's environment 1997*, Wellington, Ministry for the Environment and GP Publications.
- Te Ohu Kai Moana, 2001, Submission from Te Ohu Kai Moana to the Ministerial Advisory Committee on Oceans Policy, Wellington, Te Ohu Kai Moana.
- The Boston Consulting Group, 2004, *Auckland regional stormwater project: An action plan to deliver improved stormwater outcomes*, unpublished report, Auckland, Auckland Regional Council and Infrastructure Auckland.
- The Joint Group of Experts on the Scientific Aspects of Marine Environmental Protection, 2001, *A sea of troubles*, Nairobi, United Nations Environment Programme.
- Thompson W, 2005, 'Oyster farm plan stalled', *The New Zealand Herald*, 8 August, A13.
- Thrush S F and P K Dayton, 2002, 'Disturbance to marine benthic habitats by trawling and dredging: Implications for marine biodiversity', *Annual Review of Ecology and Systematics*, 33, 449–473.
- Thrush S F, J E Hewitt, V J Cummings and P K Dayton, 1995, 'The impact of habitat disturbance by scallop dredging on marine benthic communities: What can be predicted from the results of experiments?', *Marine Ecology Progress Series*, 129, 141–50.
- Thrush S F, J E Hewitt, V J Cummings, P K Dayton, M Cryer, S J Turner, G A Funnell, R G Budd, C J Milburn and M R Wilkinson, 1998, 'Disturbance of the marine benthic habitat by commercial fishing: Impacts at the scale of fishery', *Ecological Applications*, 8(3), 866–879.

- Thrush S F, D Schultz, J E Hewitt and D Talley, 2002, 'Habitat structure in soft-sediment environments and abundance of juvenile snapper *Pagrus auratus*', *Marine Ecology Progress Series*, 245, 273-280.
- Tibbetts J, 2005, 'America's oceans: A blueprint for the future', *Environmental Health Perspectives*, 113 (2).
- Tourism Research Council New Zealand, 2005, *International visitor survey year end December 2004*, Wellington, Tourism Research Council New Zealand.
- Turner S J, S F Thrush, J E Hewitt, V J Cummings and G Funnell, 1999, 'Fishing impacts and the degradation or loss of habitat structure', *Fisheries Management and Ecology*, 6, 401-420.
- US Commission on Ocean Policy, 2002, *Developing a national ocean policy: Mid-term report of the U.S. Commission on Ocean Policy*, Washington D.C, US Commission on Ocean Policy.
- US Commission on Ocean Policy, 2004, *An ocean blueprint for the 21st century*, Washington D.C, US Commission on Ocean Policy.
- US Senate, 2000, *Oceans Act of 2000*, Report of the Committee on Commerce, Science and Transportation, May 23.
- Vallega A, 2001a, 'Ocean governance in post-modern society – a geographical perspective', *Marine Policy*, 25, 399-414.
- Vallega A, 2001b, *Sustainable ocean governance: A geographical perspective*, London, Routledge.
- Vince J, 2005, 'The South East Regional Marine Plan: Implementing Australia's ocean policy', *Marine Policy*, in press.
- Wallace C, B Weeber and S Buchanan (eds), 1998, *Seaviews marine ecosystem management: Obligations and opportunities*, Wellington, Environment and Conservation Organisations of New Zealand.
- Watling L and E A Norse, 1998, 'Disturbance of the seabed by mobile fishing gear: A comparison to forest clearcutting', *Conservation Biology*, 12(6), 1180-1197.
- Westcott G, 2000, 'The development and initial implementation of Australia's 'integrated and comprehensive' oceans policy', *Ocean & Coastal Management*, 43, 853-878.

- Westcott G, 2004, personal communication, 24 March.
- Williamson B, G Lewis, G Mills and B Vant, 2003, 'Contaminants on the coast', in J R Goff, S L Nichol and H L Rouse, *The New Zealand coast: Te tai o Aotearoa*, Palmerston North, Dunmore Press.
- Willis G, 2003, *Oceans management at the local level*, research report for the Oceans Policy Secretariat, Auckland, Enfocus Limited.
- Willis G, J Gunn and D Hill, 2002, *Oceans policy stocktake: Part I – legislation and policy review*, Wellington, Oceans Policy Secretariat.
- Willis G, J Gunn J and D Hill, 2004, *Oceans policy stocktake: Part 2 – international and conceptual approaches*, Final Draft, Wellington, Oceans Policy Secretariat.
- Wilson K, 2004, *Flight of the huia: Ecology and conservation of New Zealand's frogs, reptiles, birds and mammals*, Christchurch, Canterbury University Press.
- WWF-New Zealand, 2005, *New Zealander's views on threats and protection in the marine environment: Results of a Colmar Brunton national survey*, Wellington, WWF-New Zealand.
- Young O R, 2003, 'Environmental governance: The role of institutions in causing and confronting environmental problems', *International Environmental Agreements: Politics, Law and Economics*, 3, 377–393.
- Zann L P, 1996, 'The state of the marine environment report for Australia (SOMER): Process, findings and perspectives', *Ocean & Coastal Management*, 33, 63–86.



New Zealand is a maritime nation, settled by some of the greatest seafaring peoples in the world. The country's ocean area is vast and has abundant marine life and mineral resources. New Zealanders are passionate about their coasts and oceans. New Zealand has been a world leader in some aspects of oceans management. In other critical areas it has fallen behind.

This publication makes a major contribution to the debate on how New Zealand could better manage its oceans. It reviews international thinking about oceans governance. It looks at how Australia, Canada and the USA have developed national oceans policies and assesses New Zealand's performance in light of this overseas experience. It sets out what a world leading oceans governance system for New Zealand might look like and how an oceans policy might help the country to get there.

"Raewyn Peart's detailed analyses of the historical record, recent attempts to restart the oceans policy debate and possible contents of such a policy are 'must' readings for elected officials, government policy planners, regional councils, NGOs, academics and their students in law, economics, marine science and ecology."

Malcolm Bowman, Professor of Oceanography, State University of New York

Raewyn Peart is a Senior Policy Analyst for the Environmental Defence Society. Her recent publications include *A Place to Stand: the Protection of New Zealand's Natural and Cultural Landscapes*, *The Community Guide to Coastal Development under the Resource Management Act 1991* and the *Landscape Planning Guide for Peri-urban and Rural Areas*.